

Math 230 Foundations of Statistical Thinking

Part 1

Course transcript with lecture-scoped numbered paragraphs

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1. Thinking Statistically

At a glance

Statistical thinking turns data into evidence about populations. This lecture introduces populations, samples, parameters, statistics, variables, levels of measurement, study design, sampling, bias, and the ethical obligation to collect data with integrity.

- 1.1 I am so excited that you are all here to learn **statistics**. Hopefully, you want to learn **statistics** and are at least a little bit interested in what it is, because a lot of times we enter into a subject matter; maybe we don't even understand why in the world we're learning it. Because sometimes we have misconceptions about what it is, what it isn't, whether or not it is fun, boring, or even useful, honestly.
- 1.2 The best thing about statistics is that it's not a math class, which is excellent for you because it's not all about formulas and doing these hardcore computations. We have programs that do that. It's really an understanding the world around you, and that makes you more powerful as a consumer and as a person who's going to interact with your world, whether it be voting or purchasing or marketing. All of these things revolve around understanding something called "data."
- 1.3 You say, "What in the world is data?" I swear when I was a kid, I thought data was a guy on Star Trek: Next Generation, and I went, "Wow, this android is super powerful because he controls everything," is what my statistics teacher said. Turns out data is not an android—he is, but not what we're talking about at all.
- 1.4 You see, what statistics is all about is understanding how data affects the world around you. It's organizing, it's collecting, it is analyzing, and it is interpreting this data so that you can make decisions. We might be asking, why are these things so important? Collecting data, the way in which we do this, can be very, very productive to understand the world. It can also be very, very counterproductive if we collect it completely the wrong way.
- 1.5 You probably have understood that if you surround yourself with a lot of your friends, they might think exactly the same way that you think. If you ask them a question, you might be getting your own thoughts regurgitated back to you. This makes for absolutely terrible statistics because it influences the way that we present this information to the world in what we call a biased manner.
- 1.6 Collection of data is super important. Organizing this data in a way that we can see it is so important because, honestly, when we look at raw data, it's going to be overwhelming. In the next lesson, we do this. We take a look at raw data and you're going to get bored within about five seconds. We do not want this. Organizing it in something like a spreadsheet makes it at least able to look through.
- 1.7 Analyzing the data—what does it look like as far as its center or how it's spread or the distribution? These things are going to make sense when we go through them. Then finally, interpreting the data. How do we get this raw data to actually make sense so that you can make decisions that matter? It helps you understand patterns, to get a picture of what it looks like overall, to make predictions.
- 1.8 Honestly, the whole thrust of statistics is to make decisions under uncertainty. Why would we ever want to make decisions under uncertainty? Well, the fact of it is that you're never going to get a full picture of every member of a **population**—which we just use the word, I'll explain in a minute—of a **population** that you want to study.
- 1.9 Why don't we do this? Why don't we see a population of everyone we want to study in medicine or in hiring or in sports, all these things where you see statistics often used? Well, they're enormous. When you think about the population of maybe the United States of America, there's hundreds of millions of people.
- 1.10 To collect all this information about all of these people all at one time and then gather that information and then analyze it, organize it, and interpret that, you can probably imagine that's going to take five minutes. No, it's going to take a tremendous amount of time to actually do that—time and money and effort. And by the time you get all of this information together, it might actually be irrelevant.
- 1.11 So do we ever collect data from all of our population, from every single element in the group that we want to study? Yeah, of course we do. In America, we call that a census. So you might have seen a census worker. Maybe you have been a census worker, and you've gone out and collected data; you knocked

on every door. It takes a number of years to actually do this data collection and analysis before you even get it out to the public.

- 1.12** Is it important? Absolutely. That's why we do it. It's been done for literally millennia. I can think of King David in Israel is the first one who actually is written down as having done a census, and he collects information about everybody in the nation of Israel. We do that in America so that we can get a real look at what is going on for our citizens, for our country, so that we can make decisions about infrastructure, all sorts of things.
- 1.13** What's the main point of that? Oftentimes, you are not going to have time or energy or money to do a census all the time. What do we do instead? Well, we take something called a "sample." A sample says, "I'm going to take a specific group of people selected from this population and collect information about them."
- 1.14** Now, you might have this thought in your head: "Wait a second. If we are taking a sample, is that sample going to perfectly represent our population?" Well, no, it can't. There's no way that you five people would represent 300 million people in America. There's no way that that's possible. It's not going to perfectly represent every single subgroup, every category, every demographic, every socioeconomic status. We can't do that.
- 1.15** And so a lot of people, when they enter into statistics, they say, "Why are we doing this? If this sample is not going to perfectly represent our population, is it even worth collecting?" Oh, absolutely.
- 1.16** See, when we start to apply these statistical methods and when we start to gather a sample with integrity so that it actually represents our population, when we apply these methods with integrity—and we're talking a lot about that at the end of our lesson today. When we start to apply these methods with integrity, using them when they're appropriate, not using them when they're not appropriate, we can say a lot of things about the population with just a small sample or a relatively small sample.
- 1.17** What we're really trying to do is make a claim about what's called a "parameter." A parameter—really, what it means is just a characteristic of the population that you want to study: age, hair color, favorite ice cream flavor. It doesn't matter. But we want to find out this parameter.
- 1.18** Now, in your minds right now, I want to build that you are not often going to know what the parameter is. Why? Well, you're not collecting the entire population. Does this make sense? You can't really gather everything to know everything about the parameter. So, what we are going to do is gather a sample and take information from the sample and make claims about the population. Take information from a sample statistic—that's where this whole class comes from, is a sample statistic—and use it to make claims about a population parameter. Why? Because we're not going to have the whole population. We deal with samples.
- 1.19** A couple of things as we move forward so that you are prepared for this is that population parameters, they always use Greek symbols. They look a little funny because we don't really speak Greek all that much anymore. But the funny-looking letters that look like Greek are actually Greek. Every time you said, "Man, that's all Greek to me." You're right. Now, this is perfect. They're Greek letters. The sample statistics, those are going to be English letters. We'll have funny letters like mu, sigma, rho for our populations, and I'll refer to them as we go. We'll have things like $\bar{X}$, $\hat{P}$, and I know those all sound funny, $\hat{Y}$. Those are going to be our sample statistics that we use to estimate our population parameter.
- 1.20** That's a key word for us, this word "estimate." You see, if you don't actually know your entire population and you're trying to say this statistic points to this parameter, you're going to estimate it. We can do a really good job of that, but it's never, ever going to be perfect.
- 1.21** That's why this is not a typical math class, which is why most people love it way more than math. Because you're like, "Wait, I can estimate and it's okay?" Absolutely. In our last lesson in part two of statistics, we talk about Bayesian thinking, and it's literally using what you believe to be the case as a probability. You're going to love that. We're not there yet.
- 1.22** Where we are right now is learning how to make these statistics estimate a parameter. We start with two different understandings that we're going to have: descriptive statistics and inferential statistics. All descriptive means is probably what you think it means. It's just describing your data. It's taking a look at

a summary of it, maybe a measure of center. It'll give you a graph, it'll organize, it'll give you a spreadsheet.

- 1.23** A descriptive statistic only really describes the data. It's what you can see—what's right there in front of you. You've collected it; you're looking at it. What we're going to do is take those descriptive statistics and make inferences, make claims about a population. This is making decisions about the things that you actually can't see, and that's pretty impressive. That's what this whole class is all about. Two classes, really—what we have is understanding how statistics is used to represent populations, how statistics is used to estimate parameters well. That's what we're looking at.
- 1.24** For example, just to get our brains wrapped around it. If I told you that the average GPA—and I'm using some very non-statistical terms here because we haven't talked about mean or median or mid-range or IQR, any of those things. We're going to use an average for now, but I'm going to flesh that out as we go. If I told you the average GPA is 3.1, and that was from 50 students, that's not making an inference about a population; that is describing a sample of 50 students. This would be a descriptive statistic.
- 1.25** Now I say, okay, we did a large sample, it's a random sample, and we collected this data, and we infer from that that 90% of people, plus or minus, like 2%, prefer Peterson Academy. Now that's an inference. That says yes. Now I'm taking this sample. I've done appropriate statistical methods with integrity. It's collected well, it's analyzed well, and now we have a great inference that if this is true—hopefully it is—that we have now an inferential claim. That's what we're doing.
- 1.26** Let's take an example. If you wanted to study how many hours a typical college student studies, what's your population? You'd think, wow, that's a surprising amount of people because there's a lot of people in college right now. You'd say, well, the population is all the people who are actually in a college class.
- 1.27** Are you going to go out and collect information from every single person who's in the college class right now? Now, that would be silly and absurd and honestly, the question is not that important to do. Sometimes they are. That's why we have a census once in a while. But we're not going to do that.
- 1.28** So, the population is all the college students. Let's say you go out and you select 120 randomly selected college students. That would be called your sample. We're going to use that sample to make inferences about your population. Your parameter would be the true amount of hours studied by your college students. Notice why I said "the true parameter." Are you ever going to actually know that? No, not unless you take a census.
- 1.29** What we're going to do is take our statistic—the average from the 120 students—as a point estimate, and then build what we call a "confidence interval" around it and say, "Hey, we're pretty sure that this interval represents our population." That's really the whole point of what we're doing.
- 1.30** Before we get into that, we have a lot of different types of data that you need to understand how to use. The reason why we like to talk about the different types of data is as we go through our course, there are a lot of different methods that we have, a lot of different tests, a lot of different practices, but they only work with certain types of data. Even organizing them is really important to understand how to organize these things.
- 1.31** Probably the biggest difference that I want to get across to you today is the difference between what we have called "categorical data" and what we have called "quantitative data" or numerical. We use those interchangeably: numerical, quantitative. That makes sense because we have quantities of things. They're usually represented by numbers.
- 1.32** We have qualitative data, which is categorical. We say, okay, categorical data is very important data, but it doesn't necessarily have to do with numbers. You could probably think of some categorical data that you can't assign values to, but still super important. Gender. Gender's categorical data. Eye color, categorical data, things that you can only put in categories.
- 1.33** You could not probably put those things in order. Some categorical data you can, but a lot of other data is numerical in nature: height, age, number of cups of coffee that you drink in a day. All of those things are going to lend themselves to certain methods that we use, how we analyze this, and it's super, super important for the statistical software that you're going to be using in this class. I'll show you that at the end of our lesson.
- 1.34** Within numerical data, you have a couple of different types of that, too. This is going to make sense because you know that all numbers aren't the same. You know that when you roll a standard die that has

six sides, how many outcomes are there? Thirty-four? No, not 34. There's only six. You have one or two or three or four or five or six. You can't get 5.7. You can't get nine.

- 1.35** That is called "discrete data." What does "discrete" mean? It means a finite amount that's available, or that you can count it. One, two, three, four, five. Or there's a fixed number, like the number of sides on a die, the number of sides on a coin, if you'd like.
- 1.36** Continuous measurements or continuous data is different. Continuous data is like a scale. Think scale, think age, think height, think weight. It's something that you would have an infinite number of possible values between any two of those measurements. For instance, if someone has a certain height, they're not probably going to be exactly 70 inches tall.
- 1.37** Maybe they're somewhere between 70 inches and 71 inches, but there are an infinite number of possibilities, even between those two. You're not just 70 inches, you're 70.01, or you could be 70.0001, or any number in between there because that's a real number line and we can be any number on that. That is where continuous data comes from. We have this idea of a scale, and that's something we'll use for our software in just a little bit.
- 1.38** Discrete, think counting, think finite; continuous, think you're measuring something, and there's an infinite number of possible values in any interval that you have there, which blows our minds when we think about it. But it's very important, and here's why. As we get onto this—and I know I typically will front-load a lot of the lessons we have later so that you understand where we're going. When you deal with discrete data, there are different distributions we can use.
- 1.39** You see, you can either have a one or a two or a three or a four on a die, which means there's only really six categories that we can put our counts into for that. That builds a completely different distribution called a "discrete distribution." But when we talk about height or weight, for instance, there's an infinite number of values. You can't say you're going to fall just in the 71-inch category. You could be anything in there, and that's going to build a continuous distribution.
- 1.40** They're both equally important. We're going to move from discrete to continuous, but they are used for the rest of our course because there are two different types of numerical data that we use. Within that, within both of these, really—categorical and numerical, which are the words I'll be referencing here—between categorical and numerical, we also have these levels of measurement. What a level of measurement is, is that it determines what math you can do on it, for instance.
- 1.41** Let's talk about the **nominal** level of measurement. If you know any root words, "**nominal**" sounds like "name," really. That's honestly what it means: **nominal** is just by name. They're categories, but you can't order them. For instance, we talked about gender. Gender would be a **nominal** category. You're not going to put female, male, male, female in any way that you can discern that one is higher or lower than the other. That doesn't even make sense to do.
- 1.42** Now, let me ask you a question. Can you actually find an average of gender? Can you add them up and divide by the number that you added? No, that doesn't make sense. Certain math operations only make sense at certain levels. Nominal means you can put the data in categories, but you can't really do much other than that. You can't order it, you can't add, you can't subtract, you can't find ratios. None of that makes sense.
- 1.43** The next level of measurement would be **ordinal**. This is still categorical. You still have these categories, but now you can order them: low, medium, high. Mood level, maybe: low, medium, high. Amount of caffeine that you've had this morning: none, a little bit, or woohoo. Just kidding. **Ordinal**, you still can't do a lot of math on those most of the time. Sometimes you can trick the programs and add them and find means. I'll show you that in just a little bit. But in general, they're just categorical. You just put them in order.
- 1.44** Interval is different. Interval is the data that has a scale to it, that has equal intervals between your numbers. However—and this is weird for a lot of people—it doesn't have a true zero. What do you mean it doesn't have a true zero? Where zero doesn't mean the absence of everything. Think temperature here.
- 1.45** When you have zero degrees Celsius, it doesn't mean that there's absolutely no heat whatsoever. That's like negative 273 degrees Celsius, like zero Kelvin or something. That doesn't mean no heat. It's just an

arbitrary zero, and I'll prove it to you. Because zero degrees Celsius is the same thing as 32 degrees Fahrenheit. Well, 32 degrees Fahrenheit doesn't mean no heat. It's an arbitrary zero.

- 1.46** Here's what that means. You can still add these numbers and get something valuable. You can still average these numbers and find something valuable, but ratios don't work. What it means is if I took 70 degrees and divided by 80 degrees, it doesn't mean that 70 degrees has 87.5% of the heat of 80 degrees. That doesn't make sense. Why? Because zero doesn't mean zero. It's just a scale.
- 1.47** Very much like a satisfaction survey. Zero satisfaction doesn't necessarily mean you had literally none. It's just, I can't go any lower than zero, so I'm going to put that number there. Just like five doesn't mean the most satisfaction a person could ever have, ever, and the ratios between there don't work.
- 1.48** When does it work? Well, with ratio data. With ratio data, like the name, ratios make sense because it has a true zero. You can think money. If you have \$0, you have \$0. Seventy dollars divided by \$80 means you have 87.5% of the money that you had at \$80. Really, if you took anything out of the levels of measurement, it's that certain math works at certain levels.
- 1.49** Also for our programs, a lot of them work this way. They group the last two and they say, "Okay, we have nominal." Nominal means just by their name. You can put them in groups. You can't really order them. Ordinal says mostly categories. You can put them in order. You can't really do a whole lot of math there most of the time. But the last two, they group, and they call it "scale." For much of our computer programs that we use, we have nominal, we have ordinal, and we have scale.
- 1.50** Let's take a look at some of this. You go into a coffee shop and you decide that you don't want to stand there all day and collect data from every single person who's in that coffee shop. So, you randomly select 12 people and you put it in a chart like what we have here. Is this your population? Hopefully not. That'd be a pretty weak coffee shop.
- 1.51** Let's say the population is everybody who had coffee at this particular coffee shop on that day. You have randomly selected 12 of them. Let's go ahead and let's find out what our levels of measurement are here. Let's take a look first at drink type. Drink type data. Can you put that in order? Well, not really. It's just the names of the drinks.
- 1.52** Now, I could put them in order for me—a preference—but that's not going to give any order in general that we could do. You can't average drink type. You can't add it. A latte plus an americano equals probably disgusting. You don't want to do that. These are not really addable, orderable. They're just nominal level of measurement.
- 1.53** Now, you go on to size. If you look at size, that's definitely ordinal data. It is categorical. You're saying it's small, it's medium, it's large, whatever. You could put that in order. You still can't add it. You still couldn't find an average of this data.
- 1.54** Price you can, can't you? You take a look at price and go, "Oh, yes, yes, yes." Price is different. We had drink type, definitely nominal. It's categorical, it's nominal. We had size, that's ordinal. It's categorical, it's ordinal. But when we get to money, see, money has a true zero. Money you can add, money you can average, money you could find a standard deviation for. We'll see that later. Money is ratio. Zero means zero. This is ratio-level data. In a computer program, you would call this "scale."
- 1.55** Satisfaction. What's the satisfaction? This is a really good example of interval-level data. So interval, you can still find an average. Zero doesn't necessarily mean true zero, that your satisfaction was absolutely nothing whatsoever. It doesn't work like that, but the levels between them are equal.
- 1.56** The last one is customer. I left that for last because this blows people out of the water. Sometimes you go, "Wait a minute, Professor Leonard, that looked like numbers to me." You go, "Yes, it does." It does look like numbers. However, do the numbers mean anything? You see, instead of using Kim or Abel or Sally, I use numbers 1, 2, and 3 so that I would not personally identify these people. That should make sense.
- 1.57** We're hiding someone's name, but these numbers, they just represent names. Could you add these and actually do anything with this? Does Sally plus Johnny equal Abel? That doesn't make any sense at all, just like these numbers. If you average them, it doesn't make sense. Even though we're using numbers, this for a customer is nominal data. You cannot order it. It doesn't matter when the sample was taken; the order in which it was taken does not matter at all.

- 1.58** That's really what we have here: customer, categorical, nominal. Why? Why is it that? Well, the labels are customers. They don't measure anything at all. And drink type, categorical, nominal, for sure. We can't order these things. Size, honestly, it should be pretty apparent right now that we got ordinal because you can go small, medium, large, not a problem.
- 1.59** Price, we talked about—it's continuous, it's numerical, it's ratio, it has a true zero. The differences are meaningful. You can find a percentage of this that makes sense. You could average them. Satisfaction is not that; it's discrete because there's only a finite number that you could select, right? Here we couldn't do 4.7538—I would do something like that because I am me, but a normal human being would probably not do that. You'd pick four or five because that's what they would give you. This is an interval level of measurement.
- 1.60** Now, how do you actually get this data? There's really two ways that we collect data in statistics. One of them is with an **observational study**; the other one is with an **experiment**. Now, we hear these words a lot, don't we? We say, "Oh, yes, okay, we're going to go do an **experiment**." Most people do **experiments** very, very, very poorly. We'll talk about how to do an **experiment** here in just a little bit.
- 1.61** Observational studies, on the other hand, they don't actually do anything. You're just collecting data. These are very powerful, but if you want to show cause and effect, you probably need an experiment. I'll say that with a caveat. There are many times with integrity that you do not want to perform an experiment, for instance. If you wanted to prove that a drug actually had an effect on a person, you would probably do an experiment. We'll talk about control and blinding, all these things in a minute.
- 1.62** But if you wanted to prove that texting while driving causes accidents, you should not do an experiment. Because what you would have to do is, you'd have a group of people over here and say, "Tell you what, you're going to drive without texting. See how you do. You're going to drive and all you can do is text while you're driving." Well, that's horrible because that's going to damage these people's lives, and we do not want to do that as much as possible to intentionally affect someone's life like that.
- 1.63** There are a lot of times when observational studies are great, when we cannot do an experiment. Observational means you're observing something. You have no intervention. You are not performing any action upon them. You're giving them no treatment. You're just looking basically at who these people are, selecting them with integrity, making sure that it is random, and then seeing what they did.
- 1.64** That's different from an experiment. The experiment says the same thing—collects the people, makes sure it's random, and then you do something to them and you measure the effect of that. That's what an experiment does.
- 1.65** Let's see if we can determine whether this is an observation or whether this is an experiment. You go out to a local college and you randomly select out of the students at the local college, so your population is every student at the local college. You say, "I do not want to take that much data just to determine the number of cups of coffee they drink. That's crazy." So we're going to select 10 people. We're going to put them in order. We're going to randomly select them—we'll talk about sampling several lessons from now—and we get 10 of them. And then we collect some information from them, and this is it.
- 1.66** Did you perform an experiment? I hope not because that's terrifying for me to be chosen as a history major, and you said, "You are now a history major. Let's see how much coffee you drink." "Not as much as engineering, I promise you that." This is an observational study. All that's happened is you selected your 10 people here; you collected information. What did you collect?
- 1.67** Maybe in your mind right now, you can look up there, look at major, and think about what that would be. Is that nominal data? Is that ordinal data? Is that numerical? Is it categorical? Well, it has no numbers on it, so it can't be numerical. It's got to be categorical.
- 1.68** Now, could you look at those categories and put them in an order that says this is the lowest and this is the highest? I have a personal preference here, but no, the answer is no. No, this would be a nominal level of measurement. You can't put these in any discernible order.
- 1.69** You look at the next one and say, study hours. That's different. In your mind right now, you should be building up that this is probably not categorical because it has numbers, and those numbers actually make sense. I could add these together and get a total number of study hours that's telling you that it's numerical. It's probably not discrete, even though there's whole numbers here, because you could study 9.5 hours or anywhere in between there. This is continuous data. This would be not interval, but ratio

because you could literally study zero hours, and I have had many students who have tried that. It doesn't work.

- 1.70** The next one, class year. Class year has no numbers. It has to be categorical. Class year has some categories. Could you put them in order? Yes, this is categorical data, but you can order it. That's ordinal.
- 1.71** Cups of coffee. Cups of coffee, well, typically, mine's not up there. That's usually about five cups of coffee. But this would also be a type of scale we would use for a computer program. This would be ratio. You could actually compare percentages of how much coffee someone drank. Zero means zero. If you had no coffee, you had no coffee. This would be numerical, and this would be ratio data.
- 1.72** How about the student? You probably know now. How about the student? Even though it's got numbers, does it actually mean the numbers that they are? Could you add them? Could you do anything with it? No, this is just nominal, categorical and nominal.
- 1.73** How do you conduct an experiment? We use experiments to determine cause and effect. We already talked about when you might do this and some cases where you probably do not want to do this. When it's dangerous to a person and you know it, you're not going to do an experiment. That would be very disintegrative and harmful. We don't want to do that.
- 1.74** But if you do want to determine cause and effect, there are certain ways that we go about doing that. First is, we have to have something called "control." You have to have a group that you're really doing nothing to and a group that you are. How do you select these people?
- 1.75** Well, this is where randomization comes in. You say, "What in the world does 'random' mean?" We'll talk more specifically about it. But a lot of people really think random—the way we use it in our common vernacular— is, oh, it just means weird. That's random. That's not at all what statistics means. See, in statistics, it means that every single person has an equal likelihood of being selected.
- 1.76** Why do you need that? Because if you want real data that represents your population—remember, that's what we're trying to do: take our sample statistics to represent our population parameters. If you want that to actually be accurate, you can't try to make the data say what you want it to say. We see this all the time where people are trying to make the data say what they want it to say.
- 1.77** Randomization helps prevent bias. Bias is making the data say what you want it to say if you're intentional about it. There is also bias in the way that we collect it, an accidental bias. Not everyone is trying to go ahead and take this data and completely misconstrue it. That's not what people do. But still, we unintentionally inject bias into our sample sometimes.
- 1.78** The best thing we can do is randomly select our people, randomly select our participants, randomly select whatever we are doing the experiment on, making sure that there is absolutely no bias in what we want to do with them.
- 1.79** After that, we might consider blinding. Blinding comes from the phrase where people used to wear blindfolds, and you would put two things in front of them and say, "Which one is which?" Well, that's basically the same thing. What blinding refers to is, in your group that you are going to perform your experiment on, they don't know whether they're receiving the treatment or not. We call that a "sugar pill" sometimes—don't we—to say they don't even know. No one knows. Double blinding means no one knows. The people who are performing the experiment and the people who are receiving the treatment have no idea which is which.
- 1.80** Why do they do this? Because the human mind is absolutely amazing. There's something called a "placebo effect," where you think you're getting something and you're not, and yet your body responds in a certain way. It's phenomenal. We have to identify the placebo effect. The worst thing that we can do is to give an experiment to not identify this placebo effect and then say, "Because we have a statistically significant result, this works." We don't know that.
- 1.81** See, with experiments, you try to control every confounding variable that you have. What in the world does that mean? A confounding variable is something that's really affecting your data, that's affecting these people that you haven't identified yet. That's where the control comes in. Try to limit every variable that we possibly can.

- 1.82** Then what we do after randomly selecting them is we say, "You're going to think you're getting something and you're not. You are going to think you're getting something and you are." Let's see if there's a significant difference between you. That's what the identifying of the placebo effect does.
- 1.83** Why do we do this? Why do we even collect data? At the beginning of our lesson today, we talked about how we have these things called "populations" that we really, really want to know about. And yet, we're probably never going to know most of it. We only do a census once every 10 years in this country because it's so laborious and costly—my goodness—and slow. These things are incredibly slow. So only every 10 years.
- 1.84** Well, we're going to take these samples and use the statistics to estimate and make decisions and inferences about this population. How do we do that? How is that even possible? You probably thought earlier that if there's going to be a difference between our statistic and our parameter, how can this ever fully represent this? Oh, well, this is because of patterning.
- 1.85** See, patterns emerge when you collect really large pieces of data. As we collect lots of samples or large samples, these patterns start to develop. And based on the patterns, like a sampling distribution—what in the world does that mean? A sampling distribution says if you took a whole bunch of samples and you found the statistics for each one of those samples and you put them on a graph, there's going to be some very, very distinct patterns there.
- 1.86** We can build inferences on those patterns. Why? Because they're predictable. What you're going to learn about statisticians is that they love predictability, where you can look at data and it will predict what happens in the future dependably and reliably. That's what happens with our data.
- 1.87** Patterns emerge in aggregates when you group our data together. One of those ways that patterns emerge is called the "law of large numbers." It says if you collect enough sample data, it's going to start to point to your population data. The other one is the central limit theorem.
- 1.88** When we get there, I'll spend a lot of time talking about how in the world this data makes sense from our samples to estimate our populations, why we can build a confidence interval, what a standard error is, how the spread of this data even looks, why it makes sense that we can do this in a way that is equitable to the data itself, that we're not misrepresenting it.
- 1.89** Two of the most important things that we have are right up there right now: the law of large numbers and the central limit theorem. These things will emerge in aggregates through these two theorems. It lets us do sampling distributions, which I talked about a little bit. It lets us talk about how the data is spread out. This is called "standard error." It's the basis for doing everything in confidence intervals and hypothesis tests.
- 1.90** When we get there, we have a little bit of legwork to do. We have to learn about sampling distributions. We have to learn about the central limit theorem. After that, everything we do in statistics, for the most part, will involve a confidence interval and a hypothesis test. Even the little stuff at the end will always include this. Why? This is the whole message of statistics.
- 1.91** You have to have a way to use this sample data to estimate your population data because you're going to have this, but you won't have this. That's what confidence intervals and hypothesis tests even do.
- 1.92** Before we get into our computer program, and I have to talk a little bit about it to show you how it works, to show you what you're going to be using. It's a free program. It's an amazing program. It's very, very user-friendly and very, very easy to use, which is nice for everybody involved, including me, because I don't spend a lot of time coding, which is great.
- 1.93** Statistical versus practical significance. You're going to find out that some things have statistical significance that aren't actually practical. They don't matter. Because statistics is not really just about memorizing formulas and crushing data, we're always answering the question, is what we're looking at different statistically? Is it statistically significantly different or not? Then secondly, does that even matter? Does the difference actually make a change in your life?
- 1.94** Sometimes we collect data and we never—oh, my gosh, this happened so much. When I was teaching for 18 years, we would have someone come in and say, "Hey, we're going to do a survey of your students. I want you to collect all of this data." For a year, we just collect data and data and data. The next year, it's like, "What happened to the data?" "What do you mean?" For nothing, because when you're going to collect data, it has to be actually significant, statistically, and practically significant. It's got

to make a difference. Otherwise, we're really just wasting everybody's time and money and effort. That's one thing we also look for.

- 1.95** The last thing I'm going to leave you with before we get to our program is this statement. We're actually going to start this course with this statement here, and we're going to end part two of statistics at the very end with this very same statement because it's such an important point that you need to get. Correlation does not equal causation.
- 1.96** This is a major kind of thinking that we see. We go, "Oh, yes, because two things are correlated, they are causal." Very hard to determine without an experiment. We can predict with them, but correlation does not equal causation.
- 1.97** What's that even mean? Correlation means that two things are related, or multiple things are related—correlation. As one goes up, the other goes up. As one goes up, the other goes down, or maybe there's no correlation. A lot of times, you'll see this. You'll see this on the news, you'll see this with people. We have this assumption that correlation means causation.
- 1.98** Well, there's two things that are correlated that are not causal. As the amount of ice cream consumption goes up, drownings go up. Oh, my gosh, I'm going to stop eating ice cream. Well, unless you swim in ice cream, that's probably not the reason why people are drowning. No, it's because more ice cream is consumed during hot seasons and more people also swim during hot seasons. So naturally, they are related, but they are not causal. They completely miss each other. They just happen at the same time. So sometimes we see a lot of instances where two things are correlated but not necessarily causal. We have to do a lot more research to say, is this really a causal relationship or are they simply mimicking each other in some way?
- 1.99** Now, let's take a look at one more data set. We'll talk a bit about correlation. I'll show you how to do regression briefly just so I can make this point. But I also want to do it to introduce our statistics program so that when we get to our further lessons, you understand what's going on. Otherwise, it's going to just go right over your head. I don't want that for you.
- 1.100** So sleep and test scores. You can probably look at this right now, and in your heads, hopefully, you can identify whether some of these are categorical variables. You can probably identify whether some of them are numerical variables, whether nominal, ordinal, ratio, or interval, hopefully at this point.
- 1.101** Student: nominal. Hours slept: you could sleep zero hours, and that would be literally zero hours. That is interval—sorry, that is ratio-level numerical data. It is continuous. It's not discrete. You can sleep any number of hours in the day if you'd like. Then we have test score, which is also continuous, and it is also ratio. We have these levels of measurement. Now we're going to see if these two things are actually correlated, co-related. Let's take a look.
- 1.102** The statistical program that I have up is called JASP. It's JASP for short. We'll be using it for the remainder of these two courses. If you ever wanted a stats program that works well, that's free, and you can actually understand how to use it, this is it. I'll spend some time developing this through the course so that you're not just flat left in the dust.
- 1.103** First, we see that this looks a lot like the table that we had. In fact, up here, just so that we can understand how this works, this will highlight your levels of measurement. When you see the little three circles, this is talking about the nominal level of measurement. "Hey, that's exactly what the student was." It's basically a number taking a place for a name. This is nominal data. It can't be put in order. It's just identifying the person itself.
- 1.104** Over here, we notice that we don't have ratio or interval, which is why we didn't spend a ton of time discussing those two because most software programs group them. It groups them into something called "scale." It says, "Hey, can you measure it? Is it numerical? You have scale data." We can do a lot of math on these programs with scale data.
- 1.105** For us, this is very good because we do not have to remember hardly any formulas. It's just about understanding the concept. That's exactly what I'm trying to do for everybody, so that you understand statistics. You're not worrying about memorizing formulas. You don't have to anymore.
- 1.106** A long time ago, oh, man, when I first started teaching, yes, we had tables and formulas and calculators, and we had to know everything and do all the little work, and it's just extremely tedious, and we miss the

forest for the trees, really, because we get into the formula, we miss what's going on. That's not what's going to happen this time.

- 1.107** We have hours slept. Hey, that's scale. There's a zero in there that means zero. We have numbers, they mean something. We can do math on them. Test score also is on scale. If you needed to change this, it's very easy to change. If you wanted to look at something like it was ordinal data, or if it was nominal, if it was ratio or interval, you can change these. I will show you on a couple of cases where that's very useful to do. In fact, next lesson, we're going to be doing that.
- 1.108** Here's our data. It's just raw numbers. We have taken a look at it. It says, "Okay, this looks pretty good." What we're going to do is go over to something called "descriptives." Hey, we talked about that. Descriptive statistics simply describe the data. This is what you can see from this. I'm not going to do a ton of stuff with this right now, but we will be when I introduce mean and median and mode and standard deviation and all these things that might not make sense. It's okay, we'll get there. We'll be here a lot.
- 1.109** What I'm going to do right now is move over to something called "regression." Remember, the main point that I'm trying to show you right here is that correlation does not necessarily equal causation, but we can find some strong relationships, even predict these things.
- 1.110** Descriptives, we have some choices. We'll look at some of these. T-tests, we will be in t-tests quite a bit in the second part of this statistics course. ANOVA sounds super fun, doesn't it? We'll check out what ANOVA means. It means analysis of variance, and I'll show you exactly how it works. We'll be in frequencies. I'll talk about a binomial test and contingency tables and multinomial tests. It will make sense when we get there.
- 1.111** But right now, we're going to look at correlation. Here's how JASP works. When you have your data on the opening screen and we click on a test or something like what we're doing here, it's going to allow you to manipulate your data in a variety of ways. How to get our data where we want it to go? Let's take a look at our variables.
- 1.112** Our variables are not student. That was just the people that we collected, so that's not a variable. But the variables of these students were how much they slept and their test scores. We can simply double click on this and it's going to move it over to our variables. We go, "Oh, this is fantastic. This is pretty easy." Or you can highlight this and click this little arrow. It will move it there. If you make a mistake, you can always move it back. So double clicking, get it over there. You can drag it and drop it. It's pretty user-friendly.
- 1.113** But here's what correlation is going to do for you. It's going to take these two variables and it's going to ask the question, are they actually related? Does it make sense that they have some relationship whatsoever, whatever they are? I will also tell you one thing about this is that you can always find some level of correlation, but whether it's statistically significant, that is going to be a different test that we answer at the very end of this class.
- 1.114** Okay, so right up at the top, we have this correlation. I'm going to scroll over here. We say, wow, what is all this stuff? We're going to take a look at what Pearson's r is at the end of our class. Here's what I'm going to tell you right now, just so that we can move forward. These are extremely, extremely positively correlated. They're almost on a line.
- 1.115** Now, you say, "Professor Leonard, how in the world can I see this line?" Oh, well, that is a fantastic thing about JASP is that it will give you any plot that you want on the spot. So scrolling down here, we have a scatterplot. A scatterplot shows you a relationship of two continuous variables.
- 1.116** We had a continuous variable. We had hours slept, we had a continuous variable, we had test score. It says, okay, these are paired up. In fact, one student had both hours slept and a test score, and student two had both hours slept and a test score. This is paired data. Oh, this works great for paired data. It says, does the first one relate to the second one across these however many people we had? I think we had 10 or so. Does it actually relate? How does it relate? Is it going to look like a shotgun blast on a piece of paper, or does it follow a line? Does it follow a line going upward? Does it follow a line going downward? How does this make sense? Does it make sense?
- 1.117** What do you think? That's on a line. Now, what does that say? It says, as hours slept increases, test score increases. This may be the most intuitive result of all time that if you sleep more, you're probably

going to do better on a test. Now, very strongly, positively correlated, that as these students' hours of sleep went up, their test score also went up.

- 1.118** Is it causal? If you sleep 12 hours a day, are you guaranteed to get 100% on your test? It's not causal. There are other things going on behind it. This is why we're doing this example. Those are called confounding variables. What they mean is there's something that's affecting this that you haven't identified.
- 1.119** Hours of sleep is definitely one contributing factor for a test score. If you get zero hours of sleep, you're probably going to perform fairly not great on that test. As that goes up, well, your test increases according to this, but there's other things that are going on, just like with the amount of ice cream that's eaten in America and the number of drownings are highly positively correlated also. There's other things that are going on. This is the first impetus for figuring out and doing more research: now that we've identified a relationship exists, if we scrolled up, we'd see that it is a very, very significant positive correlation, statistically significant.
- 1.120** Now that we understand that that is significant, we do more research to say, is it causal? What else is going on? Should we do an experiment? Oh, could we do an experiment? Could we gather a whole bunch of people who are randomly selected, that are not our friends, that we don't know anything about? Can we control this group that we give them all the same test at the same time where some of them get five hours of sleep and some get seven and some get nine, and then give them this test and see, with no studying and no background whatsoever, if these people perform differently? You can do that with an ANOVA, and you can judge their test scores on three different groups and see if one performs significantly differently than the others with an ANOVA test. We're going to be doing that, not with this specific data, but with similar data.
- 1.121** So, it gives us our first cause to really dig deeper and understand if there's a causal relationship, then we might consider doing an experiment. Are they related? Absolutely. Strongly and positively where hours went up, test score went up. Is it causal? Not necessarily.
- 1.122** The last thing we want to talk about in our lesson today is integrity. You see, there are lots of ways to be very integritous with statistics, and there's lots of ways to be disintegritous with statistics. It starts with the collection of our data, it ends with our inferences, and using the tests properly.
- 1.123** In order to use data ethically and responsibly, you've got to collect it well. That's where randomness comes in. That's where not doing volunteer response surveys comes in. Convenience samples where you're just gathering the data that's easy to get. Easy data is not always good data. It starts with collection.
- 1.124** Then are we organizing? Are we intentionally leaving out some of the data because it doesn't say what we want it to say? That's a pretty big no-no if you're going to be an integritous person.
- 1.125** Avoid misleading statistics. What in the world are misleading statistics? Oh, lots of things. Graphics can be very misleading. They say a picture says a thousand words, and it does, and that could be a thousand bad words if you're using poor graphics. What are poor graphics? Poor graphics can be 3D graphs that overinflate things. They can be truncated axes, which we'll talk about next time when we actually visualize our data.
- 1.126** But from our data collection randomness, past that, past our graphics, understanding the limits of what statistics can do. You see, every single test that we have has some underlying factors that have to be addressed. Like your data has to fit this shape in order for you to do this test. You cannot do this test unless you have five successes and five failures in each category. You should not do this test if it's not homoscedastic—and it's okay, we're going to get there—or it fails linearity or homogeneity. It fails these things. Well, you can't do the test.
- 1.127** Now, here's the point. Statistics software will still let you do the test. It will not prohibit you. It will not say, "Oh, no, you should not do this test." Oh, no, it will let you do the test. That's up to you to make the decision of whether you actually should do the test. That's understanding the limitations of both your data and the methods that you are using.
- 1.128** The last one. Mark Twain is famously quoted—maybe not so famously; it's famous for me because I'm a mathematician—as saying, "There are three types of lies in this world. There are lies, there are damn lies, and there are statistics." This is the whole point.

- 1.129** One of the most important things that you could do in your life is be an integritous human being, to actually do what you say you're going to do, to represent what you say you're going to represent. The same exact thing is true in statistics: to do what you say you're going to do, to represent what you say you're representing.
- 1.130** The data does not need you to speak for it. The data will speak for itself through statistics. As soon as we start to manipulate the data to say what we want it to say, that is very dangerous. That is where we get a lot of things gone wrong from news to marketing to sales, to sports, to hiring. All of these things get very skewed when we want to say a certain thing going into our study, so we collect data a certain way because we know we're going to get the answer. We say, "Grandma, what do you think?" Grandma always answers what I want to think. We get her data, it just says what we want, and then we say, "Now, this sample I've collected here, this represents the population."
- 1.131** Now, listen, your data can look that way. You can use the right test, but if it's poor data, you're going to give the wrong answer. The integrity of statistics, I cannot overstress how important it is that we are going to be integritous in this class.
- 1.132** I'll show you every different way that I know of to collect data in a manner which is random. We'll talk about when to use every test, why you would use every test, how to use every test through our computer. We will talk about integrity quite often as we move forward. I hope that you have enjoyed this lesson.

2. Describing & Summarizing Data

At a glance

Data summaries reveal pattern without replacing the underlying observations. The lecture develops frequency tables, graphs, measures of center, proportions, notation, skew, and responsible visual display, emphasizing faithful rather than manipulative representation.

- 2.1 I hope that you are enjoying statistics so far. We've done exactly one lesson and learned a little bit about data. We've learned that we're collecting data not on an entire population because that is insane. There's so much to collect in most populations. You're going to start asking a question, do we ever use **population notation** then? We do, and we're going to see that today.
- 2.2 The reason why we do that is sometimes we do deal with populations, we do collect a census, and when we get to probability distributions, we will have absolutely every single outcome that we have possible, and therefore, we'll treat it like it is a population, for instance. When we talked about sides of a die, the only things you could possibly get out are one, and two, and three, and four, and five, and six. That's the sum total of everything that could happen. We treat that like a population because it's all that can happen.
- 2.3 And so what we're going to be seeing in this lesson is not only how we describe data, so not just collecting it, not just putting it in a spreadsheet, but starting to really see what's going on inside the data. Remember, we're not inferring yet. We're not making any claims yet, but we have to understand it in order to start making those claims, those inferences. And we'll start describing the data in ways that we can understand it, so we can present it to people in a way that they can understand it.
- 2.4 And we will be using some of the population notation to get very comfortable with that. So today, it's not going to scare you. Don't worry. We're going to learn about some Greek symbols. At first, you're going to go, "That's a little weird." We'll be very used to this within just a couple of lessons. There are not that many of them. I can think of three or four that are very, very important.
- 2.5 Other than that, we'll have some English letters that represent these. Remember, our statistics are going to represent our parameters. Our sample is going to represent our population. These English letters are going to point to our Greek letters. That's what we're going to start talking about today. In order to understand that, in order to start getting to describe our data, we have to answer the question, what in the world are you going to do with this data? What's it look like?
- 2.6 Suppose we went out and we collected data on 250 adult males, and we said, "All right, we're going to gather a whole bunch of information about you. We're going to gather your height and your weight, body mass index, whether you're a smoker or not, what your mood level is in the moment, how many cups of coffee." We're going to organize all of that into a big old spreadsheet. Well, that's what we talked about last time, that JASP and other statistical programs do this very easily for us. We're going to take a look in just a moment.
- 2.7 But let's look at height for just a minute. There's heights of, I think that's 114 out of our 250 adult males, just their heights. That in itself is a little overwhelming. I'm bored like two seconds looking at that. That's just a fraction of our data. Getting a clearer picture of how this is organized is very important, and we can look at JASP.
- 2.8 One of the best things about JASP is that it organizes our data very well and very easily. You already see the categories coming up. It's nice and easy to see. We have an identifier for every single type of data that we've collected for these 250 adult males. We have ID, and we have age and height and weight and body mass index. We have number of cups of coffee, smoker habits, and mood.
- 2.9 Up at the top, we also see that we have the levels of measurement. Maybe as you would expect, we have ID as nominal. We understand why. These are basically just numbers that are taking the place of someone's name. We have age. Well, that's scale the way it should be. We have a measurement of how old someone is. Same thing with height. That should be scale. We have scale for weight and body mass index.
- 2.10 We have smoker habits as nominal. Well, those are categories, but you can't really put them in order. That's just yes or no. Mood is definitely ordinal. We have high and average and low. But you might be wondering about cups of coffee per day, and you think, "Professor Leonard, that looks like something

that you would measure." You can have zero or one or two or three or four or Professor Leonard-size five number of cups of coffee, which is why I talk so fast sometimes.

- 2.11** There's a reason why I've treated this as ordinal. It should be scale. But in JASP, the way that it constructs a **bar chart** and a **histogram** are different. Sometimes we can trick our program to get the type of picture that we want, not manipulating the data, but manipulating the graph to represent the data a little bit more clearly.
- 2.12** Even though this is discrete, counted data that we have here, we're going to treat it as ordinal so that we can make what's called a "bar chart." I'll contrast that with a histogram when we get there. I will also show you that in the moment, you can change a level of measurement to create some different statistics that we want to find out.
- 2.13** So ordinal, we're not going to be able to find a **mean** from. Ordinal data does not work that way. They're just categories. But if we want to find a **mean**, we can change it to scale in the moment and it'll allow us to actually compute some of the things that we want. So this looks really pretty. But honestly, when we scroll down and we look at 250 different adult males and all the statistics that were collected on them, that's a lot of data. Probably right now, your brain is getting numb because there's just so much of it.
- 2.14** There's really no way for us to look at all of this and say, "Hey, where's the center of the height? What's the average weight? What's the spread of body mass index?" There's no way to do that. You see, this is organized very well, and we can point to the level of measurement. We can point to individual entries. We can actually pair them up if we want to, but we can't analyze it because it's not even summarized. It's not described at all. We have to have something a little bit more. That's what we're going to be learning today: how to describe this data. We're going to be talking about measures of center, like what that even means, a frequency table, and then some of the graphs that we can produce that give people a really clear picture.
- 2.15** You probably understand this, the phrase "a picture's worth a thousand words." For most people, looking at data like this is mind-numbing. It is for me. I get bored in five seconds. I have to have something that gives me a clear picture that we can look at a snapshot and say, "Oh, yes, this makes great sense." That's exactly what we're going to be talking about. Now that we have it organized and we want to describe it, one of the first things we'll be looking for is what's called a "measure of center."
- 2.16** You see, you ask this question, what are you supposed to do with all of this data? It's only a portion of the values that we just looked at. It was a very small amount. We took a look at maybe the first 50 rows or 60 rows. We're not going to spend a whole lot of time digesting that data because the computer is going to do it for us. We need to learn how to describe that data. Well, what that means is that we want to find patterns. What patterns exist, if any, in the data? We want to talk about what is typical.
- 2.17** Well, we use this word "average" a lot, don't we? We say, "Oh, that's average." Well, what does that actually mean to be average? Sometimes in our normal language, we somehow think that that's subpar. Well, that's not at all what average means. Average means right in the middle. We really want to identify what is typical. Then we want to know, okay, is everybody typical or are some people out here very not typical or some people way out here? We want to talk about the spread, the spread of our data. We want to be able to compare individuals.
- 2.18** We want to talk about all of this data in a way that is essential to gather, essential to organize, but really hard to digest. It's not interpretable. We start with summarizing our data as we just have. So spreadsheets, they organize our data. We just saw that. But summaries start to reveal what's actually going on inside of it.
- 2.19** Here's some of the summaries that we'll be talking about. First—and today—we'll talk about the center of your data. Most of us are familiar with the word "average." We say this word; we probably have an idea about what it means, but there are different ways that we describe the center of our data. We have mean, we have **median**, we have a few other ones that we'll talk about to understand where it is. That's what we'll focus on today.
- 2.20** We also want to talk about the spread. We talked about typical. It could be here, but if everybody's right in the middle, that has a distinctly different spread than if some people are over here and some people have data points that are way out there; we can call those outliers. Well, we want to identify that. Any unusual features, the shape of it.

- 2.21** Does this make a uniform distribution? What is a uniform distribution? Looks like a box. Say, these are evenly distributed across every possible outcome that you could have. That would be uniform. One is not more likely than the other. That's not really real life most of the time. Most of the time, we have a spread of data that can look like a curve, like a bell-shaped curve. You've probably seen a bell-shaped curve before. Well, that's describing the shape of this data.
- 2.22** We can talk about a **skew** and whether it's a tail to the left, a tail to the right, where some people are way out here on the left-hand side of height, and some people are way out here on the right-hand side of weight, and what that does, what that does to our statistics and our features and how we describe this.
- 2.23** We're going to start with measures of center. Probably the biggest one that if you said the word "average" to someone, they would think of this one, and you probably know what I mean. That's going to be funny later. I promise it's going to be funny because it's the mean. You're not laughing. That's okay. No, the mean is what most people mean by the average.
- 2.24** The arithmetic mean, it's not hard to calculate. Arithmetic is this word "arithmetic," and that just means you add stuff up. The arithmetic mean means that you're going to take all of your data entries. You're going to add them all up, and then you're going to divide by the number of data entries that you added. Don't worry, you have to do none of this by hand. The computer will do it all for you very, very fast.
- 2.25** You see some symbols. Okay, what in the world do those mean? Well, they mean the mean. But you'll see a couple of different ones up there, don't you? You see one that looks not like an English letter at all. You see one that looks like an English letter with a little bar on top of it. Those are the two ways that we symbolize the mean.
- 2.26** All it means, all these mean, is that you're going to add up the data entries and you're going to divide by the number of data entries that you added. What's the first one? The first funny-looking symbol looks like an M and a U mixed together. That's called mu [μ]. No joke, I swear. It's mu. It's like an M and a U. It's a Greek letter. Greek letters represent population parameters. That's exactly what we spoke about last lesson, that this represents the population mean.
- 2.27** Now, how do you do it? Well, there's another funny little symbol, and it's this big capital letter sigma [Σ]. It's a Greek letter sigma. Now, that's math notation. All it says is you're adding. It's a Greek letter sigma, starts with an S. For a sum, you're just going to add stuff up. If you wanted to read this in English, it says, "If you want to find the mean of your population, you're going to add up all of those data points." The X is your variables. That's all it is. You're just adding stuff up, and then you're going to divide by nu [N]. It's a capital letter N in [Latin]. It's nu. It means the population size. That's all we're doing.
- 2.28** Now, I want you to think about something right now before we go any further. Does the mean take into account every single data value that you have? Yes. The mean is going to add up every single data value that you have. Please get this. This is important. It's going to take the really, really little ones. It's going to take the ones that are at the center. It's going to take the really, really big ones. It's going to add all of those up.
- 2.29** Tell me if most of your data is here, but you have one person way over here, is this one person going to affect your mean? Definitely, definitely, absolutely. The mean is very strongly influenced by what we call "outliers." We'll talk about outliers a little bit later. There's these things that are away from typical. It's going to affect the mean because you're adding it. It's got to be part of that data set. Now, could you conveniently excuse that outlier and get a better mean? Yes, but that's not integrous. The mean takes all of our data values, adds them all up, divides by the number that you just added. It's also going to include your outliers.
- 2.30** The next symbol that you see is this x with a little bar. We call that x-bar [$\bar{x}$]. I will tell you right away that statisticians were not super inventive when it comes to naming things and saying things, which is convenient for you because it will explain what's happening. But we call this x-bar. It's an English letter with a little accent on it, and it's a bar. Typically, our sample statistics, so the information that we get from our samples, our statistics, they are English letters, and they have some sort of an accent. The only one I can't think of that has that as R for a correlation coefficient.
- 2.31** But this is going to be called x-bar. It's done the same way. It's still got a sigma. It still says you're adding things. In fact, it still uses X for a variable. You're adding up all your data pieces and you're dividing by n, a lowercase letter n. It just means sample size. It's an English letter. It means sample size of our sample.

- 2.32** Now, the next measure of center that we have is median. I asked you a little while ago, would the mean be affected by these data values that are way, way out here? Yes, absolutely they would be. We don't want to just get rid of them. It's going to affect it. The mean is going to be pulled toward your outliers. Your mean is going to be pulled toward any tail that you have, these values that are away from center. Why? Oh, well, it adds up all your values. It's got to take that into account.
- 2.33** We have another measure of center called median. You might have heard this. I bet you've heard it on the news or something like that, where it says the median household income is. The median home value is. The median—why do they have that? Well, here's what the median does. What the median does is it takes all of your data values, whether you're in a population or whether you're in a sample, and it lines them up. It starts at the very smallest and puts them in order, boom, boom, boom, boom, boom, all the way to the very largest. It does one very, very simple thing. It finds the middle. That's all it does. It finds the middle.
- 2.34** You say, "Well, wait, Professor Leonard, what if there's an even number of data values? How can it possibly find the middle there, smart guy?" Well, what it would do in that case if there's two, like an even number of data values, it'll take your two middle values and then find the mean. You'll find the middle of that. It's not hard to do. A computer will do it for you. But why would you use it? Remember, this class is all about the "why," understanding the concept here.
- 2.35** Why would you want to use the median rather than the mean? Well, imagine this. Imagine you've lined up all your values. You have the smallest one over here, you have the largest one over here, and you find the middle number. Now, let's just pretend that you got this one wrong, that this guy's salary wasn't \$1 million a year, that this guy's salary was \$1 billion a year. Is this middle value going to change? No. Would it change the mean? Yes. But the median is a locator. It's locating what the middle value is. So it is very, very robust against outliers. It's not going to change.
- 2.36** This is why we use the median for things like household income, because we all know that most of us are here in this little range. And there's some people that are way, way, way over there. Well, if we just average them according to what most people think of as an average, like the mean, these incomes are going to make it look like we are all over here. Well, you're going to miss the true center. Why? Well, the mean is pulled toward your outliers. The median is not. So, we have these two different statistics, these two different measures of center or parameters if we're talking about a population.
- 2.37** There's something I'm going to preface now that you're going to see in just a little while that comparing your mean and your median will tell you if you have a tail on your graph. You see, because the tails pull your mean and they don't pull your median, if there's a difference between them, it will show you if you have a tail and where. I will see that in just a little bit.
- 2.38** The last measure of center that we're going to talk about is called the "mode." We don't honestly use it very often. It's telling you what is the most common occurring value. That's all the mode does. It says what happened the most.
- 2.39** Now that we have our data, we understand a little bit about our measure of center. JASP has organized all of this for us, which is fantastic. You want to know how to do it because the formulas—I don't really like those so much.
- 2.40** But we did talk about descriptives last time. What the descriptives are going to do is exactly what we want to do in this lesson. We want to describe our data. Now, you're going to see a lot of things that can be overwhelming. You go, "What are all of these different computations and statistics I can do?" Don't worry about it. We're going to get to the ones that we need as we go. We are going to click on Descriptives, go down to Descriptive Statistics, and you say, "Okay, I have all of these." This is each particular category we had in our spreadsheet. Let's talk about weight. I'm going to take our weight and I'm going to move it right over here into Variables.
- 2.41** What's going to happen is that on the right-hand side, this is going to start pumping out a lot of statistics for us. It says, "Okay, you had 250 valid entries. You're missing zero of them." What does that mean? Let's suppose that you collected 250 entries on people, but for some reason, one person didn't give you their weight. "I'm just too anxious about that." Well, then you would have one missing value and 249 valid ones. It will take care of all the missing data entries that you have.

- 2.42** Then it will start to find some statistics. One of them is what we would call the "average" in normal speech, the mean, the arithmetic mean. This is the mean. What this has done is taken all 250 of those weights, it has added them up, it's divided by 250 for you, and it's done.
- 2.43** Well, I want more than that. That's going to be found down under your statistics. We'll take a look at this. Since we're not missing any, we really don't need to know that. But check this out. As soon as you click on one of the statistics you want to find, it's going to pop up just to the right. It'll also give you an explanation if you need one. It says, "More than one mode exists." Yeah, because there's a lot of data in here, and a few of them are showing up more than one time. It's going to be a multimodal data set. It doesn't really matter all that much for us. But we also want to know median. We're going to click on median.
- 2.44** Now, remember what the difference between these is. The mode tells you the most common occurring value. The median tells you the middle value. The mean has taken all of them. Look at your mean and look at your median. Do you see how your mean is slightly above your median? It's not hugely above it, but it is slightly above it.
- 2.45** Why? Well, look at your maximum. At the very bottom, it tells you a maximum. There's a 288.8 pounds. That, we're going to find out, is pretty far away from typical for this particular data set. That is pulling the mean because it's taken that and averaged that in there. It's added all up and divided by the number that we added. Then the median says, "No, this one value is not affecting me. Whether it was here or here doesn't matter." That is static, so that's not going to move. Just by judging right here, we can see this is going to have a little bit of a tail to the right. This is nice.
- 2.46** In fact, if we wanted to find another group of statistics just for fun, let's pull over height, and it's going to calculate all of the same stuff for us. We'll move this over so we can get a little bit of a better picture. But now we can start to compare these two data sets. All right, well, let's look at height, let's look at weight. We had 250 people in each. We have a mode for each of them. They're both multimodal, which means that the same value is occurring over and over again, and it's not necessarily the same value in each data set.
- 2.47** We've got a median for both of them, and we also have a mean. Take a look at the median for your height. Take a look at the mean for your height. The median is 69.70, and the mean is 69.74. Median, 69.70. Mean, 69.74. Are those very close? Those are very, very, very close. What that means is that we probably don't have a significant tail on one side or the other, or it's equal on both sides.
- 2.48** We start to make some of these, not necessarily inferences about a population, but at least we're able to summarize this in a way that people can start to see. That's really the point of doing this. There are some other measures that we have down here that start to describe some things that we're going to get to in the next lesson. We'll start talking about skewness. I'll explain exactly what that is. And kurtosis. We'll explain what kurtosis is. We'll talk about standard deviation, IQR. We'll talk about variance, coefficient of variation. All of these things in future lessons will start making sense.
- 2.49** But for right now, we just really want to focus on the measure of center. How do we do that? Well, we let JASP do it for us, but we understand it. The mean takes all of the data values. It's a measure of center that includes all of them. It is not robust against outliers. Median puts them in order and finds the middle one. It is robust against outliers.
- 2.50** What do outliers do? Firstly, I've used this word so many times, but I've never really explained it. What an outlier is, is a data value that is away from your typical, that's away from your measure of center. In fact, in the next lesson, we'll talk about a way to mathematically determine whether a value is an outlier. It's not that hard. The computer will do it for you, but we need to understand what they do.
- 2.51** So outliers can occur, these data values that are way different from the measure of center. They're going to pull your mean, which is what we were talking about just now. They're going to pull your mean. They're not going to do the same thing with your median. That stays relatively stable compared to the mean. But what they can do is really, really distort your graph. So if everyone's here and you get this one guy way over here, that scale, that axis is going to be really, really stretched out. So they can really distort your graphs. They can distort your mean. They don't do the same thing with your median. They also affect your standard deviation and your range. We'll talk about what that is when we get to spread.

- 2.52** Remember, right now, we're just really starting to talk about describing this data. We've done measures of center. But next lesson, we'll talk about the spread. Standard deviation and range are two ways that we start discussing that.
- 2.53** Now that we have organized our data, we have a spreadsheet, and we've started to really summarize it. We have a measure of center. We want to know what it looks like. We started the whole lesson with, "Man, we really love a picture so that people can get a snapshot of our data and actually start to understand it." That's where we're going to start building this out.
- 2.54** Why do they matter? Well, they're more than just pretty. It's not just another pretty face. This is a tool. It's a tool on how you can express what's going on in a way that people can understand it. Isn't that the whole point of this? And so that when you look at another graph that you understand it, that makes you powerful as a consumer and as a citizen to understand what is being fed to you by media or wherever you're getting your information, if it's been collected well or not, if it's representing the population well or not. What does the graph say? Can you even understand it?
- 2.55** That's what we're trying to get across here. It's that you are powerful in the way that you're presenting information. You're also powerful in the way that you are absorbing information. That's what this tool, these graphs, can do for you. They reveal patterns, of course. They show you trends. They'll show you where data is grouped. They'll show you if we are on a line like we saw with a regression, correlation. They'll show you outliers. They'll also show the shape of a distribution.
- 2.56** We're going to talk a lot about the shape of a distribution in this class. What we're going to learn is that the shape of a distribution determines how we can actually function in some of the tests and some of the methodology that we have. It has to be a certain shape. It can't be too spread out. It can't be always uniform. Sometimes we have to have what's called a "normal distribution," which we'll talk about in just a little while today.
- 2.57** Why? Well, two data sets can be completely different and have exactly the same mean. Does that blow you—that might actually blow your mind. Wait a minute. Two data sets can have the same center and one can be right here all in one space, be completely uniform, and the other one can be completely spread out. Absolutely. And graphs and spread and shape will show that to us.
- 2.58** Let's talk about our first one. A **frequency distribution** is one of the first ways that we can really get across organizing this data and then starting to summarize it in ways that make sense to most people. We're back to our data set, and I want to look at the cups of coffee that are consumed per day by these 250 adult males. There are a lot of them. We're not going to scroll back down. What we are going to do is take a look at another Descriptive Statistics branch here, and we're going to move over our cups of coffee.
- 2.59** Now, you're going to notice something right away. This did not give you a lot of the same data, a lot of the same statistics that we had from our height and our weight. You might be asking, why? That's because we have this labeled as ordinal. It's not going to compute a mean. It's not going to compute a median. It's not going to compute those things because we have it labeled this way.
- 2.60** If there's a mistake, this is really easy to come over here and now we change our level of measurement and it will compute for you some of these statistics you might want to know. Why did we leave it as ordinal? That's because if we come down here, we talk about a frequency table, a frequency distribution, it will categorize this very nicely, and it will give us what's called a bar chart shortly afterwards.
- 2.61** A frequency table does this. You probably have actually seen a frequency table at some point in your life. All the frequency table really does is organize your data in a way that counts up how many are in each category or each bin, this group, this grouping for ratio-type data, for measurements. When we look at it, is this easier to see what's going on than our spreadsheet? Oh, for sure. We can see immediately that only seven people had zero cups of coffee. Seven people had five cups of coffee. What was the most common occurrence of coffee consumed? Well, that would be three cups of coffee. Eighty-nine people consumed three cups of coffee.
- 2.62** You see right after that, there's a percentage. What in the world is that percentage for? That's called a "relative frequency distribution." It's telling you percentage-wise what percentage of your data fell into this category. Same information, it's just divided it for you. We can see that 35.6% of this particular sample drank three cups of coffee. Just a way to organize it and then summarize it so we see what's going on.

- 2.63** The cumulative, all that does is it adds it up as you go. It's saying, "Hey, for your first, let's say zero, one, two, and three." If you were to take what percentage of people had between zero and three cups of coffee, you would have 84%, and you can see it right there. It's right out in front for you. The beauty of a frequency distribution—or some people just call it a frequency table—is that it shows you a picture of your spreadsheet in a way that most people would really understand, which is what we have here. Same picture just on our PowerPoint so that we can all see a little bit more clearly.
- 2.64** Now, what do you do with it? Some people really don't like that. They say, "I want to see a picture of this. Can you give me a picture that illustrates this?" Of course, that's called a bar chart. There's a difference between what's called a "histogram" and a bar chart. Say, why do we care about this? Oh, they're used for different data.
- 2.65** Now, you know this because you know there is discrete data, there's categorical data, there is now continuous data. Histograms are used for continuous data. They look different. The bars are touching. Why? To symbolize that you're on a scale, that there's no real gap between your values. That's why we use a histogram. We'll see one in just a moment. It's for continuous data.
- 2.66** Bar charts, that just counts stuff up. Bar charts are for categorical data. Say, how many people with brown hair, blonde hair, red hair do we have? It would just count that. How many cups of coffee did you have? Something that's countable, something that is discrete. Bar charts are for discrete data, and they are super easy to find on JASP and Microsoft Excel. I'll show you how to do both of them, actually.
- 2.67** Starting from the same table, if we wanted to make up a bar chart, all we have to do is scroll down to one of our Basic Plots and click Distribution. What a distribution plot is, is probably what you think it is. It's going to show you how your data is distributed. That's what a frequency table does. It's the same stuff. It's just giving you a prettier picture for it. That's what this is. It's just telling you, okay, hey, three. Three was the most commonly occurring number of cups of coffee that was consumed. We had the same levels at zero and five. It gives you a picture. That's all we have. You might ask a question, why did you leave this as ordinal and not just do your ratio data?
- 2.68** I'm going to show you why. Just in case, if you leave this as ratio or scale or interval data, the picture that you get looks funny. Here's why it looks funny. This is important. It's not just looking funny, it's important because it's trying to fit discrete data into a continuous graph. Histograms, it's looking at your scale and saying, "Okay, that's got to be a histogram," but then you don't have continuous data that you're trying to fit into it, so it's not going to look right. Identifying this appropriately makes your graphs look better.
- 2.69** What cups of coffee really is going to function better as, well, that's going to be ordinal, because the computer is going to know, "I'm going to put this in specific areas. I'm going to count it up." It's for counted data or categorical data. That's what this is for, which is exactly what we see. Same thing. It is for categorical counted data. The bars are separated because you don't have a scale measurement here. What's the difference in a histogram? Well, if one is for counted, discrete, categorical data, the other one is for continuous data. How do we create that?
- 2.70** Honestly, JASP does not do a very good job of this as easy as Microsoft Excel does. I'm going to show you how to do this in Microsoft Excel to get a really pretty picture. We'll take a look at the data set. Same data set, maybe a little less pretty because it doesn't tell you your level of measurement, but we still have our identifier, age, height, weight, body mass index, cups of coffee a day, smoker, mood level, and we're going to create a distribution for our weight, so a weight distribution. Let's take a look.
- 2.71** What we're going to do is start with some upper class limits. What this means is the maximum value that would fall in each class. So 120 would be the maximum for the first group. Between 120 and 140, it's going to fall in the second group. We're going to take this and drag and drop it. The numbers are fairly small, but we have all the way through 300. Why am I going through 300? Well, if you remember, the highest value is 288, so I want to make sure I include that guy. Now, how are we going to actually do it?
- 2.72** In Microsoft Excel, you have to tell it what to do with an equal sign, and that tells it, "I'm going to give you a function. Here's the function I want you to perform for me. I want you to count up the frequencies," and it's going to show you your function right below. Tell me my frequencies. It's going to ask you, where is your data? Oh, well, I want to point to my data. I'm going to point to the entire column of weight. It's going to look at all of this and it's going to count up what data fits where, all of our bins.
- 2.73** Okay, this is my weight, and now it's going to ask me, where are my bins? In other words, where do you want me to organize this data? What are your cutoff marks? Well, I've already told it. I've told it 120 for

the first one, 140 for the second, 160, and so forth and so on until we get to 300. So I'm going to point to here, put a colon. I'm going to go to one less because Microsoft Excel does something very interesting. It counts for your next bin. I don't know why, but it just does.

- 2.74** This will tell you immediately—that's faster than any of you could count, right? Could any of you look at 250 data entries and say, "Oh, yeah, there's exactly 28 of them between 140 and 160 pounds." There's no way you could do that. But this gives you a very nice frequency distribution for scale data. Why couldn't you do this as a bar chart or in JASP? Oh, well, because there's too many actual entries. It doesn't know what to do with them. You can't count that many. There's so many there that this had to figure out how many fit in a certain interval.
- 2.75** If we want to make it prettier, we'll go into our charts and click Recommended Charts, and it's going to ask you, what type of a chart do you want? I would really love a bar chart, but you see, I don't really want it to look like a bar chart because we know something about this. This is scale data.
- 2.76** I want this bar chart to really look a little bit better. I want the bars to be touching. That looks much better. I actually want a border on this. I'm going to go to Border and I'm going to say, yes, I want a solid line and I want the color to be a black border. That gives it a much better look for us. We could put in chart titles and axis labels and everything. But doesn't that give you a very nice picture of what's going on for your weight distribution?
- 2.77** After we manipulate this, we get something that we can put in any publication. Say this is a weight distribution. We have an axis at the bottom that's called weight. We have a frequency up the side that tells us a count for every bin width between zero and 120 and 120, 140, and I'll tell you what those values are.
- 2.78** Do you see your outlier? Easy to see now, isn't it? Say, wow, there's no one in the 280 value, but way over there, the 288 is counted between 280 and 300. You can see it. It's telling you, this is the reason why your mean was pulled to the right. It was that guy right there. That did it. That was what a histogram does. It's for scale, continuous quantitative data; the bars touch to show you that it's measurement-type data.
- 2.79** Now, the reason why I showed you this on both JASP and on Microsoft Excel is for the histogram, but it's also for the frequency distribution. You see, if we were to try to do this with weight, and we just bring over our weight, it is not going to give you a frequency table like we like to see it. It says that weight has more than 10 distinct values. This is what it's telling you—I'm going to get rid of our coffee real quick. We don't need that anymore.
- 2.80** It will not give you a frequency table. It will not do it. It says, "You have more than 10 distinct values here." You can adjust some bin widths and you can do it that way. It's much easier on Microsoft Excel, which is why I showed you. But it will give you a very similar distribution plot. This is the same picture that we had with Microsoft Excel. It'll still do it, but it won't give you the frequency table like we'd like to see.
- 2.81** Now, there's one more thing that I would love to teach you today. It's about scatterplots, how to make them, and how to find that trend line that we saw. Remember, I'm prefacing correlation, regression, which is probably one of our biggest lessons later in the second part of our class here. But getting it out in front and saying, how can we get these equations, is very, very valuable.
- 2.82** So height and weight. Let's take just a small sample from our sample, just a little collection of people—12 people from our male data set. I've selected 12 of them at random, and we see that they are paired up. We have person one who has a height of 62 inches and a weight of 120 pounds. We're going to take a look at how these things relate in a scatterplot, making that scatterplot look really nice, and then finding an equation that matches that. Why would you want to do that? Well, we'll talk about that in just a minute.
- 2.83** Let's go back to our Microsoft Excel. We have our same data set. We have our 12 people, we have their height, we have their weight. Say, I really, really, really want to make this look better. Now, a histogram isn't going to do it for us. Why? A histogram only compares one data set at a time for one particular variable at a time. We could look at height, we could look at weight, but we can't look at them together.
- 2.84** One of the best things we have to do is a scatterplot. You've seen at least a brief glimpse of the scatterplot last time, but now I'm going to show you how to make an equation for it. Let's go ahead and grab both of these. We're going to go and insert a particular chart. It's going to be a recommended chart,

and right here at the very top it says, "Do you want a scatterplot?" Oh, yes, I do. But this looks terrible because everything's clumped together.

- 2.85** Now, at the very end of our lesson, we're going to talk about something. We're going to talk about how you manipulate data and how you manipulate axes, whether that is integrous or not. We can cut out certain portions of our axis that don't have any data available. What we can't do is truncate them so that it cuts the data out. We don't want to do that. Why? Well, because that's starting to make the data say what we want. We really don't want to do that.
- 2.86** We're going to cut out part of this axis from zero to 55 as far as our height in inches, and we're going to cut out zero to 100 as far as our weight in pounds because no one's there. None of these men are between zero and 55 inches. None of these men are between zero and 100 pounds. What that's going to do is let us take a better look at our data without affecting the actual data itself.
- 2.87** How do we do that? We'll click on our axis. Up here at our minimum, we're just going to put in the smallest amount that we want. I'm going to choose 55. That's going to spread this out a little bit better so that we can see it. I'm going to click on my vertical axis. I'm going to adjust this to 100 pounds. Then I'm going to close this out. That gives us at least a little bit better of a look at what's really going on.
- 2.88** Now, think in your heads. What's happening as your height increases? The weight's increasing. Oh, for sure. Yeah, as the height goes up, the weight goes up. Is this predictable? That's what we answer with two questions. Number one, is the correlation statistically significant? That's much later. We have to answer that question much later because there's a specific test we do at the very end of part two of this.
- 2.89** Is it predictable? We're going to have a test for that as well, but I'm going to show you how the prediction line works now. If you want to get a prediction line—what that's called is a "trend line." Now, how you find trend lines is by adding a chart element. Right up here, we can add axes, both primary and vertical. We have that.
- 2.90** We can start to add a trend line. We want a linear trend line. It's going to give us a model, exactly what we're looking for. It says this data fits this line pretty well. It'll even give us the equation for that line. We get this. We can adjust all of our axes. We can find our equation. You go, "Oh, my gosh. I didn't know we were going to have equations like algebra." This is exactly the same thing as an algebraic equation.
- 2.91** Why do we have that? Well, you are on an X, Y coordinate plane. What that means is you have some X values, that's just your height. You have your Y values. That's just your weight. In fact, these were paired up, weren't they? So that for every height, you had a certain weight. That was how this data was paired and we got this very nice equation from it.
- 2.92** But if it's statistically significant and if it actually matters, it's now predictable. What that means is that if you were to plug in a height for your X and you used that equation, so you took a height of, let's say, 73 inches, and you multiplied by 5.9159, and then you subtracted 248.87, it would give you a predictor for that person's weight.
- 2.93** How accurate that is depends on the statistical significance. It depends on how good our model is. It'll also forecast a little bit for you. Now, we'll talk about ARIMA, which is a different type of forecasting that uses a lot of different data. It's at the very, very end of our class, but it can predict and forecast.
- 2.94** Now, here's something to understand before we end today. While it is a prediction for something, it doesn't guarantee that it's going to be exact for someone. You can't tell me that a person who's 73 inches tall is going to be exactly this weight by this equation. That's not going to happen. But it's going to be, hopefully, relatively close, depending on how close this is to the line.
- 2.95** The other thing that it doesn't do, it won't tell you what a person is already. So these data points don't have to fit exactly into that model. Remember, this is the overall theme of it. It's not like I'm going to plug in my height and then it's going to give me out my exact weight according to that model or anyone else on there.
- 2.96** It's taking all of this data. It's saying, what fits best? Then from what fits best, can you predict others? That's exactly what statistics are supposed to do. Take what you have, make it predictable, and make it fit the data so that you can predict and make inferences about something else. That's exactly what we've done here.

- 2.97** One thing to avoid. Graphs can be very misleading as we have all probably been misled at some time in our lives. We look at it, we go, "Oh, this makes exact sense." And someone used a 3D graphic, and that makes it look a lot larger of a difference than it is. Because when we start talking about volume of graphics, sometimes pictures can look way more inflated than the actual differences of our data. So we avoid things like this.
- 2.98** Manipulating axes scales. What we mean by that is cutting out some parts that are relevant that we don't want to show, making it so that our scale is zero to 10 here, then 10 to 30 here. Well, that's going to start grouping things in the wrong bin, in the wrong segment category. We don't want to do that because you're going to start to inflate certain categories to say what you wanted to say rather than letting the data speak for itself.
- 2.99** We try not to use 3D charts. We try not to manipulate our axes so that they're on a different scale, one to the next. We don't want unequal bin widths. We're not going to group people between zero and 140 pounds, and then between 140 and 150 pounds. You would group them incorrectly. The chart that you have is going to be way inflated for one group and not the other.
- 2.100** Now, why would you do that? Well, if you wanted to market in certain ways, you might want to show that you have more of a result here, but it's not integritous. This all goes back to you don't have to make the data say what you want. The data is already there. Collect it integritously, represent it integritously, and you will have reliable results. We don't want to cut these axes off. Nothing overly decorative, even though they look very pretty. It all goes back to integrity: saying what the data says, not what you want it to say. We're going to end there.

3. Understanding Data Spread

At a glance

Measures of spread describe how observations vary around their center. This lecture develops range, interquartile range, variance, standard deviation, z-scores, outliers, and distribution shape, explaining which summaries are robust and how spread affects interpretation.

- 3.1** Now that we've decided what in the world the population is and what a sample is, and taking these statistics to estimate and make decisions and inferences about our populations, we've talked a little bit about how to describe data, just a bit. But mostly we focused on the center of the data, just where the center is.
- 3.2** Today, we're going to talk about how that data is spread out and how we can use that to make more inferences about a population later on. We'll talk about describing data a bit, focusing mostly on measures of spread. It will be brand new for some of you. You might not have ever heard of **standard deviation**, **variance**, coefficient of variation, what a box plot is, although you will see it. You'll see a box plot in nearly all research that wants to show a spread of data. We'll talk about that and how to read it so you can interpret what these people are trying to tell you.
- 3.3** Remember, part of this is trying to make you more powerful as both a consumer and as someone who wants to put forth information to the people that you care about or even to the world in a way that builds integrity around what you're doing so that you're not taken advantage of.
- 3.4** We spoke a little bit off camera about the influence of influencers and how even when we try to make informed decisions and we want to do what's right for our families and what's best for them and our kids and our parents that we're taking care of as they're growing older, well, we're influenced in so many different ways. A lot of that is data-driven. A lot of that is research-based, and some of it is assuming that it's research-based when maybe it's not. The advent of influencers makes us actually very vulnerable to that. If that's not research-based and we're influenced in a negative direction, it has a profound impact on who we are as humans and the people that we care about.
- 3.5** This is what we're trying to avoid. Part of that is understanding the data; part of that is collecting it. A lot of it is describing the data well and accurately. We've already talked about a measure of center like mean and median and mode, which we don't use very often. But today, when we get into measures of spread, it's according to where that center is, how is the data around that? Is it very far apart or is it right near the center?
- 3.6** We'll also talk about visualizing data. In fact, we're going to start with this talking about a box plot. What is a measure of spread? What is that talking about? It's talking about how far away from typical your data tends to be, whether it's a long way or just a little bit away—not just that measure of center, but how this is spread out around it from what's typical.
- 3.7** One easy way to get a handle on this is by what's called a box plot. You might have actually heard of this: a box and whiskers graph. Maybe you have. If you haven't, that's okay. It's not that hard, but it will start showing you at least an introduction to how data is spread out.
- 3.8** You see, what we really want to define is variability. Variability plays an enormous role in how reliable your data is from your sample to estimating your population. If your data set has a lot of variability, it's not going to be a very good estimator for your population. We judge variability, or at least we take that into account when we're starting to make these inferences. One way in which we do this is called the standard deviation.
- 3.9** The flow for our day today—we're going to start taking a look at what a box plot is. We'll talk about a five-number summary. We'll say this is very good and very nice, but it doesn't do everything for us. Here's why. Much like a median only looks at one value, not all of it, the box plot only looks at five values, not all of it. We have to have a measure of spread that takes into account all of this stuff, all of our data values, and that's what standard deviation—that's what variance is going to do for us. We'll talk about that as we get there.
- 3.10** It'll bring us into skewness. Do you remember when we looked at the mean and the median? We said, "Oh, this mean is apart from that median." That's going to lend itself to understanding skewness—the tail of our graph, whether it's on the left, on the right, or we don't have any significant tailing or skewness,

whether it's positive to the right, negative to the left, or neither. We'll take a look at that. And lastly, we'll end up with something called **z-scores**. What in the world is a **z-score**? That is not for you to worry about right now. We will get to it at the very end. You will perfectly understand it, I promise.

- 3.11** Why spread matters. The measure of center alone can be really, really misleading. I'm going to show you this in just a moment. But you can have two data sets that have exactly the same mean that have a dramatically different spread, and you would treat them completely differently. One can be perfectly uniform where every data value is exactly the same. One can be completely spread apart like crazy and ends up with the same exact mean.
- 3.12** Because if we add all the numbers together, you don't know how they're spread. You don't know where they're coming from. You're just adding them. They could all be the same number, or they could be dramatically disparate, way, way apart. So the spread really gives us more teeth to understand how the data is shaped and how it's going to work. It helps with consistency, of course, reliability, and as we mentioned, shape.
- 3.13** We're going to start with this box plot like we talked about. The box plot starts with a five-number summary. Naturally, we're going to take 13 numbers—I'm just kidding. We take five numbers, and they're numbers that are really actually fairly easy to find. We start with our minimum, the smallest value. We have one that maybe you're not super familiar with called the first quartile. It's called Q1. We'll talk about that in a minute. We have the median, which you are familiar with. We have Q3, which is our third quartile. And lastly, we have our maximum.
- 3.14** Well, the first one and the last one, we all are pretty familiar with what a minimum and a maximum is. We might know what a median is, but you might not know what the first quartile and the third quartile mean. Here's what this is. I already prefaced it by saying what a box plot is going to do. It's going to take these five numbers and put them in a graph. Now, what we talked about was that while that will show a certain aspect of spread, it's not going to be great because it really only takes into account these five numbers that you're looking at right now, the minimum, Q1, median, Q3, and maximum.
- 3.15** So here's how a five-number summary works. It's not hard, but it starts the same way you would start with a median. You take your smallest value, you line up all of your values until you get to your largest value. The largest value, of course, that's your maximum, and your smallest value, that's the minimum. The number that's right in the middle, that particular value, that's your median. If you have more than—well, if you have an even number of them, you'll take your two middle values, and you'll find the mean of that. It's literally the middle value.
- 3.16** How the first quartile works. If you take your first half of your data—so, understanding that the median is representing the bottom 50% of your data—you take this bottom 50% of your data and you find the median again. That is your first quartile. It's basically a median in the first half of your data. Then your third quartile is the median in your last half of your data.
- 3.17** Well, now, what's that going to do? You have now one, two, three, four segments, four quarters, where we get the word "quartile" from. All this really does is tell you the minimum value, the value that represents the bottom 25% of your data, the value that represents the bottom 50% of your data, the median, the value that represents the bottom 75% of your data, the third quartile, and then your maximum, that's 100% of your data.
- 3.18** This is going to take and give us a spread. All it does is it groups together the first and third quartile and says, "Show me that **range**." What's the spread of your middle 50%? Remember, 25%, 25%, 25%, 25%. Between Q1 and Q3, you should have your middle 50%: 25 here, 25 here, middle 50. If that's really close together, that gets us a very good indication of the spread, that it's not very spread out.
- 3.19** This is extremely robust to **outliers**. **Outliers** are not going to affect this very much between Q1 and Q3. They're way over here. They're way over here. It's not about the value like a mean, it's about the order and picking the middle number, the median of the first half, the median of the last half.
- 3.20** We take a look at what's called the **interquartile range**. It's **IQR** for short, very easy to find. We take our Q3, Q1, and we just subtract them. That's going to give us a range for the middle 50% of our data. If that's really spread out, that gives you a pretty good indicator just right off the bat that your data set's fairly spread out. If it's really close together next to the median, you know that that's pretty well measuring towards central tendency. It's right around the median.

- 3.21** The median is always going to be between Q1 and Q3, and it's very, very resistant to outliers. These things aren't going to change. Think the Elon Musks. If we wanted to find a five-number summary for all the income that's made, we'd have some very low values. We'd have most of the—50% is fairly close together, and we have some very big guys over here, some outliers over here. Those aren't going to affect this for the way that five-number summaries are created, which is nice for us.
- 3.22** Now, you should have this thing in your head. How in the world are we supposed to actually identify these outliers? Is it something that's just subjective? Do we say, "Well, that looks like an outlier, so I'm just going to kick it out?" We talked off camera a little bit ago about how you can't just throw out outliers, not just because you want to. You have to have some way to at least identify them, then talk about if they're relevant to your data set in a way that builds integrity into what you're doing.
- 3.23** We don't just kick out outliers for no reason. We don't just identify them because they look bad for our data set. We don't just say, "Well, it looks big enough or looks small enough, so I'm going to consider this an outlier." You first have to identify it. Then from that question, you say, "Do I need to include it or not? Is it relevant or is it just a one-off that happened by some fluke?"
- 3.24** Well, we start with identification. There is a mathematical way to identify what an outlier is. We build these things called fences. What a fence is, is it says, if you make it below or above this mark, we're going to consider those to be outliers. How do we do that? The lower fence does this. It's not hard to do. The computer software will do it for you. It'll actually do it automatically when we get to box plots and we will identify them.
- 3.25** But here's what it does. It takes your IQR, your interquartile range—so your third quartile minus your first quartile, that range, and then it multiplies by one and a half. If you take that value and you subtract it from Q1, it's going to give you a very low margin. Anything below that we would consider to be an outlier. If you add that same amount to Q3, that's going to give you a very high margin. Anything outside of that, we would consider to be an outlier.
- 3.26** This gives us a better way to say "this is way out there" than "it looks way out there." Remember, you're not trying to read into the data what you want. Say, "That looks really big; that's not okay enough." We have to have some mathematical way, and this is the way that we choose, at least for box plots. The points that are beyond our fences, we consider those to be potential outliers. We would look into them.
- 3.27** Do they still need to be included in the data set? That depends on what we're doing. Sometimes those outliers are very relevant. Sometimes they are just random occurrences. Sometimes they're data that is miscalculated or that is put down incorrectly. Oh, that happens. Those are errors that happen. And so we want to take a closer look at those.
- 3.28** So what we do from there, after we collect our sample and after we organize this as a five-number summary, we take and we build a box plot from it. What a box plot does, it just takes this data, these statistics, and it puts them in a graphic that we can understand. You probably see where the word "box" comes from. I told you, we are not inventive as statisticians when naming things. We don't call this the cool plot or something. No, this is the box plot because it looks like a box.
- 3.29** Now, what's the box mean? Well, the box is 50% of your data. It's the middle 50%. You see, the lower line on the box is your first quartile. Below that's 25% of your data. That middle line right in the middle, that's your median. It should always be captured by the box. It has to be because if this is the bottom 25% and the third quartile, that top line up there is the bottom 75%, giving you the middle 50, your median has got to be somewhere in there, 25, 50, 75.
- 3.30** This will show you a little bit about how spread out your data is. If your box plot is very narrow, that's going to tell you it does not have very much of a spread for 50% of your data, the middle 50%. If it's really getting up there, well, that means that it's fairly spread out, and we'd check variability with some other measurements.
- 3.31** You also notice up there that we have some whiskers—is what they call them—a whisker at the top end and the bottom end. What this is going to say to you is what your maximum and what your minimum value are that do not make it past your fences. Now, we said a lot of words real fast. This is how box plots work.
- 3.32** It's going to calculate your minimum, first quartile, [median], maximum, third quartile, put them on this box plot, and then it's going to adjust for those potential outliers. It's going to measure it for you. It's going to

take $IQR \times 1.5$. It's going to create a fence above your third quartile, below your first. Then it's going to mark any values outside of those as outliers. It'll show you what they are. They'll show up as little dots above or below. In this case, we see there's just one little dot above. We'll see that on some of the box plots that we're going to build in just a moment.

- 3.33** In fact, let's take a look at one right now. Let's go ahead. We're going to take a look at how to create a box plot in Microsoft Excel that shows exactly what we want it to do. Now, if you had to line these up on your own with all 250 of our adult males and put them in order, that would be absolutely crazy. This is going to do it for us in a short amount of time.
- 3.34** We do have to select all of them. We're going to go ahead and select all of our 250 items just like this, and we're going to look for a very specific chart. We're going to Insert. We're going to insert a recommended chart. Under Charts, you're going to find right below our histogram, which we've talked about in this class already, a box and whisker chart.
- 3.35** If you click on this, it's going to give you a very, very—actually, not so pretty yet. We're going to make this prettier. This is a box and whiskers, and it has some of the features that we already see. One thing we might do is adjust some axes. If we double-click on this, there's no one below 100. Maybe we make this 100. I think 110 would be a good choice. Let's adjust that so we're not cutting off data. We're not just arbitrarily saying what we want here. We're just making this so it's a little bit easier to see.
- 3.36** Then we're going to add some data labels to this so we can identify what's going on. We're going to add data labels. Let's put them on the right. In fact, you know what? We're going to highlight this and we're going to adjust our font just so we can see it a bit better. That's already a much better-looking chart. Maybe even our fill, so we can read that middle number because that's an important number. We want to choose—let's see what won't explode our cameras today, maybe that one.
- 3.37** This is a typical look for a box plot. You see your minimum value. In your head right now, you could think, is there an outlier at the bottom? Is there an outlier that's a low outlier? No, there's no dot below there. This says that your minimum is your minimum. There's nothing below that. It did not make it below the lower fence, so we have no low outliers.
- 3.38** If you remember our weight data set, we had most of the stuff here. There's some over here, and then we had some tails off to the right. That was our weight data set. It was from a couple of lessons ago. Even last lesson, we talked about that. Now we move up a little bit, and we see the bottom line of our box. This is your first quartile. It's done it for you. It said this value that represents the bottom 25% of your data is 167.65 pounds.
- 3.39** Then you have the box, and the box has three numbers there for you. It's got the first quartile; it's got the 183.85. Well, that's your median. In fact, if you went back and you looked at one of the lessons we did, the second lesson, we found the mean and we found the median. Those are the same numbers right there. That is the median, that 183.85. It's showing you where that's at, and it also shows you your third quartile, 200.85 pounds.
- 3.40** The question you should have is, what in the world is the 184.3? Why is that there? Why is that in the middle? And what does that do right in the middle of our box? That's your mean. That's your mean. It is slightly, slightly above the median, exactly like we saw in our descriptive statistics. But this gives you a decent picture, doesn't it? It's not too far above that. It's just right there saying, "Yes, you have some values that are high that are pulling that mean up just a little bit, but it's not too dramatic."
- 3.41** Now we move up. We see our maximum value is 242.4. Now, you know something. You know the true maximum value was 288, 288.8 pounds. Why isn't that the maximum value? Well, those values that you see up there as dots make it past your upper fence. It says, "Mathematically, we would consider these to be potential outliers." We would check out 285.1. Is that a mistake or is that truly there? We'd check out 288.8. Is that truly there? Do we want to include that? Why or why not? But we don't make that decision now. We organize our data now, describe our data now, and this would tell you what your outliers are. This is how box plots generally work.
- 3.42** Now, going from there, we've got to start talking about a more mathematical way to determine spread. That picture was great. We have a pretty clear understanding that most of it's right in the middle. We have 50%; it wasn't crazy like this. It was pretty close together. But we didn't have a number or a statistic that took all of our data values and used them.

- 3.43** You see, median to mean is analogous to a box plot or IQR—really, in a box plot—compared to standard deviation. We want to find a value or a statistic that takes all of our values like a mean does, calculates something like a mean does, and gives us a measure of spread. Median gives us a measure of center, but it only includes that one value. It does not take into account all the data values. Mean does that. It's affected by outliers.
- 3.44** Likewise, a box plot takes into account only five numbers. It has outliers. We discount those. The IQR is not really affected by those outliers, but it's only three numbers. We want something that includes all of our data values like a mean that gives us the same information like a box plot. However, you can probably think this, is it going to be affected by outliers? Definitely, because it's taking into account all of our data values at one time.
- 3.45** This is the start of what we're going to be learning today about standard deviation. It's not hard, but it's important that you understand it. Let's start with some very simple differences. One way we can start talking about spread is with range. You probably understand range a little bit. Range is just the big number minus the small number, the max minus the min.
- 3.46** Can you think for just a second if this is going to be affected by outliers? Maximum minus minimum. Would that be affected by outliers? Oh, greatly. Absolutely. If you have things that are way out here, it's only taking into account two numbers and giving you an estimate for spread. If any of those is way out beyond the center, it's going to greatly, greatly affect the range.
- 3.47** We don't really work with range a whole lot, even though it's very, very easy. It is so affected by these outliers. It's almost unusable for us in any sort of mathematical methodology moving forward statistically. Is it important? Sure, to understand a general feel for things, but not really to work with. Let me show you the range.
- 3.48** But notice how nowhere does that range include all of our values. Now, we only have three of them, but you can probably extrapolate if we had 250 and one was way out there, our range is going to not really represent the spread of most of this. We have to have a different way. You think, "Well, what about the mean? The mean is fantastic. Why don't we just take that?"
- 3.49** Well, let's look at that for a second with just these three groups. What if we took group one and we found the mean? Now, the mean is not hard to find. I put three numbers here, so maybe we could do it in our heads. If we did $7 + 7 + 7$, we get 21, and $21 \div 3$ would be 7. Our sample mean $\bar{x}$ would be 7.
- 3.50** However, group two does the same thing. You see, $4 + 8 + 9$ is also 21, and $21 \div 3$ is still 7. And $1 + 6 + 14$ is also 21. And $21 \div 3 = 7$. Your means are exactly the same, even though these are very, very dramatically spread differently. That's crazy.
- 3.51** How do we come up with this number that not only takes into account every value, maybe it's affected by outliers, but it will give us a spread for the data like a box plot does? That's where our standard deviation comes in. It has this thought. What if we averaged the distances from the mean? Now, we know how to average things. We're talking about a mean here. But what if we started finding the difference between that?
- 3.52** We're going to take a look just at group three. Group three has the numbers 1, 6, 14. We had a mean of 7. If we started finding the differences from the mean, what we'd take is our data values like X , we would subtract the mean. Yes, it's got to be in that order because what we want are data values above the mean positive and data values below the mean negative. We want to show whether they're above the mean or below.
- 3.53** That's a pretty big problem because you would get that the spread of the data is 0. We know it's not 0. Listen, this is going to happen with every data set ever if you do this because the mean represents 50% of the value. If you're going to take 50% of your value, subtract it from the mean—or the mean from it, sorry—and 50% of your value and subtract the mean from that, well, half of it's going to be above the mean and half of it's going to be below the mean. Half of it's going to be positive value-wise; half of it's going to be negative value-wise. If you add it, you're always going to get 0.
- 3.54** So, we have to have something different. Two things come to mind that we could actually use. We could use absolute value, which makes everything positive all the time, or we could square. We could square the amounts. See, squaring something takes a number times itself. A positive times a positive remains

positive. But a negative times a negative becomes positive. If we square these values, it will become positive.

- 3.55** We have a choice to make. Do we use absolute value or do we square these deviations? This is what they're called: a distance from the mean. We start with absolute value, and this is something that people do do. It's called the mean absolute deviation. All it really does is it takes—it looks crazy. Actually, you know what? It might make you mad. I don't like math, and this is really—that's not what it stands for.
- 3.56** But the mean absolute deviation does this. Look at the inside of your absolute values. Here's what it says mathematically. It says, take your data value, subtract your mean. No problem. We just did that. Now take the absolute value. It'll keep the positives positive. It'll make the negatives positive. Then it will add them. That's what the summation does. It says you're just adding stuff up, and it will divide by the number of items that you added.
- 3.57** It would come up with this estimate for a mean absolute value deviation, distance from the mean. This is great. This is actually a statistic people calculate. It measures the average distance from the mean. There's only one problem. Absolute value does not behave well for a lot of the formulas and methods that we use later on, so we have to have something else. Not friendly.
- 3.58** We're going to talk about variance and standard deviation. This is the alternate route. Hopefully, you understand why we need something at this point. We can't just add the differences. They're going to be 0. We can choose absolute value, but that's not going to work later. So, we have to have something else.
- 3.59** What we've chosen to do is to square those deviations. What a variance is, is the average, mean, squared—we're taking it to the second power, it's going to make everything positive—deviation. We're going to average the deviations squared.
- 3.60** The standard deviation is the square root of that variance. You're going to have a question in just a moment. Why do I need both? Why do we have both of these things? I will answer that when we get there. You see, something will happen. I guess I'm answering it now. Something will happen when you square something with the units, and we really, really, really want those units to match our data.
- 3.61** How do we do it? Well, we're going to find a way to prevent these positive and negative values from canceling out. Let's take a look at the formula. I told you that we're going to have population formulas. We're going to have sample formulas. Honestly, we don't need to know really how to calculate it. The software is going to do that for you, but you do need to understand the actual terms that we're using here.
- 3.62** For population variance, hopefully in your mind, I'm going to put up a Greek symbol here real quick, right? Because population stuff, we have Greek symbols. Population variance is going to have this funny little symbol. It looks like a cannon over there, and it has a power of two, which means it's squared. That makes sense. It's just a square in things.
- 3.63** That is the Greek letter sigma. It's like a lowercase letter s in Greek. It's the Greek letter sigma [σ]. Why don't you have V, Professor Leonard? Why not for a variance? Because we use standard deviation a lot more. That starts with an "s." Sigma squared is our variance. Look at how this is calculated. It says what you're going to do is you're going to take your data values, you're going to subtract your mean.
- 3.64** Oh, it's a population mean. Yes. Why? Because you're talking about population variance right now. Do we ever have that? Of course we do. Sometimes we do a census. Other times we want to use our sample statistics, our sample variance, to estimate this population variance that you're seeing right now. We take our X values, our data values, we subtract our mean, we know some are going to be positive, we know some are going to be negative. Value-wise, half the value will be positive and half the value will be negative.
- 3.65** Oh, but we've got to square this because those negatives are going to cancel our positives. We square them, we add them, and then we divide them. It's an average squared deviation. This is what population variance does. Population standard deviation—here's the answer to the question—says, "Okay, let's take a square root." Why would we want to take a square root, and why isn't population variance good enough?
- 3.66** Population variance is a great number for the spread of data. It's going to take all your data values. Oh, look at that. All your data values are there. They're all going to be subtracted by the mean. They're all

going to be squared. It takes into account all of them. It's highly susceptible to outliers for sure, but they're all there.

- 3.67** Why would we want to take a square root of that? Well, when we square things, you also square the units involved. If we took the variance of, say, weight, for instance, we're going to square each and every single one of the differences of those weights; you're also going to get not pounds, but pounds squared. You wouldn't get feet; you'd get feet squared. Well, we don't want to talk about feet squared, not unless we're laying carpet or something. What we want to talk about are units that are relevant to our data set. How do we get those? We take a square root.
- 3.68** What the standard deviation does is it puts this statistic back into the same units as our mean, back in the same units as our data set, so that's more usable. When we're talking about feet, our standard deviation is in feet, not feet squared like variance would be. It still does the same work. It takes our data values minus our mean, we square them, we add them, we divide them, we take a square root, and this is our population standard deviation.
- 3.69** We will be working with this for the rest of our class because it gives us a very good number for the spread of our data. The larger the standard deviation, the more spread out your data is. The smaller the standard deviation, the less spread out your data is, and it's incorporated every single one of those data values. Now, of course, you're saying, "But wait, we don't always know the population parameters. We don't always know the population values. We don't take a census all the time." No, of course. We have sample formulas that illustrate this for us.
- 3.70** So sample variance, it's going to look similar formula-wise. There's going to be something that's going to drive you nuts, and I know it is, and that's okay. We'll talk about it in just a minute. Sample variance doesn't use a Greek letter. It uses an English letter. It uses s^2 because we're going to get the standard deviation that has an "s" in it. So s^2 for our variance.
- 3.71** It has the same-looking stuff. We have our data values minus our sample mean. Sure, we're in sample land now. We're dealing with sample data. We take our sample data, we subtract our sample mean. Some of those are going to be positive, some are going to be negative. So we're going to square them. It's going to make them all positive, but it's going to mess our units up, isn't it? It's going to take all of our units and make them units squared. We don't like that very much.
- 3.72** Sample variance is a point estimate. It's going to estimate our population variance, just like sample mean is a point estimate. It's going to estimate our population mean. And sample standard deviation, much like population standard deviation, is going to be in the same units as our mean, the same units as our data, and it gives us a great look at the spread of our data overall.
- 3.73** Well, sample data tends to cluster around sample means, not population means. Sample data clusters around sample means, not population means. What that means is that variance and standard deviation, when it's constructed from a sample, tends to underestimate the spread of a population. It tends to underestimate it. So here's the Bessel correction. What the Bessel correction does, it says, "I'm going to take 1 and I'm going to subtract 1 from the denominator."
- 3.74** You see, if you subtract something from a denominator, you're dividing by a smaller number. It makes the fraction larger. So it makes it so that we don't inherently underestimate this value for our population. Using n underestimates population variance; we subtract 1 from that sample size, makes the fraction larger. Denominator smaller, fraction larger, so we don't inherently do that.
- 3.75** Now, the last thing we do before we get into some of our work on the computer is talk about the coefficient of variation. Why we have a coefficient of variation is because sometimes we love to determine whether one data set is varying more than another, but they have different units. See, it's really hard to compare two standard deviations with different units. What varies more? A standard deviation of 6.2 pounds or a standard deviation of 4.3 inches? You can't compare because one is in pounds and one is in inches. So, we have to have some way to get rid of those units.
- 3.76** That's what a coefficient of variation does. It's a very, very simple calculation. It measures relative variability. We can compare two data sets, and it takes your standard deviation and divides by the mean. For a population, if you have the population standard deviation, we would divide by the mean. If we have our sample standard deviation, we would divide by the mean. Same calculation, and it gives you a percentage, some sort of a ratio.

- 3.77** Let's take a look at this on our computer. Taking a look at some of our data, our big data set that we've had for a while now, let's look at our height and our weight, and we're going to find our standard deviation. We're going to find our mean so that we can calculate the coefficient of variation.
- 3.78** We're going to open up Descriptives. We're going to go to Descriptive Statistics. We're going to click over our height. Remember that we can drag and drop this. We can use our arrow right here. It'll move it over. We can also just double-click and it will move this for us. Once in a while, you're going to find this, too, that JASP has one little quirk that sometimes it does not do what you want and you have to do this. Click Data, click Analysis, and—for some reason, I have no idea why—this works. It works.
- 3.79** Let's find the two things that are relevant for us to calculate our coefficient of variation. We see here that we got a mean for our height. We've got our standard deviation. Oh, my gosh, look at that. That's a standard deviation. This is quite incredible, really, when you think about it, what this computer program has done right now.
- 3.80** Because if you had to do this for—I think we're on height—if you had to do this for the height, what this means is that you would have taken every single one of those 250 height data values. You would have subtracted 69.74 from every single one of them. You would have squared every single one of them. Then you would have added them. Then you would have divided by 249. Then you would have taken the square root. That's what that's given you like that.
- 3.81** That's crazy. That's amazing. Trust me, you don't want to do this by hand. I did it for years. It's incredibly frustrating. If you make one little error, it's completely off. This is what it's done for both the height and the weight for you. We look at our height, we have a mean of 69.74. We have a standard deviation of 2.399. Our weight, we've got a mean weight of 184.3. We've seen that before. We now have a standard deviation of 25.04.
- 3.82** This is what we're talking about, which has more spread? Right now, we cannot say because one is in pounds and one is in inches. This is why we did standard deviation and not variance. Variance would be in pounds squared and inches squared. This is not conducive to dealing with our data set, talking about the mean, or building any inference from. We have to have our units be the same. That's where this is coming from.
- 3.83** Now, let's compare them. All the coefficient of variation does is one very simple thing. It's going to take your standard deviation, it's going to divide by your mean. That's it. It's going to take 2.399, it's going to divide by 69.74. It's going to take 25.04, it's going to divide by 184.3. It's going to create a ratio for you.
- 3.84** Let's take a look at it. After we have found our standard deviation and our mean, we're going to compare them for height and weight. We just found these numbers out. We do use sample data. You probably understand why. Two hundred fifty is not all the men in the world. It is just 250 of them, and we have sample-driven data. So, we're going to be using sample statistics. We have sample standard deviation. We have sample mean. Let's divide them.
- 3.85** What we just found is we had about 2.4 inches for standard deviation. I rounded there. We have a mean of 69.74 inches. Our weight was 25.0 pounds. It was 25.04, but we rounded. And a sample mean of 184.3. All that happens is we divide: $2.399 \div 69.74 = 0.034$, and $25 \div 184.3 = 0.136$.
- 3.86** Think about what's going on here. Think about if you were to divide 2.4 inches by 69.74 inches, the units are going to cancel. There are no more units here. It's a unitless measurement. It's just telling you a measure of how much this is varying. The same thing happens with our weights. The units are canceling. It's just a measure of variation.
- 3.87** We look at this, which one is varying more? Height or weight? Weight. Weight has a lot more variation in general. That's what the coefficient of variation does for you. It's going to give you this unitless measure of variation that says, overall, this is how you can compare disparate variations among data sets. We calculate this number; it's a snapshot of what's really going on.
- 3.88** Let's get a little bit deeper. Even in JASP, when we go down to Statistics, this will calculate not only the variance for you—you'll see it pop up in just a minute when we click on Variance—it'll also calculate the coefficient of variation. The same numbers that we got just a moment ago are right here, the 0.034 and the 0.136. Very quick, very easy, but this is where it's coming from.
- 3.89** You see, if I don't show you the math behind it, these numbers are meaningless. You don't know where they're actually coming from. You don't know why we're dividing these two things, why we need the units

to cancel, why we're using standard deviation, not variance. The numbers are meaningless if we don't understand these concepts. We're using standard deviation because it has the same units as our data, as our mean. We're using standard deviation on a box plot because it incorporates our entire data set, every single value. We saw that in the formula.

- 3.90** We're doing coefficient of variation because it cancels the units, and you can compare two data sets with different units and still get something tangible. They say, "Ah, one varies more than another."
- 3.91** Now, let's talk a little bit more about standard deviation and how we do it. What are we trying to do? What we're trying to do is connect the shape and the spread of our data. What we're going to find out in just a moment is that they are strongly connected—that if we have a symmetric distribution, our mean and our median and our mode are all going to be exactly the same. You see, we saw that the mean was pulled toward a tail. If there are no tails, your mean is going to be right next to your median. Your mode is also going to be there, the most commonly occurring value.
- 3.92** We also talk about a skewed distribution, how some of these have tails and the mean was pulled toward that, so whether we have a skew or not, and how outliers distort the mean more than the median.
- 3.93** Now that we have an idea about some of the spread of our data, let's get a look at it. We already have height and we already have weight over here on the right-hand side, and we have some of our statistics that we like, but let's get a picture of what's going on. I want to come down and we want to click on Basic Plots, and we are going to click on a Distribution Plot. We also want to display density. This is what this is doing for us. This is taking our height, showing a distribution, and overlaying a curve to it.
- 3.94** Here's what we're going to be talking about in just a moment. We're going to be talking about something called a normal curve. It's going to be bell-shaped. It's going to be symmetric. It's going to be predictable. We love predictability because we can make inferences that are reliable because of that. This right here shows us a fairly symmetric plot, but we have some weirdness going on. Do you see the weirdness on the tails of these graphs? We have this nice bell, and then all of a sudden we have two tails. We have some values out there that are very far away from center.
- 3.95** This sort of a distribution is important to understand, and this sort of a distribution is important to understand—our weight. Look at our tail on this graph. We've been talking around it for a long time how the mean is pulled toward the tail. That's the tail. This shows you where your distribution is, that shows you that curve and where your tail lies.
- 3.96** Now, we take a look at this and we say, "This is funny." In fact, with our height, we have maybe a few values that are over here to the right-hand side. We have a few values that are over here to the left-hand side. They don't look that great. On our weight, we have maybe one or two that are over here. How do we identify them?
- 3.97** This is where we're going to connect our box plot to these graphs right here. If we go down, we click on our box plot, it's going to be under a customizable plot. We're going to click on box plot, and we are going to label our outliers.
- 3.98** Before we go any further, a lot of connections should be happening right about now. We should be making the following connections: that what a distribution does is it just shows where your data is. That's all these things are doing. What our statistics are—are there estimates for a population? Sure. But our distribution shows how our data is spread out. What we should be seeing is that our median for our height is really, really close to our mean. That's what this graph says. This is a fairly symmetric graph. Right down the middle, it's spread pretty much on both sides. Now, there's some weirdness in our tails. We'll talk about that in a minute. We'll talk about why that is in a minute.
- 3.99** We look at our weight, and our median is below our mean. Our mean is pulled to the tail of this graph. That's exactly what this is showing you. This is showing that your mean should be pulled to that tail. This is called skewed right or positively skewed. This graph for our height doesn't have any skew to it. Now, it has some oddness, but it's not skewed.
- 3.100** How do we tell what's what and how do we tell what these values are that are making this look funny? That's where our box plots come in very handy. Our box plots give us a picture of this. Our height says, "Oh, my gosh, we now know how to read box plots, don't we?" We look at this and say, "Okay, in the middle, 50% of our data for height is very, very, very close together." That's what this picture is showing you. This is very, very, very close together, right in the middle, there's not much spread there.

- 3.101** However, we have some very weird tails. We got maybe two or three values on the left, maybe two or three values on the right that are way different from what we would call typical. That is exactly what these values say. They say, "Okay, you have three potential outliers above. You have three potential outliers below." That's your upper and your lower. That's your right and your left tail.
- 3.102** Now you notice something funny. You go, "Wait, wait, Professor Leonard. No, no, no, no. There's no one in the world who's 244 inches. That's impossible. They're not that tall, and this is height. So where's this coming from?" JASP does something different than Microsoft Excel. It doesn't show you the actual value of the outlier. It will show you the identity of the outlier.
- 3.103** This says—and I did this on purpose so that we could see it easily in a moment—the 241st, the 242nd, the 243rd, the 244th, 245th, and 246th entries—people, they are the outliers. It's going to identify those people for you, which is very convenient if you have to go back and check your work.
- 3.104** The same thing happens on our weight. Oh, look at that. There's two outliers. It says this is a little bit more spread out—there is a little bit of spread there. It does not have any outliers at the bottom end. There's no left tail, but it does have some outliers at the top end. That's exactly what this graph is showing you. In fact, there's two of them, not just one. There's one right over here, and there's one right over here. That's what that's saying to you.
- 3.105** Now, let's go ahead and identify what they were. Remember, they were 241 through 246. If we go to our data, we go all the way down to the very bottom. Here are the 241st, 242nd, 243rd, 244th, 245th, and 246th entries. These are your three lowest heights. These are your three largest heights. The box plot has identified what your outliers were, what people your outliers were, so you could go do maybe a little bit more research on them.
- 3.106** Since we know that our distribution and our mean and our median are connected to our shape, our skewness, let's talk a little bit more about that. We have some different terminology when we deal with skewness. We don't just say it's skewed. Well, we want to talk about how this is actually skewed. This is what skewness looks like, what we refer to in statistics as skewness.
- 3.107** A symmetric curve—we're going to call this a normal curve in just a bit—a symmetric curve is going to have your mean and your median and your mode all in the center. It's perfectly symmetrical. It looks like a bell. It looks great. Very, very predictable. The other things that can happen are what we've been talking about for pretty much the whole time. We can have a skewed left or a skewed right distribution. What's that mean?
- 3.108** Skewed left means you have a tail on the left-hand side of your data. You have some smaller-than-normal values. Your mean is going to be pulled toward those smaller-than-normal values. So your mean is going to be left of center than your median. On skewed right or positively skewed, your tail is on the right. You have some larger-than-normal values. That's going to pull your mean toward those larger-than-normal values. Your mean is going to be greater than your median.
- 3.109** This is how this really looks for our skewness: positively, to the right. You're going to have tails to the right. You're going to have outliers potentially to the right or skewed to the left, skewed negatively, tail left, possibly some outliers on the left-hand side of your data. Now, what do we really want? What's the magic button, the easy button? We all love that, don't we? We want something that's always predictable. We want something that is relevant, that we can make decisions and it holds true. We want things that are usable.
- 3.110** Skewness really affects that. We don't typically like to see skewness in our data sets. We definitely don't really like to see it in our populations. What we like to see is a **normal distribution**. A **normal distribution** is what we would call a symmetrical graph—we saw that a little while ago, but now we're giving it a name—where the mean and the median and the mode are all in the center. It is symmetric. It has a nice bell shape. Mean equals median equals the mode.
- 3.111** Interestingly, a whole lot of natural phenomena follow that curve. When we get there in two lessons, two lessons from now, I'll show you the actual formula for that curve. There's a formula for it, and it has a funny little number called e . "Wait, you've lost your mind; e is not a number." There's a few numbers that are so common that occur in nature like π . Another one is e . It's a transcendental number. It's a number that's not rational. It has no end. It's not patternable. It's not a square root of any number. It's a transcendental number that occurs in nature. It's a phenomenon.

- 3.112** That curve is built on that number. I'll show you exactly where it comes from. Don't worry, I won't teach you calculus, but we'll talk a little bit about calculus on how that is actually created, which will be hopefully interesting to you, and I won't put you to sleep. But a lot of natural phenomena follow that because it follows the natural number. And in nature, it's found remarkably almost everywhere.
- 3.113** Now, how do we use it? We use it because it is so predictable that we have a rule for it. You say, "You're breaking your own rule. You say the statisticians don't come up with cool names. This is the empirical rule. Sounds super cool." Yes, but we also call it the 68, 95, 99.7 rule, so we break all that. We call it that because of what this does.
- 3.114** You see, the normal distribution is so predictable that if you know the mean—hey, have we talked about the mean? Oh, yeah. If you know the standard deviation, oh, we just talked about that. What's the standard deviation? A standard deviation is this statistic—or parameter if we're talking about population—this statistic that has the same units as the mean that shows the spread of your data set that incorporates every single value from your data set that is now relevant to us in a normal distribution.
- 3.115** If you know your data is normally distributed, here's what happens. Your mean is going to be right in the middle. We know that already. Your spread is determined by standard deviation, but very, very specifically. In fact, if you took your mean, right in the middle, you added this standard deviation that you now know about—it's a number that represents the spread and variability in your data—and you added it to your mean, and you subtracted it from your mean, you are now going to have an interval. That interval is going to contain 68% of your data. It has to. That's what normal distributions say.
- 3.116** If you take your mean, you add your standard deviation, you subtract your—I just got the math chills, guys. This is going to be exciting. You add standard deviation, you subtract the standard deviation, you're going to have an interval that will contain 68% of your data. Now, I'm going to make a jump here. I'm going to make a jump. When we talk about our tests, we're going to talk about probabilities. It also says this: if you were to randomly select a person from this population, there's a 68% chance they will fall in that interval.
- 3.117** Why is that important? We're going to get to tails in a moment. I'm going to ask you a question. I'm going to ask you, are there still people who would fall in this tail? You're going to say, "Yes, yes, because it goes on forever." I'm going to say, "Okay, what's the probability that that would happen?" You say, "Oh, it's going to be very small." Exactly. If the probability of something happening is very small, it's probably not happening by random chance. Something else is going on. That's the fundamental basis for pretty much all of hypothesis tests. That's it.
- 3.118** Let's keep going. If we added another standard deviation, and we subtracted another standard deviation from the mean, so we're now two standard deviations left, and we're two standard deviations to the right, so within two standard deviations of the mean. It's a farther spread, isn't it? It's a larger interval. It's going to contain 95% of your data.
- 3.119** Why does it get smaller? Well, most of it, most of your data is around the center, isn't it? Most data is centered around the mean. Why? Well, that's the whole point of the mean, is that it captures the center. In a normal distribution, 68% is within one standard deviation of the mean, 95% is within two, and 99.7% is within three.
- 3.120** Here's the question. Would it be strange, odd, weird to get a data entry further than three standard deviations from the mean? Would it be strange? Yes. Within three captures 99.7% of your population. There's only 0.3% on the left and right combined. That's 0.15% on the left and 0.15% on the right. Those are very, very small probabilities. This is a probability distribution. It shows you a distribution of percentages or probabilities.
- 3.121** We're going to use this. When we capture something that says, "Oh, my gosh, you are like three standard deviations from the mean here," that's very strange. That's probably not happening by random chance, because if it was random chance, 68% of that would be right in the middle, right next to the mean within one standard deviation. It'd be very strange to get something outside of that. It's the fundamental basis for our hypothesis testing.
- 3.122** Now, how does this look? Well, 68% within one and 95% within two, and 99.7% of your population within three means that we can use this to predict a lot of stuff. In fact, if we took a look at this for height and weight, where our standard deviation for our height is 2.4 inches, and our sample mean for our height is 69.74 inches, and for our weight, likewise, we had 25.0 pounds for our standard deviation and 184.3 for

our mean, if this represents our population, if this is normally distributed, we can make the following claims.

- 3.123** Hopefully, you can identify your mean for your height. Maybe you could do that now. Look up there and say, "Our mean for our height is 69.74." What are we going to do? Well, let's add 2.4 and let's subtract 2.4. That's our standard deviation. Sixty-eight percent of men in this population, if this is normally distributed and this represents population data very well, 68% will fall between 67.34 inches and 72.14 inches. Likewise, if you were to randomly select a person, there's a 68% chance they will fall within that range.
- 3.124** Let's add it again. Let's add 2.4 again. Let's subtract 2.4 again. We're adding 2.4 twice. We're subtracting 2.4. It's going to give you a larger range, isn't it? Oh, this makes sense. If you want to capture more of your population, you should have a broader interval. This is going to lend itself to something called confidence intervals. The more confident you want to be, the larger that interval is going to be.
- 3.125** Ninety-five percent says, "Yep, we're going to be between 64.94 and 74.54 inches." If you were to randomly select a person from this population, there is a 95% chance they're going to be within that range. How about the next one, 99.7%? Are there very many people outside of 99.7% of people? No, there's very few people outside of that.
- 3.126** So according to this, 99.7% of this population of men is going to fall somewhere between 62.54 inches and 6 foot 4 and 0.94 inches. Are there a whole lot of people out there who are taller than 6'5"? Not really. A whole lot of men shorter than 5'2"? No. In fact, 99.7% of men are going to fall somewhere between 5'2.5" and about 6'5". That's what this says.
- 3.127** Now, are there the people who are very, very tall? But are they rare? Are there people who are men who are very, very short? Yes, but they're rare. That's what this is saying. When we come up with those data values, could they happen by random chance? Yes, they could, they could, but it's not likely. Let me ask you a question. If we have some assumption that's built on a claim and we say, "Okay, the probability of this claim happening is really, really, really, really rare." What's it say about your claim? It might be wrong. That's how hypothesis testing is going to work when we get there.
- 3.128** We can do the same thing for weights. If we take our weights and we subtract 25 pounds, which is our standard deviation, we subtract that from the mean, if we add it to the mean, we're going to get a range. It says 68% of men should fall within this range, and 95% of men should fall within this range if we add another standard deviation.
- 3.129** Ninety-nine point seven percent of men, if we are three standard deviations from the mean, three above and three below, we're going to capture 99.7% of data: 109.3 pounds to 259.3 pounds. Now, think back. Do you remember the maximum? Do you remember the maximum weight for our one guy? It was 288. It falls outside of that range. That's another indicator that we have an outlier.
- 3.130** Now, the last thing we're going to talk about—sorry, I lied. This is not the last thing we're going to talk about. The second-to-last. I always do that. I see the last thing. This is like the last last. We have five more lasts. I'm so sorry. You have the question, "Normal distributions are great. What if it's not normally distributed?" There's another rule. It's seldom used, but I thought I would share it with you. It's called a super fun name, Chebyshev, Chebyshev's rule.
- 3.131** Now, the last thing. I promise it's the last thing. The last thing we want to talk about is a way to use this for individual data entries. You see, what we now know is that we have these intervals that contain some data values, but we have no way to determine where our data entry is.
- 3.132** This is what a z-score does for you. What a z-score does—it sounds funny, it sounds weird, but here's all a z-score does. A z-score will show you where your data value is on a normal distribution. In your head, you should have a question. Does a z-score work with every distribution? No, it's just a normal distribution, but it is absolutely remarkably powerful.
- 3.133** What a z-score takes is your data value, your mean and your standard deviation and calculates a number. Let's think through how this number is calculated. Well, the formula—remember, we're not all about memorizing formulas here, but understanding them is quite important. The formula takes your data value and subtracts your mean. Now, that looks really familiar because that's how standard deviation and variance was started, is we took these data values and we subtracted our mean. The ones that were above the mean were positive, weren't they? The ones that were below the mean were negative.

- 3.134** That's what happens here. This is going to show you that when a data value is above the mean, it is positive. When a data value is below the mean, it is negative. It gives you a distance. That's all this is, is a distance. But then it does something else. Don't write yet. You've got to wait for it. It divides by the standard deviation.
- 3.135** Here's what a z-score does. It finds a distance from the mean, positive or negative, and figures out how many standard deviations fit in that distance. It will give you the number of standard deviations from the mean. That's it. That's all a z-score is. It's the number of standard deviations that a data value is from the mean. If it's above the mean, it will be positive. If it's below the mean, it will be negative.
- 3.136** Now, you're wondering how to use it. Oh, this is remarkable. This is fantastic. Let's actually calculate a z-score in just a moment. Number of standard deviations from the mean. I just said all this stuff. Positive is above the mean and negative is below the mean. I should be clicking my clicker faster. I'm so sorry.
- 3.137** Do you understand that's exactly what a z-score is going to do in just a moment? That's all it's going to do. It's going to subtract these two numbers. Then it's going to try to fit how many standard deviations—the spread of your data, the one number that we have that shows spread of your data—how many of those fit into that distance. It's going to give you a number.
- 3.138** This is precisely what a z-score does. Data value minus mean, that's just a distance. Positive is above the mean. It finds out how many standard deviations fit. This tells you a z-score of 1.00. But understand it this way: this data value is 1.0 standard deviations above the mean. One standard deviation above the mean. If 50% of your data is below the mean, 50% is above, and you have a data value that you now know is one standard deviation above the mean, you are going to add 34%. Half of 68 is 34%. You're going to add 34%.
- 3.139** You now know that your score is better than 84% of people. You've just found a percentile. You are at the 84th percentile. That is what your test score did. So 50% is halfway. We had one standard deviation better than that. You did great. Eighty-four percent, that's half of 68—sorry, 34, half of 68. We've added that. Now we're able to use z-score to find a percentage and a probability and how much it did better or worse or even between. This is what we're getting to. That's exactly what our z-score tells us. It's just the number of standard deviations that a data value is above or below the mean.
- 3.140** We'll end with one. We're doing this for a reason, so you understand that there's a necessity for more. Let's say now you have a different test. It's got a different population. You scored a 70% on it, or 70 out of 100. The average for this test, for the whole population of people who took this test, is normally distributed and it has a mean of 75. Did you do better or worse than the mean? Well, you did a little bit worse. In fact, you scored five points lower. If you were to subtract that right now, you should get a negative in your head, shouldn't you? Seventy minus 75 should give you a negative.
- 3.141** But you have a problem, don't you? Because all we know with the empirical rule is mean ± 1 , ± 2 , and ± 3 . This is half. We have to find a way to deal with that.
- 3.142** Here's the nature of the rest of the next two lessons. We start talking about probability distributions in the next lesson—how to really organize our probabilities so that we can use them to make determinations. We're going to do this with discrete data first. We're going to say that that's not discrete data.
- 3.143** How are we going to do that? We're going to build out the concept of a normal distribution with this z-score, change it to a probability distribution, and figure out a way to determine the area that's to the left or right of any z-score that's on that graph, not just one or two or three standard deviations from the mean. I'm going to end there.
- 3.144** Is it true for, like, skewed data, it's better to use the median and the interquartile range for measurements, and then for symmetrical, mean standard deviation?
- 3.145** That's a great question. For skewed data, is it better to use the median versus the mean? I'll say it this way. It'd be a little bit nicer because you can't really estimate using standard deviation and mean if you have a skewed data set normally. With a normal data set, very predictable. Very predictable. This stuff works. As soon as we get away from that and things start getting tails on them and outliers, it's very hard to make predictions because they don't follow the model.
- 3.146** There's some ways around that. If you want to—I would say it this way. I'd say the median is much, much more applicable when you have strong outliers, whether they are to the left, to the right, or both. That will represent the center of your data better, and so will IQR. Standard deviation is very strongly affected.

3.147 We saw this actually in the data set. When you would take your data value minus your mean, if that is really a large distance, now you're squaring that. Your variance is strongly affected by this. Then we divide, okay, but that's going to be really pulled one way or another. It's not as nice. IQR for stuff that has those outliers would give you at least the 50%. Looking at that is going to tell you whether that's spread or not, regardless of outliers. Did that answer your question? Good question.

4. Random Variables & Probability

At a glance

Probability models connect random variables to possible outcomes. The lecture distinguishes discrete and continuous variables, develops probability rules and expected value, and introduces major discrete distributions including binomial and Poisson models.

- 4.1 It's time to make a jump to something. We've only really been describing our data. We've only really been able to talk about the measure of center and spread. We've just barely talked about really the distribution of things, and we touched on probability. But today, we're going to really dig into it because we're going to talk about something called a **random variable**. It's incredibly important to understand what a **random variable** is and why we're going to have two different types of distributions.
- 4.2 You see, you know this from our types of data that we have both **discrete** data and we have **continuous** data. Well, they're not the same. We're going to have two types of **probability distribution**s. We're going to have **discrete probability distribution**s, and we're going to have **continuous probability distribution**s. Why do we have both? Well, **discrete** has categories, doesn't it? Like nice little bars, and those little bars have probability. There's a distinct difference between zero and one and two. It's countable. It's very easy. **Continuous** is different. **Continuous**, you can't say how many people fall into exactly 72.3489 inches. That's virtually no one. There's not a distinct count for that particular entry. We're going to have two types of **probability distribution**s based on two types of random variables.
- 4.3 I'm going to start prefacing some of these things now. As we go through, some important concepts to get are what a random variable is, what that even means, that it's not chaos, it's not weird; we use the word "random" in common vernacular, that it means something. We also want to talk about what **expected value** is. From that, I'm going to build the idea of what's called a sampling distribution and preface our next couple of lessons. I'm always going to be front-loading for you. How we're going to do that is by talking through what's called a binomial probability distribution. It sounds super fancy. It's not hard. None of this is going to be hard. We're going to take a look at it in Microsoft Excel. It'll do it for you. But again, you have to understand the concepts.
- 4.4 Let's start with what a random variable is. The word "variable" means, well, of course, we're going to have a value that changes or that we're going to input a different value into this or somehow get a value into this one identifiable variable like height or weight or for discrete, the number of counts on a die or talking about successes or failures.
- 4.5 We're going to talk about that right now. What is a random variable? Random doesn't mean chaos. We sometimes use it that way like, "Whoa, that was random." What we mean is that dude was super strange. That's sometimes what we mean by it. But that's not what random means. Random means something determined by chance. That's what random means.
- 4.6 So what's a random variable? A random variable is a variable whose outcome is determined by a chance process. Now, listen, this is pretty key. What those chance processes are determines whether we have discrete data or continuous data. That's why we have two different types of random variables. If we are dealing with a chance process that deals with discrete outcomes—rolling a die, successes or failures on gambling, something like this—there's only a few categories. That's discrete. That's going to lead to a certain type of random variable.
- 4.7 If we have measurements, that doesn't do that anymore. We're going to have to have something different. It's still determined by chance, randomly selecting a person in your population by chance. Then where are they going to fall? Well, they're not going to fall into a certain bin when we're talking about height or weight. They're going to be an individual value, and there's infinitely many of them that you could possibly select from. It's completely different. Two types of distributions because we have two types of variable. Both of them are determined by chance processes. It assigns numerical values as outcomes of chance processes. This is the foundation for pretty much all of probability and statistics.
- 4.8 Now, this is a statistics class, but we have to understand a little bit about probability. Typically, if you went to a college statistics class, you would talk a lot about probability. In fact, you'd have chapters on how probability works and permutations and combinations and all of these ideas and fundamental axioms

that lead to probability. I'm going to teach this as we go—an intuitive approach to probability so that you can grasp the concept of statistics.

- 4.9** We're not going to talk a whole lot about combinatorics. We're not going to talk a whole lot about permutations, though I will reference them, show you what they are, and talk about what they mean. But we're not going to get heavy into you calculating your own combinations and permutations and things like this because we can talk about them as we go through our **binomial distribution** and **Poisson distribution** and talking about our normal curve and how that is its own probability distribution for continuous random variables.
- 4.10** We just talked about it. We're going to talk about it again. You have two variables; you're going to have two distributions: discrete versus continuous. What a discrete random variable consists of is a variable that has specific values. They are distinct. There's only a finite number of them. This should make sense at this point, where you can only categorize them, counting for categories, in a certain number of places. As we do that, we're going to get probabilities for certain categories.
- 4.11** Those categories don't exist when we're talking about just continuous levels of measurement. It doesn't work. We can't just categorize them as how many people, how many 2-year-olds weigh exactly 32.12864 pounds. How would you do it? There's an infinite number of possible values between 32 and 33 pounds. If we divide one value by infinity, you're going to get zero. It needs a different distribution.
- 4.12** What we talk about now is discrete, but it's going to lend itself to continuous probability distributions in the next lesson. Discrete values are those that are countable. They are finite. They're 1, 2, 3, 4. They are a certain number of categories. There aren't infinitely many of them, which means our probabilities can be calculated for individual categories or individual groupings or individual counts. Continuous, on the other hand, is any interval: height, time, weight, body mass index—not number of heads on a coin or something. Sorry, you want to go back? There you go. Not number of heads on a coin or something or number of faces on a die.
- 4.13** One concept as we go is this concept of expected value that we're going to build out across these distributions and across a continuous probability distribution. Only there, we're going to call it the mean. You see, the expected value is a mean. It's an average. It's what you would expect to happen long run. Expected value is simply a long-run average outcome of repeated trials, what you would expect long run.
- 4.14** Now, there's a super common misconception that people have because we're going to calculate expected values, and they're going to think, "Wait a minute, that's not one of my bins. How in the world can you get an average or expected value that isn't a possible outcome?" This is where the idea of a sampling distribution and collecting different samples comes from: your average doesn't actually have to be in your data set. Your average doesn't have to be in your data set.
- 4.15** For instance, if you had a weighted die—don't do a weighted die, they'll kick you out of Vegas, okay? But if you had a weighted die and you managed to somehow smuggle it in and it gave you more threes and fours than any other number, and you rolled that weighted die, would you sometimes get a one? Sure. And would you sometimes get a six? Sure. Maybe you'd get a two and you get fives every once in a while. For the most part, you might get threes and fours.
- 4.16** Now, let's roll this 100 million times. That's not even close to infinity, by the way, but 100 million times. We write down all the times we get one, and all the times we get two, and all the times you get three, and four, and five, and six. How many outcomes do you have? Just possibly six total outcomes. That's the entire population of the number of outcomes you could get. There's only six of them.
- 4.17** You now start averaging them. You say sometimes we get one, sometimes we get six, most of the time we get three and four. You add up all those 100 million times that you rolled it and you divide by the 100 million, you might get a number like 3.2. Can you roll a 3.2 on a die? No, you can't. It doesn't have to be on the actual die. It is a long-run average, not a specific indicated value. It's not a prediction of a single outcome. It is what you would expect long run if you did it hundreds of millions of times.
- 4.18** Now, what that's called is an expected value or a mean. Why are we using population notation? Oh, for what we talked about, there's only a finite number of values, and you know them all. When we build a probability distribution, you're going to have all the possible values. You're going to have all the probabilities for those possible values. It's the complete picture. For that reason, we use population notation to describe that, where the mean is μ , and the standard deviation—talking about the spread of these probabilities around that expected value—is still σ . It still gives you the spread of

the data, which is quite impressive. Why do we use population notation? Because we will have all the possible outcomes and every probability from it.

- 4.19** We're going to take our first look at a probability distribution and what they mean. I've talked a little bit about teaching you probability as we do this. One concept in probability that you do really need to know is that probability can never be more than one, and it can never be less than zero.
- 4.20** You see, zero means it is not going to happen. It is impossible. There's nothing less than impossible. And so we say there is zero chance of something happening. That means it's impossible. One means it is certain. It is absolutely going to happen. So there are no negative probabilities, and there are no probabilities greater than one. Don't confuse this with increase. You can absolutely have a 150% increase. That's fine, but not when we talk about probabilities. They're always between zero and one.
- 4.21** Every probability distribution is going to do this, because we're dealing with all of the possible outcomes and a probability for every one, if you have listed down every possible outcome and a probability for each one, they're going to have to add to one because you have represented 100% of the things that can happen—the population idea. You have everything there.
- 4.22** Now we're going to do a really weird example. I've got kids. I've got a lot of kids. Actually, we're about to have another kid, and babies tend to cry. We did a survey of 25 moms and how often their babies cry at night. Here's the responses. One of them responded, "My baby cries zero times." This is the one that you go, "Good for you. I'm so happy for you." Then five responded, "My baby cries one time a night." And 12 responded that my baby cries two times a night, four responded my baby cries three times a night, and my wife says five. That's her. Yeah, that's the last one. Two responded four, and my wife responds five times a night.
- 4.23** Now, listen, there's only 25 moms. We have all the responses. We have zero through five. That's the only responses that we got. What we're going to do right now is put this in a frequency distribution that has probabilities. It's going to look the same, except instead of counts of things, we're going to translate those into probabilities. That's a probability distribution. It lists out every single response that's possible. We only had zero, one, two, three, four, and five. That's six total types of responses. Oh, categories. These are finite. This is discrete. There's only six possible categories. All of our 25 mothers responded within those six possible categories.
- 4.24** In fact, one of them out of the 25, $1/25$. To find a probability, we just take the occurrences divided by the total. There's only one out of 25. I've done the math. That's 4%, or 0.04. So 0.04 chance that one of our mothers responded with, "My baby cries zero times a night." Well, we go down. There's five mothers out of 25 who responded once, and there were 12 out of 25 who responded twice, and so forth and so on. When we come up with this and we have every possible outcome, we have a probability for each one.
- 4.25** Let's look at the probabilities. Are each of those probabilities between zero and one? Yeah, there's none that are negative. There's none that are more than one. If we were really quick at math—which I'm not all that quick at math—but if I were to add all that up, $0.04 + 0.20$ and 0.48 , 0.16 , 0.08 , 0.04 , if you added those up, those would equal exactly, exactly one. They have to. They have to. You see, you have every possible outcome. You're missing none. Everyone has to fall in there for this particular group.
- 4.26** You have every possible probability, and it has to add up to 100% of your data, has to add up to 100% of your probability, that equals one. This is a probability distribution. It's how they pretty much all look. Every possible outcome, a probability associated with every single one of them, they've got to be between zero and one, and they have to add to one.
- 4.27** Now we're going to play a weird game because you are sleep-deprived and you don't know what else to do with your time. You're just pulling your hair out. Here's your game. You say, "We are going to put \$5 into a group of a lot of different people." The bet is \$5 a night. Here's how it works. If your baby cries less than two times, you win \$25. If your baby cries two times or more, then you lose your \$5. Do you get the game?
- 4.28** How much are you actually gaining? When you put in \$5 into the group and they pay you out \$25, how much are you gaining? It's going to feel like \$25, isn't it? But how much are you actually gaining? Only \$20. Do you understand this is how casinos get you? They hook you because you put your money on the table, your money's gone. They take your money. Now, you win. They give you your winnings plus your money. It feels like you're making more than you're actually making.

- 4.29** We're going to go ahead and compute the expected value. Let's take a look at it. Now that we've talked about the probability distribution, this is the probability distribution for our crying mommy game. Actually, maybe mommies do cry—I suppose that that is true—but our crying baby game that moms want to play like my wife does because sometimes you just need that, don't you? We're going to compute the expected value from this. We have all of our possible outcomes, zero through five. We have a probability for each one.
- 4.30** Now, our game said this: if your baby cries less than two times, you make not \$25, you make \$20. Let's put in \$20. If your baby cries zero times, you make \$20. If your baby cries one time, you make \$20. However, if it's two or more, you are going to lose \$5. I'm going to try to go slowly so that we can follow along on the manipulation of Microsoft Excel. If we drag and drop this, it will copy these down. That doesn't seem so impressive when you only have six entries, but if you had 100, it would be very impressive. You would not want to do this every single time.
- 4.31** Here's how expected value works. It works very similar to an average in that we have to find out a sum and divide. In this case, our sums are built into our value and our probabilities. We're going to multiply and just add. It will automatically divide because we have fractions here that are already divided. We divided 1 by 25 to get this number. We divided 5 by 25 to get this number. It's already built in. All we've got to do is somehow figure out how to multiply and then add.
- 4.32** In order to use Microsoft Excel, we've got to tell it how to do things. We're going to do that with an equal sign. We are going to multiply this, and then we're going to add it. Let's click on the cells that we want to multiply. I want to multiply the probability of our outcome. I want to multiply by the value that we get from that outcome.
- 4.33** All this is saying is that there is a 0.04 chance that you're going to get an outcome of zero. What you'd gain by that outcome is \$20, and we're going to click Enter. That's the value of that particular outcome. Now we're going to drag and drop. What's really nice is that the formula follows it down for you and calculates this in just a moment's time. It's very, very fast.
- 4.34** What an expected value does is it adds up the value times the probability. So on expected value, remember that we have to tell it what to do. We have to put in a function we want it to add. I'm going to click on Sum. I'm just going to highlight what I want it to add. If I do that and click Enter, it's telling you 1.
- 4.35** What in the world does 1 mean? What's an expected value of 1? This is the value of the money that you just told it. You have an expected value of \$1. Oh, here's the kicker. You said, "I can't get a dollar." No, you already know the outcomes. We spent a lot of time talking about the outcomes. If your baby cries zero or one time, you gain \$20. Two or more, you lose \$5. How in the world is this an expected value of 1? You can literally not make a dollar in this game. It's impossible.
- 4.36** This is what expected value does. It's not a predictor of a single value, a single outcome. This is a long-run outcome. If you played this game 100 million times, 100 million times, some of those times you're going to gain \$20, aren't you? Some of those times, most of those times, you're going to lose \$5. However, over time, the 20s are going to outpace the negative fives. In fact, if you played it 100 million times, you would expect to make \$100 million, \$1 on average for each time. The expected value is not a predictor of a single outcome. It is a long-run average. That's why we call it the mean. This is what you would expect to get.
- 4.37** Now, let's change our game. Let's say that, well, we're actually losing too much money as the casino that's offering this particular game. The way that we build these is really nice because if we just change one value, everything will change automatically. Now you say, if you did this game 100 million times, you would lose on average \$4 a game. You'd only win \$20 in one instance if your baby cried zero times; otherwise, you're losing \$5. The 20s are not going to outpace the negative fives. The negative fives are going to outpace the 20s. You still cannot lose exactly \$4 on any single game. But over 100 million times you played this, you would lose \$400 million. Don't play this game.
- 4.38** We're going to transition to another example, and I'm going to show you why in casinos, the house—you've heard this expression—the house always wins. We're going to talk about why that is. You see, every expected value is very important. While you cannot gain or lose this specific amount, over time, you will gain or lose a tremendous amount of money.
- 4.39** Let's transition. Let's talk about roulette. Maybe you've even played roulette. I like my money too much. I don't gamble. I would lose too much if I did. Sometimes it's fun, but I care about it too much. I work too

hard. Here's how roulette works. You're going to notice that we've done something here to show you how to categorize things. I'm starting to build an idea for you right here about something called a Bernoulli trial that will lead to a binomial probability distribution. You can do a lot of things if you can categorize them in just two categories: success and failure.

- 4.40** Even though the roulette table has 18 red slots and 18 black slots, it also has two green slots. I think there's a green and a double green, if I remember right. Those green and double green, that's how they're going to win. That's how this is not fair to you, and you are going to have a negative expected value. Because if it was just 18 red and 18 black, and it was just 36 slots, you could play red the whole time, and your expected value would be zero. Half the time you would win, half the time you would lose. The greens are going to shift that for you.
- 4.41** We're going to build a probability distribution out of this, but we're going to classify this as red and not red because we're going to bet on red. You have two outcomes, don't you? You're going to win or you're going to lose. Red or not red. Here's our probability distribution. With our only two outcomes: Bernoulli, binomial.
- 4.42** With our two outcomes, you have an 18 out of 38 chance of winning, and you have a 20 out of 38 chance of losing. Do you see how you have a greater chance of losing than winning already? We change that into a decimal. Is it a probability distribution? Well, let's look. Are our probabilities between zero and one? Yeah, every one of them is between zero and one. If you added that up real quick, it looks like it equals exactly one. This is a probability distribution.
- 4.43** Let's go ahead and calculate our expected value. We know from earlier that we have to multiply our probability, so I'm clicking the cell. I'm pressing Shift 8 because that is a multiplication sign for us. I'm going to click my value. I'm going to click Enter. No surprise, it's exactly the probability because you're just multiplying by one. But now this is in terms of money. You've multiplied value times a dollar. So this is 47.37 cents. I'm going to drag and drop this so that we have both of them.
- 4.44** Now we're going to compute the expected value. Notice how our second probability is negative. You're losing that dollar. You're losing 52.63 cents. Now, you can't actually do that, can you? You can't lose 52 cents. It's impossible. You're gaining a dollar, losing a dollar. But the greens are going to get you right now, and here's how they're going to get you. We're going to add this up. The word for adding up things is "sum." We're going to say, yes, we want to have this sum. I'm going to highlight the numbers that I want this program to add, and it's just going to add them for me, and I click Enter.
- 4.45** However, if you did this 100 million times, you're going to win a dollar, you're going to lose a dollar, your losing dollars are going to outpace your winning dollars, and you will end up losing \$0.0526 a spin of that roulette wheel. That's from your perspective. Now, why does it work? Why do casinos actually make money? Oh, from your perspective, you might actually come away ahead because you have a very small sample size. You might play this 10 times and walk away. You might actually come out on top.
- 4.46** Casinos do not think this way. Casinos think, "We're going to have this machine or this tool or this whatever you're playing, whatever game, and we're going to have hundreds of millions of times that this is played." So what's their perspective? Their perspective does this. Every time you bet on red, their probability of losing and their probability of winning shifts from you.
- 4.47** Now, they're making 5 cents, but for them, it makes sense, literally and figuratively, because they are expecting to do this 100 million times. So, every time this is played, they're making effectively, on average, long run average, 5 cents a spin of that roulette wheel. They're going to come out ahead. Every single expected value in a casino is negative for you, but positive for them. Why we play it is because in a very small sample, we might come out ahead. But long run, that is not going to happen at all. The house is always going to win. This is not small-sample stuff. This is long, long range. That's what this is talking about.
- 4.48** Now that we've understood the idea of a probability distribution, there are some specific ones that are extremely important to us. We will use them the rest of the class. We're going to build continuous probability distributions from the very last one, the Poisson, which we'll do in our next lesson. Before we get into that, we have to understand what a Bernoulli trial is. It's a really weird name.
- 4.49** There's a guy with the last name of Bernoulli who started studying probability theory, and he said, "You know what I really want to do? I want to have a way to study this so that we have some parameters that are fixed." One thing he said is, "Let's talk about a single trial." In the single trial, you can only have two

outcomes—just what we talked about—red or not red, not red and black and green. That fails this. That's very hard to calculate a probability distribution from when we talk about that.

- 4.50** We want success and failure, red and not red, candidate A, not candidate A. We want to categorize them. This is what a Bernoulli trial does. It's a single trial with just two outcomes. Now, understand, you can have lots of different outcomes, but you have to be able to categorize them as a success and a failure. Every single one of those trials has to be independent. This is a probability word that we use that says one trial doesn't affect the outcome of the next one. For instance, you roll a die, you get a two. You pick up the die, you roll it again. Does the fact you got a two affect the probability of you getting a two again or any other number? No, it doesn't matter. It's a die. It resets itself. That's an independent probability.
- 4.51** How about playing cards? You pick out a card; you get the ace of spades. If you keep that card out and draw another card, your probability has changed; that is dependent. It's called without replacement. You take it, you keep it out, your probability shifts. It's no longer independent. Well, if you put it back and shuffled it, now it's independent because it's with replacement. What we're talking about in a Bernoulli trial are these trials that are independent. One probability cannot affect one outcome, cannot affect the next outcome. The probability never changes. Think roulette, think die.
- 4.52** It's a basis for something called the binomial distribution. We're going to talk about that. If we look at just the name, "bi" means two. Well, actually, you might think of it this way. We had nominal levels of measurement. Nominal was just by name only. This is two names, two outcomes. This is all it is. Two categories: success and failure.
- 4.53** So, it's a fixed number of Bernoulli—oh, we just studied that—a fixed number of Bernoulli trials. These single trials, there are only two outcomes, they're all independent. The probability of success never changes. That's why they're independent. It's a single trial with just two outcomes, but you have to be able to categorize them as success and failure. You can have lots of different outcomes as long as you can say, "These are my successes, these are my failures." That's not going to change, and the probability of the success can't change. That's what the independence means.
- 4.54** We have a special letter for the failure and the success. The probability of success, we call that p . It's just a lowercase letter p . The probability of failure, we call that q . Are those going to change trial by trial? No, they can't. That's what the independence means. It means these do not change. They can't change or we fail the whole premise of a binomial distribution. They can't change.
- 4.55** What would your expected value be? Now, we use μ [μ] because that is our mean. We know that these are long-run averages. We have a probability distribution, which is every single element in the population of outcomes we could possibly have. We have our Greek letters signifying our population notation. If you wanted to find your expected value, all you would have to do is take the number of trials that you have, your n , and multiply it by how probable it is that they're successes. It's not too hard. It's just a multiplication of your sample size or number of trials times the probability of your success, which cannot change. Standard deviation is the spread around that mean.
- 4.56** Let's take a look at an example. I researched it a little while ago. It turns out that the rate of developing lung cancer is about 5.4% in America. About 5.4%, 0.054. How we change a percent into a decimal—which we have to have for all of our formulas—we move the decimal two spots left or we divide by 100. We're going to get 0.054. That would be our probability of success. Now, understand something right now. Developing lung cancer is not a success. So don't misconstrue the idea of what a success is. It's just getting a data value that satisfies your group of interests. That's what we call success. It does not have to be good or bad. It's just the grouping that we have.
- 4.57** You're going to survey 150, let's suppose, randomly selected people. What's your n ? Well, your n would be 150. You have 150 trials. You go out, I pick you, I pick you, randomly, not because I want to—I want you and you and you. I do that 150 times. We have $n = 150$. Your x would be the number of people who have lung cancer or develop lung cancer. This would be your successes. Now, again, not a success, but this would be the outcomes that satisfy the interest group that you are looking for.
- 4.58** The probability of identifying one of those people with lung cancer is 5.4%. It doesn't change because they're independent. Whether you get lung cancer, you get lung cancer, you get lung cancer—hopefully, God forbid, we don't want that to happen. But whether one person does and the other person does are completely independent. You're not going to give it to someone. They're independent. Therefore, the

probability of failure is just one minus that. Take 100%, subtract 5.4%, you're going to get 94.6%. The probability that someone doesn't have lung cancer is 94.6%.

- 4.59** Now, we're going to find the mean, the standard deviation, and I'm going to show you how to do this in Microsoft Excel. It's going to blow your minds. It's pretty cool. But first, what would you expect to happen? If we took the 150, that's the number of trials we had, and we multiplied it by the 0.054, the probability of success, it's going to give us how many people we would expect to find in that group of 150 that develop lung cancer. That's going to be 8.1.
- 4.60** Is there something going on in your heads? Wait, how can you get 8.1 people? People aren't that little. That's not what this—remember, this is long-run averages. Here's what this says, and I'm going to build the idea of a sampling distribution from this right here. Let's say you collected a sample of 150 people and you saw how many people had lung cancer. You might have zero. Maybe you do it again. A different sample of 150 people, you might get 10. Let's do it again. Let's take another sample of 150 people, you might get eight. Let's do it again. Let's take another sample of 150 people, you might get seven.
- 4.61** Now, you do that again and again and again for like 100 million samples of 150 people. You find out how many people have lung cancer in each one of those samples. You take all those statistics, you group them together, you now have a sampling distribution. We're going to talk exactly through that again. Then you average them. That is going to be your average. It's not talking about finding a single statistic from that sample, or in this case, our population. It's not pointing to a predictor from one sample. It's long run, long-run average. We would expect on average in groups of 150, we'd have about 8.1 people who would have lung cancer.
- 4.62** Our standard deviation is computed pretty basically. We take our sample size times the probability of success times the probability of failure. We take a square root that makes our units comparable to our data and our mean, and we say this is spread about that at 2.78 people above and below, and we can extrapolate from that.
- 4.63** Let's take a look now—oh, yes, one more thing: the formula. You're going to say, "Thank you, Professor Leonard." You never have to do that formula in your life. Never, ever, because we're going to find a way to actually do this on a computer. But it does mean something. The first weird part that says you're really excited, like n , it doesn't mean that. It's not an exclamation point like you're excited. It's called a factorial. It takes a number and multiplies by every previous number till you get down to 1. So 10 factorial would be $10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. They can be dramatically large numbers.
- 4.64** Then it divides by the sample size minus the number of successes, factorial times the number of successes, factorial. What it gives you is the number of groupings you can have. That's what combinatorics give you: the number of groups you could get. For instance, if you had a population of a million people and you plugged in a million for your n , and you wanted to find out how many groups you could make out of five, you could plug in five for your x for that many successes, it would tell you how many groupings you could do. It's combinatorics. It tells you how many groupings.
- 4.65** Then it multiplies by a probability of success to the successes, probability of failure to the failures. It would give you a probability for a specific number of successes. That's all it does. The formula just to the right of that does exactly the same thing. It's just a little bit more concise. That's what this formula is. That's where what we're about to do comes from.
- 4.66** We're going to take a look at Microsoft Excel, and I'll show you how people would actually do this. Let's see how a binomial probability distribution works. Let's use our lung cancer example. Maybe you remember this from what we just did. Our sample size, the number of trials, was 150 individuals. Well, we had a probability of success. The probability of success was 0.054. That's the probability of randomly selecting one individual and they have lung cancer.
- 4.67** We're going to do one more thing. I'll move this over here. I want q . I want the probability of failure. A lot of times with Microsoft Excel, instead of just putting in the point, whatever it is, 0.946, it's really nice to build this off of previous data that if you change or previous values, if you change it, it changes automatically. We're going to use this for our next example.
- 4.68** Lastly, we're going to identify what our success is, what our x is that we're looking for, and this would be people who have lung cancer. We'll adjust our widths here. For a binomial distribution, what you're looking at is every possible outcome of successes. These are outcomes of successes. What we're going

to create, what's the probability out of these 150 people that you find no one with lung cancer, or one person with lung cancer, or two people?

- 4.69** Do you understand? If you had to do this by hand, you would do that previous formula 150 different times. You would drive yourself nuts, and by the time you finished it, it would be irrelevant data. It'd take you like—I don't know how long it'd take you. But here's how we're going to do this in Microsoft Excel. We're going to press equals telling it, "I want you to create a formula for me," and we're going to put in binomial distribution. That's exactly what we're talking about.
- 4.70** It's going to say, "Where do you want me to look for the number of successes?" Oh, well, this is our column. Number of successes is starting at A2, and we're going to go all the way down to 150. Put a colon here, all the way down to there. This has selected every single one of those. That's our number of successes.
- 4.71** We're going to put a comma, and you're going to tell it how many trials you have. Well, we have 150. We're selecting 150 randomly selected individuals. We need a probability of success, so we put it here. Then we have a question, do you want this cumulative or not? Cumulative is going to add up the probabilities as you go. Not cumulative is going to give you a point value for every single one of them. That's what we want right now. I'm going to type in false. This is going to give us individual probabilities for every one of these just that fast. This is 150 probabilities given to you in the blink of an eye.
- 4.72** Here's what this says. The probability that you select 150 people and you find exactly five of them that have lung cancer is about 8.7%. How about finding eight people? That's 14.3%.
- 4.73** Do you remember what your mean was? My mean was 8.1. It should be right around there; it should be the largest probability, and that's what this says. Is it a probability distribution? If we were to add up all of these things all the way down to 150, it's going to be exactly, exactly one. Most of the time, there's an AutoSum feature in Microsoft Excel that you can turn on. It would show you that it's exactly one. I can take the time to AutoSum this for us, but this is exactly, exactly one.
- 4.74** Now, let's say we don't want to determine just what the probability is of selecting exactly four of these individuals who are going to have lung cancer, but maybe at most four. What we can do is grab our formula from up top. We're going to copy this. We're going to press Enter, come over, paste our formula, and then do something really fantastic.
- 4.75** Right now, it looks the same. However, we had something that asked whether we wanted this to be cumulative or not, whether we wanted to add up as we go or not. We say, "Well, we didn't before, but now we do." Let's come up and we're going to change our false to a true. Now that's added for us.
- 4.76** These probabilities, these five probabilities have added to 0.0879-ish. What that means is there is about an 8.8% chance that you will find at most four people that have lung cancer. You say, "Well, what about at least?" Oh, this is even better. If we take a look, put 1 minus this amount, this says you have a probability of 91% that you will have at least five people out of this that have lung cancer.
- 4.77** We can manipulate this. Now that we have our probability distribution, let's look at what a histogram would look like. We're going to come over here. Let's insert a chart. Let's grab a histogram chart. It's really a bar chart, but I like to make this look like it's a histogram. We're going to make it with a zero gap. We're going to put a border of black around it.
- 4.78** We get a really clear picture of what our binomial probability distribution looks like. Is it super common to get out of a group of 150 people, 20 people who have lung cancer? No, that's very small. Would it be fairly common out of a group of 150 people to have maybe nine? That's fairly common. Or seven? That's fairly common. In fact, if you took lots and lots and lots of samples of 150, you're going to be all over the place. But it's going to be very rare to have more than 20.
- 4.79** Now, listen to what that means. If it's very, very rare to have a group of 150 people with 20 of them having lung cancer, and yet you get that, well, there's two things that are going on. Either that was random chance that somehow you just randomly selected those people, or something's happening. They've been exposed to something. This is different. They might be a different population.
- 4.80** Then we would want to check out, what happened to this group? Were they exposed to some chemical? Are they smokers? Whatever this is, that would give you an idea that something strange is happening. It's very far away from normal. That's what our probabilities do for us. What is normal? What is not

normal? What would be very far away? What does that threshold look like? Typically, anything under 5% we would consider to be unusual. Sometimes it's 1%, but most of the time it's 5%.

- 4.81** If you look through some of your probabilities, you can easily see some that are quite unusual. It would be unusual to have anything more than 12 individuals in a group that would have lung cancer. It would be unusual to have anything less than, looks like, three. That'd be rare. We expect somewhere between, looks like, three and 12. Why? Well, your standard deviation was very small. It was only 2.79 or thereabouts. It's saying most of that data should be right around the mean.
- 4.82** Now, let's do another example. I can never do it, but maybe you can do it and take one of those water bottles like all the kids do and flip it onto the table, and then they fall on the floor and you go—and then they do it again, and they do it again, and you snap at them, right? No, that's just me.
- 4.83** Suppose you can flip a water bottle onto a table 38% of the time, which would be pretty phenomenal. Thirty-eight percent of the time when you try that little bottle flip thing, it lands right on that table. Suppose you attempt this 15 times. What is your expected value? How often would you expect to land that water bottle on the table out of those 15 attempts?
- 4.84** Well, that's not too hard to figure out. Your n would be 15, that's 15 trials. Your probability of success, it doesn't change. Whether you do it the first time versus the second time—that's the same flip that you got going on—is 38%, or 0.38. Probability of failure is 62%, or 0.62. And success would be landing the water bottle on the table. Your expected value would be really easy to find. You could just multiply the number of trials times the probability of success. You're going to get something around [five], [five to six], somewhere in there.
- 4.85** Let's go ahead and figure out how we can change the probability distribution that we have to now fit this example. Remember, our n is 15. We're only doing it 15 times. Our probability of success is 0.38. Let's take what we have and let's change what we've done. What's going to happen is before the blink of an eye, you are going to see a lot of changes happen real, real, real fast.
- 4.86** Our n is no longer 150. Our n is now 15. Look what happened to the chart. Everything's to the left. Our probability of success is not 0.054. It's now 0.38. Our probability of failure is going to change. Our success is no longer lung cancer. It is water bottle. Our probability of failure is now 62%, not 90-something percent, and every one of these probabilities has changed.
- 4.87** Look what happens as we get past 15. Oh, those values are negligible. They're so small because we only had 15 trials and our probability of success is exceedingly large and our histogram changed with it. This gives you a distribution of the probabilities you would expect to receive for each of those categories of successes. Is it likely that you're going to flip this water bottle onto the table all 15 times? Oh, no, not hardly at all. Or zero times? No, we should be right around 5 to 6. Our expected value is probably going to be right around 6, maybe a little bit 5.75 if we had to do the math on that.
- 4.88** So this is how our probability distributions work and how our binomial probability distributions work. Even if you wanted to answer the question, all the formulas are still there. If we wanted to find out the probability of flipping the water bottle onto the table at most four times, we've got that. It's changed it for you. You don't have to redo it. Or at least five times, 73%; it's got that for you as well. That's how binomial probability distributions work.
- 4.89** Then you realize that most people don't do this, do they? They don't sit there with a water bottle and say, "I'm going to flip this water bottle exactly 15 times and count how many times." No, no, no. No, we don't do that. We flip the water bottle until it works, and then we say, "I can't do any better than that," and we stop. That's what a geometric distribution is going to answer for you—what that looks like.
- 4.90** A geometric distribution looks a lot like a binomial distribution in its setup, but it answers a different question. It's still based on a Bernoulli trial. It's still based on a single outcome having only two categories: a success and a failure. They're still independent. The probability does not change between your trials. However, what geometric says is you're going to do this procedure until you get a success. I'm going to flip it until you get a success. Notice you can't start with zero here. You have to do at least one trial. So there's got to be at least one trial.
- 4.91** Theoretically—this is me—this would continue forever. I'm never going to get that. You could literally do it forever and ever and never get a [success]. What a geometric distribution says is this. You have these

independent trials. The probability of success never changes. Now do it until you get one success, and then you're going to stop.

- 4.92** The probability distribution is going to show the number of trials you have until you get that one success. That's all it does. Your x is the number of independent trials until that first success. This is very real life. This is how people operate. Probability doesn't change, and the formula, it's not super hard. Your expected value is just 1 divided by the probability of success. Notice that n is nowhere up there. Nowhere. Why? Because it doesn't matter. You're actually calculating the n that you need. You're not going to have a given number of trials going into it. You're calculating the number of trials you need. This is a complete shift.
- 4.93** Now, let's see how this looks. When we get to Microsoft Excel, we're going to be looking for something called negative binomial distribution. It's the same thing, but it goes by a different name. Let's go ahead and change this. Let's use what we have here, but we're going to modify this from a binomial probability distribution into a geometric. Here's how this is going to change. We're going to delete everything. We're going to delete this as well. We're going to look for negative binomial distribution. Why is it not called geometric?
- 4.94** They like the negative binomial because it's the opposite thought process of binomial. Binomial says you have a certain number of trials, you're finding successes. Negative or opposite says you're finding how many trials you need for that first success. It's a backwards way of thinking. It doesn't depend on n . In fact, all these x values that you have here are the number of trials you need until your first success. It'll find the probability that you need zero trials before your first success, or that you need one trial before your first success.
- 4.95** In fact, if you really think about this, the formula for doing this is not horrible, but your probability right here in this very first cell should be exactly this probability. You have a probability of 38% chance that you're going to get a success. So on your first trial, what's the probability of success? Thirty-eight percent. It should be exactly 0.38.
- 4.96** What we're going to do, though, is we're going to say what to do. We're going to look for a negative binomial distribution. We're going to click on it. It's going to ask you how many failures you want before your—zero. How many failures you want before your first success. I want zero failures before my first success. I want to fail zero times, and on my first trial, I want a success.
- 4.97** Then we're going to put a comma in here. It's going to ask you how many successes you're looking for. We want one, so we're going to put in one. Then it's going to ask you for the probability of that success, and it's right here. We don't have to put in the q . We don't have to put in the n because it's looking for a different question. It's also going to ask you, do you want this to add the probability up as we go or not? I don't want to do that.
- 4.98** Let's look up. There's our 0.38. Here's what this means. If your probability of success is 38%, or 0.38, then on your very first trial, you have a 38% chance of getting that as a success. Now, let's drag and drop this and see what happens. If we take a look, some very crazy things happen. Then you go, "Why are all these random numbers here? It's not working." Microsoft Excel will not only drag and drop your formula, but it will drag and drop from your cells.
- 4.99** We need to tell this, "I don't want you looking at cells below here." Here's what's happened. This first number looked at this cell right here. But the second thing that says, what's the number doing there? It's looking at the cell below. There's nothing there. It's dragged and dropped and adjusted where Microsoft Excel is looking.
- 4.100** We have to tell it, "Don't do that." Here's how we tell it, "Don't do that." We click on our formula and we're going to put a little number sign, so Shift +4, and that is going to tell it, do not move your cell down. I want you looking only right here as I drag and drop this all the way down. It adjusts our probabilities for us really nicely.
- 4.101** Here's what this says. It says the probability that you flip the bottle one time and you land it is 38%, that you flip it twice and land it is 23%, or three times, and you land it on your third time, that's your first success, is 14%. If we wanted to add up, what is the probability that would take us at most three [trials]? Well, we can do something very similar that we did before.

- 4.102** We'll copy our formula, we'll put it over here, and we will just tell it—now it looks the same—but we'll tell it, "Yeah, I want you to add these up as we go." It's going to tell you that this would be the probability that it takes you at most three trials to land that water bottle, 76%.
- 4.103** What about the probability it takes me at least four? About 24% that takes you at least four trials to land that first success, and it will give you a picture of what that looks like. Obviously, it's decreasing. Your probability is very high. You probably will land this in your first, second, or third trial. It's probably not going to take you, unless you're me, 20 trials to land this water bottle. It's very, very rare.
- 4.104** The last distribution I'd like to talk about is the hypergeometric distribution. You say, "Wow, that sounds crazy." It's not that crazy, but it'll help you in your card games. You see, what the hypergeometric distribution does is answers this question. Can probabilities be calculated when they're not independent? For instance, you are dealt a card and you keep the card. Oh, that's not independent any longer. You have shifted the deck, and so you've shifted the probability of what's going to happen in there.
- 4.105** If your population can be broken into two groups—one called an interest group—and they're not independent, your samples are taken without replacement. You take the card, you keep the card, and then you get more cards. That probability has changed. This is what this will answer for you. They're not independent. I'm going to show you how to do that with Microsoft Excel.
- 4.106** Now that we're here, let's talk about playing poker. How poker works in a standard poker game is you are dealt five cards. We're going to see what the probability is that you are dealt a flush. That you get five cards of the same suit made—and we'll talk about diamonds—that's a five-card diamond flush. Well, we know that as we're dealt cards, the probability shifts. Here is our interest in our sample, that you are dealt zero diamonds, or one, or two, or three, or four, or all five, or six, seven, eight, or nine. Now, we're going to see why this is zero in just a little bit. How many are in your sample size? Well, you have five. You have a sample size of five, just five cards.
- 4.107** What is the interest that's in your population? Well, if we know about cards at all, there are four suits, 52 cards. That means there's 13 of each suit in that deck. So there's 13 diamonds out of a deck size of 52 total cards. What we're going to do is tell this that I'm going to be sampling without replacement. I want to do a hypergeometric distribution. It's going to ask me what your interest is in your sample. That'd be zero, one, two, three, four, five, six, seven, eight, nine. How many are in your sample that you're interested in? The population and the population size. It's also going to ask us if we want to add this up or not. I don't.
- 4.108** Before we go any further, we know what's going to happen if I drag and drop this. It's going to get all messed up. It's not going to be looking in the right place because it's looking below here. I have to tell these cells, "I do not want you to move," so I'm going to put my dollar signs up there very quickly so that my cells do not adjust. I'm going to drag this all the way down.
- 4.109** Here's what this tells you. It tells you that if you are dealt five cards, there is a 22% chance you will have no diamonds. There is a 41% chance you will have one diamond. There is a 27% chance you'll have two diamonds. What's the probability you are dealt a diamond flush? Less than 1%, like 0.05%. That's probably not going to happen.
- 4.110** You say, "Professor Leonard, this is not my game. I like Texas Hold'em, and it's my favorite game. Tell me how to do that." Okay, I will tell you how to do that. How we do a Texas Hold'em-type game is, let's suppose that you are dealt two cards, and they're already diamonds. You have these two diamonds. You say, "Oh, I'm doing very well. I will bet lots of money because, of course, in these next five cards that I get on the table, the three cards, then the one card, then the other card, I can build my flush from that." What is left in the deck of cards? Firstly, you have two cards. That means there's only 50 left. You have two diamonds. That means there's only 11 left.
- 4.111** Your sample size is still going to be the five cards that are laid out on the table. For a total of seven, you have two, there's five more coming. This has now adjusted your probabilities. What do you need? What do you need in order to make that flush? You need at least three of those five cards to be diamonds. You already have two. You need at least this. You need three or four or five to be diamonds.
- 4.112** Let's calculate what the probability is that there's at most two diamonds. We're going to come over here. We're going to highlight our probability and copy it. We're going to come over here, paste it. We're going to change this because yes, I do want you to add this up. This is a cumulative probability. Then I'm going to tell this cell, "I want you to subtract this probability from 1." This has added up these three. In fact, you

can see it now. There's a little number right down here that AutoSums. It's so small for you guys, and I get that. But it will add up these cells that you have highlighted. It says 6.399, 6.399. Right there, 0.06. That would be a 6.4% chance.

- 4.113** So, if you are dealt two diamonds in a Texas Hold'em game, and they just put five on the table after that, you've got about a 6% chance that you are going to get a flush. Why are these all zero? There's only five cards being dealt. You only get five cards. That's what you told this as your sample size. You can't get more than that.
- 4.114** If you wanted a distribution—we want a probability distribution from this—we can easily do it. I do like to make these touching. It looks a lot better. This will give you the distribution of being able to complete this diamond flush on a Texas Hold'em hand and the likelihood of that happening. It's only three, four, or five.
- 4.115** Now that we're here, and we've talked about our probability distributions, we have only one more place to go. We have something called a Poisson distribution. We're not going to talk about it today. We're going to talk about it tomorrow. But we're going to do something I mentioned just a little while ago. We're going to build off this idea of a probability distribution from discrete values.
- 4.116** These look very blocky, don't they? They don't look very smooth. They say you're either here or you're here or you're here. It looks like a ladder, like a little stair step. We're going to take that idea and we're going to build continuous probability distributions. The same thing, but with continuous data rather than discrete. That's where we'll start next time.

5. Continuous Probability Distributions

At a glance

Continuous probability distributions describe measurements with infinitely many possible values. This lecture moves from Poisson probabilities to the normal curve, area-based probability, z-scores, percentiles, and software-assisted calculations while connecting discrete and continuous models.

- 5.1** Welcome to another lesson. We have a great one today. Not that all of them are not exceptionally great, because we're learning statistics and hopefully at a high level that you're understanding. But today we get to make a jump between a discrete probability distribution and a continuous in a way that they should be understandable, that they represent the very same thing. They represent a way for us to find probabilities. We have two of them because we have two different types of random variables. We have a discrete random variable that has only a finite number of possible outcomes, and we've studied that a lot. We're going to do one more as a way to jump into a continuous random variable and a **continuous probability distribution**, but we have to understand where it comes from. The necessity to have these two different distributions is because we have two different types of variables. We're going to study the second one for the most part today. Learn all about it. We'll talk about the **z-score** a little bit more, and you'll get your first picture of the **normal distribution** and how to find probabilities that are not 1 standard deviation or 2 standard deviations or 3 standard deviations from the mean. Hopefully, you see what we're doing. We're trying to build this case that we can find probabilities for any variable, but you have to have the right distribution, and we're making it to the continuous distribution so that we can answer questions about our natural world because many things in our natural world are represented by that normal curve. If we can somehow understand that, a **normal distribution**, we can answer lots of questions about it. In order to get there, we're going to talk about our very last discrete probability distribution. We'll segue from that into continuous. But the last one we want to talk about is called the "**Poisson distribution**." It's a funny name. It's named after a person—not me; my last name is Leonard—but **Poisson distribution**. What the **Poisson distribution** does is it answers, can you find the probability if you have a number of events occurring over a fixed interval of space or time? Now, right now, you're probably wondering, what in the world does that mean? Let me give you an example. It's a silly one, but you're on a really long road trip, and you have nothing better to do because you've exhausted all of the conversation with the person that you are traveling with. So you just start counting squeaks. For the next 10 miles, you count squeaks. There are 100 squeaks every 10 miles. So over a fixed amount of distance or space, you counted 100 squeaks in your car. Now you hypothesize from this, how many squeaks would you get in, say, an hour? Well, if there are 100 squeaks over—sorry, let me rephrase that. How many squeaks would you get over a mile? If there's 100 squeaks over 10 miles, how much would you get over a portion of that 10 miles? One mile. Well, you'd get 10 squeaks, one-tenth of that. This is what a **Poisson distribution** deals with: a number of events occurring over a fixed interval of space or time. It's not our trials that are fixed now; it is our space or it's our time.
- 5.2** Now, we only have to know one thing in order to learn everything about the Poisson distribution. All you have to know is the mean. We see this symbol again. We see this μ [μ]. We understand that for discrete probability distributions, this is your expected value. It is your mean. It's a long-run average, a long-term average. We're going to transition from this into the actual mean when we get to continuous. They mean virtually the same thing. What you expect long term, large samples, huge population, and that's why we have that symbol. All we've got to know is that the occurrences still have to be independent. So one squeak can't cause another squeak. It's just because your door is misaligned, every time you hit a monkey, it's going to drive you nuts because you counted 100 of them in 10 miles, but they have to be independent. One cannot affect the other. Now, our formula. I'm going to put this in words because it's a little bit easier for us to understand how we calculate the expected value for a Poisson distribution is just the portion of the interval times the number of occurrences in an interval. Here's what that means. Your whole interval was 10 miles. The portion of the interval is 1 mile. Well, that's one-tenth. We take the portion of the interval, the tenth, times the total occurrences of squeaks, that would be 100, and you would get 10 squeaks per mile. That's what this is saying. So, you would expect that every mile, no matter where that is in the entire interval, if this is distributed evenly, no matter where that is, for every mile, you would get 10 squeaks in that mile. Hopefully that's making sense.
- 5.3** Then once we have that, our Poisson distribution looks absolutely nuts. We say, wow, if you want to find the probability of a number of successes or a number of events occurring in that given interval, what you

would do is take our mean, that's our μ , to the number of times that would occur that you're looking for. If you want to find out what's the probability you get 3 squeaks in a mile, you'd put in 3. Then there's this e up there again. Well, we've talked about that just a little bit, and we're going to talk about it a whole bunch more today, and we're going to talk about it in our next lesson, what that e means. That e is the natural number. It comes from a guy named Leonhard Euler. You say, "Euler does not sound like it starts with an E," but it does. It's E-U-L-E-R. Leonhard Euler is the one who this is named after. He started studying nature through mathematics, and he found that this number occurs phenomenally all throughout nature in things that you can't even really understand why. For instance, the way that a cable hangs between two poles that you see going down the road, like your electrical wires and your telephone wires, a chain between two poles, that all can be modeled by e to a function.

- 5.4** We're not going to talk about that function today, but it's based on e ; e is the base. It's a natural number. It is a transcendental number. It is not a rational number. It is a real number, so it's on the real number line somewhere, but we can't represent it with any sort of a square root. We can't represent it with a fraction. It is an amazingly phenomenal number, which is why we have a number to identify. Sort of like π [π], we know that π [π] is the ratio between a diameter and a circumference of a circle. Where that came from was a long study that we're not going to get into today, but we could. It'd be super fun. But that's not the point, unfortunately. But e is very similar to that. It's a number that we don't even know the end of. It goes on forever. A lot of nature is based on this number. We take that to a negative exponent, a negative μ [μ]. That's our expected value, long-term average, long-run average. Then we divide by the number of instances in that interval. It looks like you're really excited about this variable. "Yeah, it's an x ." It just means a factorial. A factorial is something we would cover in probability theory, but it takes any number and multiplies it by every previous number till you get down to 1. This would be remarkably difficult to calculate by hand. If you had to find it for every single one of what we're going to do, it would take you a long time. Now, we're going to talk about this on Microsoft Excel in just a bit. It's going to do it for us. It's our last look at Microsoft Excel for this course. We're going to be in JASP for most of the rest of it. But Microsoft Excel does a really good job with probability distributions because of the drag-and-drop feature that we have. Let me give you a different example. You probably get emails. Do you guys get emails? I'm sure that you get emails. If you have an email, everyone in the world has 14 email accounts, and I get spam on all of them. But for my business email account, I get a whole lot of emails. In fact, one time, because, well, I'm me, I tracked them, and I said, "Wow, I got 15 emails between 10 a.m. and 12 p.m." Fifteen of them. I don't know if that happens to you, but it certainly happens to me. I got about 15 emails in that 2-hour or 120-minute span of time. Here's my question. What is the average amount of emails I will get in 30 minutes? What's the average amount of emails I will get in 330 minutes? Notice any given 30 minutes, not just from 10:00 to 10:30, and then 10:30 to 11:00, but between 10:155 and 10:45, or 10:07 and 10:37. If this is distributed normally and we get them at equal spacing, that would be independent. You don't have something that's saying, "I get a whole group of them here because I sent out a nasty letter, and then I get nothing for the other 2 hours." This would say this is distributed across the entire spectrum of your space and time. How many am I going to get in 30 minutes? Well, the first thing we would do—this is why I put this in words for our expected value—is calculate the portion of the interval that 30 minutes represents. So 30 minutes divided by our 120 minutes would be the portion of that interval. That's just one-fourth. Thirty minutes is a quarter of 2 hours. No problem. Now, we multiply by the total number of outcomes that we got in our interval. Remember, this is like our squeaks down the road. So instead of squeaks, we have emails. A quarter of our interval times 15, you would expect 3.75 emails in any 30-minute span. Can you get 3.75 emails? You cannot do this. But we understand this. We understood this from last time that what's going on here is that for any 2-hour block, if you got 15 emails and you started taking this 30-minute period, sometimes you'll get 3, sometimes you get 4, sometimes you get 0. Call it a great day. Sometimes you get 10. But overall, long run—if you did this 100 million times—on average, you would get 3.75 emails for any 30-minute span. Now, let's take a look at a different question and see how this looks on Microsoft Excel. When we deal with a Poisson probability distribution, the only thing that we really need to know is our mean. If you remember our mean for what we're working with right now, our mean was we expected 3.75 emails in any 30-minute span of time, interval of time. Over here, we have the x 's. This is going to stand for how many emails you get in the 30 minutes, whether we get 0 or 1 or 2 or 17. We're going to find the probability that you would get 0 emails in 30 minutes, or 8, or 16, or at most 3 or at least 4. How do we do that? Same thing that we've done pretty much every time with Microsoft Excel: we have to tell it what we want it to do. We are going to tell it that we're inserting a function with an equal sign, and

we're going to start typing in what we want. P-O-I-S-S-O-N. Poisson distribution. We're going to click on that, and then we're going to tell it what we want to look for or where we want it to look for a number of successes or the number of times that this occurs in your interval. We're going to scroll down and say, "We want this looking at 0." We're going to tell it what the mean is, and that's it. There's nothing else that we have here besides telling it, "I don't really want you to add this up as I go. I would rather you tell me one by one."

5.5 Now, what we're going to do is press Enter. It's going to give us a probability. Here's what this means. The probability that you receive 0 emails in that 30 minutes or any given 30-minute span is about 2.3%. That's what that says. That's fairly low. There's something I want to build right now about what fairly low means or rare. The word "rare" means it's probably not likely to occur. Usually that threshold is about 5%. We typically do 5%. When we get to hypothesis testing, this is going to make a lot of sense because I'm going to say, "Hey, your level of significance is 5%." Most of the time, this is what we have as a threshold to say, "This is rare enough to say that your assumption is unlikely." It is very unlikely that you would get 0 emails in a 30-minute span, provided that our assumption was correct, that we get 15 in every 2 hours. This would be very unlikely. It's less than 5%. Sometimes that threshold is 1%, sometimes it's 10%, but most of the time, it's right around 5%. That's what we get.

5.6 Now, we're going to drag and drop this, and you're going to notice that when we do it, something is not going to look right. We have zeros here and here. What's happening with Microsoft Excel is that it's looking below these cells and there's nothing there. When we drag this, it's looking for stuff that's not there. It's an amazing feature that Microsoft Excel does, but we have to fix it. How we're going to fix it is we're going to put in a little dollar sign—Shift + 4—and this is going to tell Microsoft Excel, "I really don't want you to change this number." I'm going to point to this mean, the exact same for every single one of these. We're going to go down to maybe 17. If you are not familiar with scientific notation, there's an e in there. You go, "Wow, what does that mean? This is crazy." What this means right here is that you would have 3.79, and in order to find this decimal, you would move your decimal place seven spaces to the left. This is 0.00000379. It is extremely, extremely small. Really, the only relevant pieces of information we have are 0 through probably 13 or 14. We'll make a histogram in just a moment. What this says is that if you wanted to find the probability that you are going to get 6 emails in 30 minutes, it is going to be about 9%, 0.09. Do you remember what our mean was? Our mean is right up there, 3.75. Well, look at our probability for getting exactly 3 emails in 30 minutes. It's the highest percent that you have. And right below that is 4, which is the second-highest percent you have. Your mean should be right in there for sure, because those are the most commonly occurring probabilities or the largest probabilities for us. Now, let's answer a different question. We say, "Okay, I want to know what's the probability that it's going to be a really calm, nice day. No one's going to hassle me with troubles." What is the probability I get at most, at most, 3 emails in this 30-minute span? I'm going to copy my formula. I'm going to come over here and paste it. Then all we have to do is tell it, "Now I do, I do want you to add up these probabilities for me." It's going to say that the probability that you get at most 3 emails in any given 30 minutes is 48%. If I add these here, you'll have an AutoSum right down at the bottom. It's very small. You can barely see it. They don't have it really out there for you. It's the exact same sum. Now, what's the probability that I get at least 4 emails? This was at most 3, and now we have at least 4. What we're going to tell it, "I want you to take 1 minus this probability," and it will tell you that. Even if we had to change our mean, if you said, "Well, now I have a different day. For these 2 hours, I got 17 emails, and my mean is now whatever that is." I'm just going to make this up as we go: 4.14. It'll change the entire probability distribution for you at the blink of an eye. I'm going to undo that, change it back. We're going to do one more thing, and here's our segue into continuous random variables. We take a look at this, and this is very pleasing, honestly, because we know that we're going to get exactly 0 or 1 or 2 or 3, which makes these very easy to add. When you're saying the word "or" in probability, you're adding probabilities together. We would just add them up. It's very easy. It's very nice. If we wanted to get a probability distribution, let's go down to maybe 14 here. We're going to insert a chart. It's probably going to be a recommended chart. Hey, there it is right there. I'm going to make this a little easier for us to see by removing our gaps. Maybe giving it a border so it looks a little nicer. This is your probability distribution for what we just saw. We saw that 0, well, fairly unlikely. Three and four, certainly very likely. You'd say, "Yeah, I'm probably going to get something around 3 or 4 emails in any given 30 minutes," and then it tails off sharply to the right. This is a tail-right distribution. You can see that. There's some out there, but there's not many of them. It's pulling this a little tailed right for the screen, tailed right like this. That's what it would look like for you guys. This Poisson distribution, it's nice because there's only a finite number of

possible outcomes. You're not going to get 14.2875 emails. It's impossible. You're either going to get 0 or 1 or 2, which makes it very blocky. It makes it very easy to add. But it doesn't work for everything because we know in real life that you don't always get 0 or 1 or 2 of something. Sometimes you measure a person's height, you don't get just 71 or 72 inches. I'm not exactly 72 inches. I'm a little bit below that. Or if we have someone's weight, you're not going to get 27 or 28 pounds. You're going to get maybe something between. How many values are between those? That's the big question. You see, with discrete probability distributions, with discrete numbers, you don't have that option. There's only a finite number. You're either here or here. You can't be between. But with normal natural measurements, you can be. With all the scale-type data that we have, you can be. It doesn't work. You can't put our typical natural measurements in a probability distribution like this because there's no way to categorize them. If you say, "But wait a second. Wait, the first probability distribution that you did with Microsoft Excel, and you had these bins, and they were 120, 140, 160, 180, and you did exactly that. You put them there." We did. However, as soon as you do that with continuous data and say, "I want to find all the values that are between 120 and 140 pounds," and you put them in that bin, you lose every specific amount of data. You have no idea what those values are anymore. All you did was count how many were in the group. You lost the person that was 127.5 pounds. You have no idea what that is anymore. You don't know if they were all 139 pounds, if they're distributed through there. It's impossible. When we try to do this with continuous data, you'll still get a graph, a histogram. It looks pretty. It's a great way to see it, but you lose specificity. Now, we know something about specificity. If you want to calculate the mean, you're going to have to know every single value. If you're going to calculate the standard deviation, you're going to have to know every single value. We have to have something different. This gives rise to a continuous random variable. Then from there, we build out the continuous probability distribution.

- 5.7** I guess the question that should be in your head right now is, do I understand the necessity to do anything else besides what we've just done? Do you understand that? Because if you understand the two different types of data, one looks this way—we can build a probability distribution one way—but with different data, it doesn't work here. We've got to have something else. That's what we're learning now.
- 5.8** So, what gives rise to the continuous random variable is this idea: these types of natural measured variables do not fit discrete probability distributions. There's more than a finite number of possible outcomes. In fact, the reason why they don't fit discrete models is because in every single interval, there's an infinite number of possible values you could get. We'll talk about that in just a minute. This is what's giving rise to a continuous probability distribution. There is one thing I want to talk to you about, though, before we get there. When we saw that Poisson distribution just a moment ago and we looked at all the probabilities, that was a probability distribution. Well, we know something about that. The probability of a probability distribution always adds up to a certain number. It always adds up to exactly 1. If you took all those little bars we saw, even the ones that were super, super, super small, even the ones out to infinity, and you added up all the probability, it would be exactly, exactly 1. Now, this is really important. Every probability distribution has to add to 1. I don't care if you're talking about a discrete probability distribution or a continuous. They all have to be 1. Here's what's going to happen with our continuous random variables. We're saying this: they don't fit here, we need something else. But we like that this probability distribution adds to 1. We're going to have to find something else that fits this data that also adds up to exactly 1. Are you seeing the idea here? We have to have it exactly 1; otherwise, probabilities fall apart. You have to have something that's always positive. It's going to be above the x-axis. We have to have something that has some area that when we add it all up, it's going to be exactly 1 just like our probability distributions for discrete, but that work with our data like this, the continuous random variables. From discrete to continuous. The discrete says this question we just answered. What's the probability that I get 3 emails in the next 30 minutes? That would be a discrete type of question. Very easy. We can only get a finite number of emails in a certain amount of time, or at least it's countable. Continuous asks this question. It says, not is it countable, not is it finite, not if there's a certain number of outcomes, because some variables are not finite, some variables are not countable. It asks a question like this one. The weights of 2-year-olds are continuous. Continuous means they could be anywhere on the scale, anywhere, not just 27 or 28 pounds.
- 5.9** Here's the question. What's the probability that a—remember, randomly selected, not your kid that you know their exact weight, not yours, a randomly selected child, the randomly selected 2-year-old. What's the probability that that randomly selected 2-year-old's weight is going to have a weight of exactly, exactly 27.9368 pounds?

- 5.10** In your head, is that fairly small right now? Firstly, no one besides me is that particular. No one's going to actually do that. I would probably do that just because. You're probably thinking, this is probably not going to happen. Why? Why, though? Why is this so different than the previous question? Let's think about 27 and 28 pounds. How many possible values are between 27 and 28 pounds? Can you even count from 27 to 28? Well, we think we can. Yes, of course, 27, 28. Oh, but see, it's not either 27 or 28. You have a real number line. How are you going to make it from 27 to 28? Some of you will say, "I go 27, 28." Others are going to do this. I would do 27 and then 27.1. But then other people are going to say, "No, I would say 27, then 277.01." Me, I would be the jerk and say 27 and then 27.00000001, and I would do that. But what's the next number? I don't know. You could do 27.000000011, and then 111. You could do it forever. You'd never make it to 28. What's that tell you about the interval? Even just between 27 and 28 pounds, there are an infinite number of possible values. So, what this is asking is, take one of those values, divide it by the possible number in that interval, which is infinity. What are you going to get? What's 1 divided by infinity? Zero. You just learned calculus. That's a limit idea in calculus that if you take—well, one of them—if you take 1 divided by infinity, you are going to get something very, very, very, very close to 0. This is 0. You can't answer this with a discrete probability distribution because you don't have a finite number of values to divide by anymore. It's impossible. You can't do it. We have to have something else. There's an infinite number of values between even just 27 and 28, disregarding the whole scale from 0 to infinity of a real number line that pounds would be on. It doesn't fit. And so we have to start asking different questions. We no longer ask the question, what's the probability of finding that exact weight? But we start asking about intervals. So what's the probability that a randomly selected child or 2-year-old will be between 21 and 29 pounds? Between. Why? Because there's no way for us to ask it about an exact value. We say, what's the percentage that weighs more than 30 pounds? This is doable. What's the percentage or probability that we find a randomly selected child and they weigh less than 24 pounds? Notice I'm using two words interchangeably. I'm using a proportion or a percentage, and I'm using a probability interchangeably because they're going to represent the same thing. Both of them equal exactly 1 if you take the entire probability distribution that we're going to see in just a moment. Exact same thing. What weight represents the 90th percentile? So what weight represents the bottom 90% of your data. This is bigger than 90% of them. These intervals are continuous. They're on a continuous scale, and they're represented by area under a curve. The question is, what curve?
- 5.11** Now we're going to start dealing with those values and understanding how this z-score can start to have an effect on our probability to show us how to do that. There's one issue, though, and maybe you see this just by thinking about it. If your mean is different and your standard deviation is different from one population to another, you're going to have a completely different distribution, just like we had with discrete. If you change one little thing, like our mean for Poisson, it changed the whole entire distribution. The same thing happens here. You can imagine if every single mean and every single standard deviation gives you a unique distribution, there are a whole lot of distributions. We're going to take a look at a couple of them right now. We're going to open up JASP and we are going to go over to Distributions, and we're going to click on, well, probably unsurprisingly, the normal distribution. I want to make sure that this normal distribution is based on a mean and our standard deviation. We're going to take a look. I'm going to adjust this just for a moment to right here. We can see it pretty clearly. This looks exactly like what we just had.
- 5.12** Now that we've studied all about the weights of 2-year-olds, you decide to go into business. You see, you want to design car seats for 2-year-olds. You say, "Yes, I can do this now because I have a 2-year-old of my own, which means I know everything about them." It's not true. We are going to design a car seat for a 2-year-old. We know only a couple of things about it. We know that the 2-year-old population is normally distributed for the weights. We know that the mean is 26.3. We know the standard deviation is 2.5. We have it in our minds. We want to design a car seat that is safe for weights between 20 and 30, between 20 and 30 pounds.
- 5.13** Okay, so we've designed our car seat. Now we're going to market it and we're going to say, this fits what percent? What percent of our population will be able to use this car seat that's designed safely for children weighing between 20 and 30 pounds? Let's go on to our JASP and take a look at this. What this turns out to be is not just an area above or below, but between. We're going to find out the probability or the percentage that our kids are going to weigh between 20 and 30 pounds. You say, "But Professor Leonard, you didn't find a z-score here." No, we didn't. What we're going to do is we're going to come

over here and be smart about it. Programs are so cool that they do all this for you. If we just plug in our 26.3 pounds for our mean, and if we plug in our 2.5 for our standard deviation, then we adjust our range a little bit because this is going to be way out there. We're going to have maybe 17 too, let's call it 45. Let's call it 35. Here's what that looks like. We're not there yet, but now we can plug in actual real values. I'm going to refresh this just a little bit. If you remember, our neighbor's kid was 22 pounds. Let's plug in 22. Let's see what this does: 96%. Instead of actually even having to calculate a z-score, this will do it for you.

- 5.14** We go back to JASP. It's okay. Well, up to 99.5%, we know that my kid is up to 99%, so maybe we start guessing at it a little bit. Okay, I want to maybe do 33 pounds. If we do 33 pounds, well, that's too much, isn't it? That's 99.6%. That's not the 99.5% we were shooting for. I don't really want to guess and check my way around this. Maybe there's a different way. And there is. WWouldn't you love it if your teacher someday said, "Let's check this out, and see if this is possible. Nope, moving on." No, this is totally possible. We're totally going to do it, but we need something else. WWe're going to open up something new. Now, we're going to tell this column what to do. We're going to do the only coding that we have to do in the entire class right now. We're going to tell this, "I want you to find an inverse normal function." What that does is it's going to look at the percentile you want and give you a z-score or a value from that percentile. I want this to be computed. Tell it yes, we're going to use our code, which is a statistical programming code. This particular one is qnorm. Because it's coding, just a little bit of coding, it's very particular on notation. We want qnorm, we want a left parenthesis, and all we're going to do is put in the percentile that we want, 0.995. We're going to close that percentile, and then we're going to tell this, "I want you to compute this." Compute the column. If I close this right down here, that value is the z value, the z value that would represent 99.5% to the left of it. This says 99.5% is going to be to the left of that value. I say, "But wait, that's not a weight. That's not a weight at all. Can we do this with a weight?" Absolutely. If we give it a little bit more information, so we're going to double-click, we're going to change this a little bit and say, "No, I don't want a mean of 0 and a standard deviation of 1. In fact, I want a mean of 26.3. I want a standard deviation of 2.5. Compute it again. Computer, that's what you're made for." I'm going to press Compute again, and now it gives us a specific weight. It says this is the weight we're looking for. You've waited long enough: 32.74-ish pounds. If we design our car seat for 32.74 pounds, it will be safe for 99.5% off our 2-year-olds. Just to be sure, to make sure we're doing the right thing, let's put 32.74, and we see that we find the percentile that is exactly what we were looking for: 99.5%. Where do we go from here? So far, we've talked about population parameters a lot. In fact, we have a whole normal distribution built on that. Our probability distributions for discrete variables were built on that; we had all of them. We've talked a ton about population parameters. But in the back of your head, you understand that this is not a parameters class. This is a statistics class, which comes from samples. We have to somehow make a jump from these populations, which we don't always know, to our samples, which we're going to try to use to estimate our populations. We've discussed μ and we've discussed σ . No problem. We know all about those things now. But real data doesn't come from populations, not normally. Sure, we do a census every once in a while, every 10 years in this country, but usually they come from samples. Samples are not usually perfectly normal. Populations, a lot of them are, yes. Samples are not usually perfectly normal. What that means is that that normal distribution isn't going to represent our sample perfectly unless we have some very strong constraints on it to understand how it works. Some populations we sample from are not normal. Can we use a z-score when our population is not normal? Yes, but we have to understand some things about it.
- 5.15** We've got to understand sampling distributions and the central limit theorem. We've been talking our way around sampling distributions just a little bit. What happens when we take samples? What happens when we collect this data? What if we took a whole bunch of samples and organized that data? What would it look like? Is there something that can take us from imperfect to making good inferences?
- 5.16** That's what we're going to take a look at next time: how sample means behave, how sample proportions behave, how sample variances behave. Our central limit theorem and law of large numbers are going to tie a lot of this together for us.
- 5.17** There seems to be a connection between the Poisson and binomial distribution. Is there a way one can be used to model the other? Are they similar in that way? The question is, is the Poisson distribution similar to the binomial? Can one model the other? They're very similar in that they deal with discrete information. Poisson asks a different question, though. Binomial asks, how many successes do you expect for a given number of trials? You have 15 or 30 or whatever, a finite number of trials, how many

successes do you expect. Poisson is not asking about trials. It's how many of those instances do you expect to occur in a given interval? It's a different type of question. Now, one thing that is going to happen that is super cool is—I'm going to ruin the whole thing here and ruin the suspense, but my opinion is education shouldn't have suspense, honestly. You should know what's coming. Here's what's going to happen. We're not going to use the Poisson very much because it doesn't ask a super relevant question for proportions, but binomial does. However, how can we use binomial with some of the things we're using? What we're going to learn is that the binomial probability distribution can be approximated by the normal. So wait, wait. You're telling me a continuous random variable, distribution of a random variable, can model a discrete structure? Yes. The binomial probability distribution will be modeled by what you just learned: the z curve, the normal distribution, **Gaussian curve**. That's going to be pretty impressive. I'll show you exactly that, but I can't remember when. It's today, though. It is today.

6. Sampling & Statistical Inference

At a glance

Sampling makes inference from a sample to a population possible. The lecture examines random sampling methods, sampling bias, sampling distributions, standard error, the central limit theorem, and the logic that supports estimation and later hypothesis testing.

- 6.1** This is one of the most exciting, exciting lessons that we're going to have because we get to make a jump on why samples are used to represent populations. So today we're going to talk about sampling, how to actually collect the data. We haven't done that yet. We've talked about a sample and we have an idea about what that maybe means. We haven't talked about any really integritous ways to collect one, and we're going to do that today.
- 6.2** Then we're going to take our sample data and talk about why in the world we can use it to estimate population parameters. And for the next two lessons after this to end our course here with Statistics Part 1, we're going to be doing that. I'm going to show you what a confidence interval is that we can estimate population parameters with these sample statistics that are collected well, and this is the start of doing that. This is really what we call the inner workings of inference, why the inference works.
- 6.3** Before we get there, we've got to talk a little bit about samples, how we collect this data. Why sampling matters? Well, we know this. We've been talking about this for the first five lessons now that we're rarely going to actually have access to an entire population. Even if we do, the collection of those populations when they're large is really laborious, takes a lot of time and a lot of money. We typically don't do it.
- 6.4** A well-designed sample is going to lead to well-designed research, which is going to lead to quality inferences, and we want quality inferences, stuff that reliably inferences our population. So poorly chosen samples are not going to do that. They're going to distort your results. We've probably seen that a lot of times. We've talked about influencers some of the times in here that sometimes we can't trust that things are reliable in life. Sometimes we can't trust that the research has been collected well. If a person is not willing to put forth their source, tell you how this is collected, then we pretty much discount those results.
- 6.5** This is why every psychological research paper is well-documented on: here is what we did. Here is how the sample is collected. Here is the methodology we use. Here is how we corrected for anything that did not fit the methodology, and here's the results, and here's why they matter. All of that is incredibly important to make you less vulnerable to someone else selling bad information with poor statistics or poor samples. To the best of our ability, we have to collect our samples well. It's not just about selecting people. That's not what we're doing. We're choosing correctly.
- 6.6** One other thing that often happens is when people get poor results, they just get more data. They say, "Oh, yes, if I just collect more of it, it's going to be better." If you collect data poorly, you're going to get a wrong answer. If you collect more of it, you're just going to get a wrong answer more precisely. You'll be more precisely wrong. This is not the point of statistics to just collect tons and tons of data so you can be wrong quicker or more exactly wrong. No, we want to collect them well.
- 6.7** It all starts with this idea of random sampling. Random, of course, we learned, does not mean chaos. What random means is that every element in your population has an equal chance of being selected. There's ways to do this and ways to not do this, things to avoid, because it's the foundation for everything that we consider unbiased. When we start to violate random sampling, we start introducing bias into our samples.
- 6.8** Now, bias doesn't have to be intentional. Like, "Oh, I'm biased toward that person"—that's not what this means—"or this group of people." It means that you are not giving equal representation to your population. What random means is you are doing that. What bias means is that you are limiting the ability to select each and every person or randomly select each and every person that introduces this bias into our sample.
- 6.9** What does a simple random sample mean compared to that? They're two sides of the same coin. What random means is that you and you and you and you and all these other people have an equal chance of being selected. And what a simple random sample means goes a step further. It says not only does every person have an equal chance of being selected for a sample, but every group of the same size

has an equal chance. This group of 30 people, or this group of 30 people, or this group of 30 people has the same chance of being selected. That's really what a simple **random sample** means.

- 6.10** Now, how we get to these simple random samples. To have a random sample, there's a variety of techniques. Some of the most common ones are called a stratified sample, a cluster sample, and a systematic sample. We'll talk about how they can be combined. We'll talk about what each one does.
- 6.11** A stratified sample, you might use if your population is broken into characteristics that are identifiable that you want to represent. Something like age group or socioeconomic status, things of this nature where we can divide our population into these layers. We call the layers strata, like a stratosphere. We can think of gender, maybe, and like we said, these other groupings that we might want to consider the characteristics of.
- 6.12** Why we would do this is if we want to make sure that our sample represents these particular characteristics of our population. We would choose to create a stratified sample. We would break it into strata; we would randomly sample from each stratum. We'd say, "Okay, I want to represent my 80- to 90-year-olds and my 70- to 80-year-olds." Maybe if my 80- to 90-year-old population, that's 10% of my population, I want to make sure that I sample 10% of my sample from that particular stratum. I would take a sample from each of these groups to make sure to ensure that we represent all the groups that are there. Because what can happen in your sample is if you just line everybody up and pick a random person, you could potentially miss these groups of interest that you want to represent.
- 6.13** This is one way when you have these natural characteristics that you want to study in your group intentionally, you might try a stratified sample. It will represent all of your groups. A cluster sample is going to seem very similar at first. A lot of students really get this confused because while these are still groups, a stratified sample breaks them into layers of characteristics; a cluster breaks them into clusters that really don't share a characteristic. For instance, your area code. You have an area code on your cell phone. That's not grouped by really any characteristic besides maybe location, but not gender, not socioeconomic status, not age. We don't give certain age groups a certain area code or a zip code per se. That's when your population is naturally grouped, but it's not by any specific characteristic. That's when we might use a cluster sample.
- 6.14** Now, what a cluster sample does, it breaks your population into these little groups, and then it samples some of those groups, but it samples all the people in them or all the items or elements in those particular clusters. That's what a cluster sample does, not based on characteristics besides those which are maybe location-based, but not a natural characteristic we would be interested in studying specifically. And then we randomly select those clusters, and we sample everybody who is in those clusters.
- 6.15** Hopefully, you're all still with me so far. So stratified—layers of characteristics you want to study. Cluster—really just groupings; we select some of them and sample everybody inside of them.
- 6.16** The last one is called a systematic sample. What a systematic sample is, is going to make a lot of sense to you. You give every item in your population a number from 1 to however many you have. You all line them all up. Everybody's got a number. Then you pick a random starting spot. You'd give me a number from whatever you think. You'd say 7. Okay, we're going to start at the seventh person in line. You're chosen. Then you choose another random number, like 5. You're going to pick every fifth person past that. You start with 7. That was chosen at random. Then you choose another random number, like 5. We're going to do 7, 12, 17, and so forth and so on, every fifth person after that, or whatever item you have. We go through the entire list. There's one drawback to this. If your list has patterns, this can introduce bias because you won't be selecting people truly at random because your list is not at random. If it does have patterns in it, we'd have a caution there.
- 6.17** The last thing we want to say about sampling is that these can be combined. For instance, you could use a stratified sample and you could break up your population into these groups by characteristics, and then you've got to sample some of the people. So you might take one of those characteristics, one of those strata, and then you might use a stratified sample within that. Or you might list out a ton of clusters and use a systematic sample within that. So we can have clusters within our stratified sample. We can have systematic within our stratified sample or cluster samples, we can combine these techniques when appropriate, if we have groups and subgroups and groups within those even. These are all ways for us to collect data randomly. Stratified, cluster, systematic—we can combine them.

- 6.18** What we don't want to do, what we don't want to do—we don't want to do what's called a convenience sample. What a convenience sample is, is honestly, it's done so often. Almost all of the surveys that you take are a convenience sample because they're based on a volunteer response. So a convenience sample is, simply put, collecting the data that's easy to get. That's really all that it is.
- 6.19** The problem is, easy data is not often good data. Why? Well, let's use the volunteer response survey, for instance. Have you ever seen a survey and you do not volunteer to respond? Sure, I have all the time. I almost never respond. I don't have time for that. No one's got time for that. Why don't I respond? Well, I don't care very strongly about it. What happens if I did care very strongly about this? Well, then I would respond.
- 6.20** With volunteer response surveys, you typically get people who care greatly about the topic, which means their responses can be quite emotionally charged at times. What does that mean? You are not giving someone who doesn't care as much the equal opportunity to respond because you're limiting their care even in responding to it. You don't have a random sample. You have only people who really care about the topic, responding to the topic. That is not random. That is specifically biased. Now, you might have equal people on both sides of the issue. That's true. But you have not considered the population.
- 6.21** That volunteer response survey that has only taken people who are maybe highly emotionally charged does not represent your population. So, if you use that to represent your population, now we have a very, very poor inference. This is why—and that's not integritous. This is why sampling is such a big deal. We have to have randomness; otherwise, we have bias, even though someone maybe is not intending to do this. We talked about that volunteer response might be emotionally charged. We feel strongly, we respond strongly; we don't feel strongly, we don't respond at all.
- 6.22** Also, getting data that's easy is usually people who maybe think like us. We usually, usually, surround ourselves with people who have the same mode of thinking or at least have the same ideologies most of the time. If we just ask those people, it's going to be an echo chamber of pretty much what we think already. That's not random either. We need random to avoid things like this.
- 6.23** You all know this last one. Grandma's going to answer however I want to say, "What do you think, dear?" "I think this, Grandma." "Me too." Then you've completely skewed all of your results. That's not at all random. You can't use that to represent people who you haven't even considered. This is the whole point.
- 6.24** Now, the types of bias we have. Firstly, we do have this selection bias. If we are not selecting randomly, we'll have selection bias. But we can have a couple other types of bias in our samples.
- 6.25** One of them is non-response bias. You might have someone who is a random selection who just flat doesn't respond. Or they say, "Yes, I don't feel comfortable sharing my weight. I'm not going to give you that portion of it." Well, that is a non-response. You've collected that person's age and their height and their BMI, but they're not going to share their weight or their smoking habits or something. So you're missing some of this information. You can start to develop some non-response bias.
- 6.26** You can also develop response bias. What's that mean? People lie. People lie. On the survey, do you ever pick your nose in public? "I never do that. No, I don't do that at all." "I just saw you do that." "That was Harold." "I don't—" People lie. Lies will skew these results. If you're calling someone on the telephone, they have no incentive to actually tell you the truth at all. And so response bias is part of that. So people lie.
- 6.27** And also the phrasing of our questions is remarkably important. I remember this when I did my master's thesis. It was about statistics, actually. I was doing a class about whether students who were just in a typical geometry class or students who did geometry with manipulatives that we built with our hands, which did better, if there was a significant difference between them. Now, I had a control and I had a sample. I taught both classes, and these just had manipulatives. And so what I wanted to do was show whether one or the other performed better, statistically, significantly, so that I could say something about the broader population.
- 6.28** We did all of this work. I had to sit down with my instructor at the time, and he said, "The questions you're asking them on the survey part of this are very misleading." I would ask questions like, didn't you think this was a better way of learning? Just like this question, do you think this decision was wrong? Well, that already prefaces the response that you want from these people.

- 6.29** He said, "You have to change all of your questions because you are unintentionally leading them to the response that you want." I said, "Well, yes, of course I do because I want this class to work better." He said, "This is not the point of statistics. This is not the point. We have to actually see what randomly selected people think about your topic. You are trying to get them to speak about what they want, not trying to get them to speak about what you want." This is the whole point. This is how we collect our samples.
- 6.30** Now, what do you do with this? First, we identify any sampling error. Understand that no matter what you do, even with perfect samples, is your sample going to differ from your population? Is your sample going to perfectly represent your population? It can't. It's impossible. The only way to actually do that is to take a census to make your sample the actual population. Then you would have no error.
- 6.31** But if you take just a subgroup, no matter how well you select it, no matter what questions you ask, the people you select are not going to perfectly represent your population. I don't know that I've ever got asked a sample about who I'm voting for in an election, ever. I'm a registered voter. I've never been asked. And yet I see these surveys all the time, 43% think this or 69% think this. I've never done that. How is that collected? Well, they're using a sample, randomly selected, of lots and lots of people and getting their opinion. It doesn't have to be mine. It may or may not represent mine exactly. It's not because they've never asked me.
- 6.32** Samples are not going to perfectly—watch, I'm going to get 30 phone calls now. Samples are not going to perfectly represent our populations. This can cause poor inferences, but we have some methodology to actually fix this a little bit.
- 6.33** Sampling variability. This error, this error between our sample and our population, this variability sample to sample. For instance, if I take a sample of these 30 people and I take a sample of these 30 people, randomly selected, everything's fine. These 30 people are not going to represent my population perfectly. These 30 people are not going to represent my population perfectly. These 30 people are actually not even going to represent those 30 people perfectly. They're going to be different. So every sample that you take, even if you collect it randomly, is going to vary that sampling variability.
- 6.34** They're going to vary sample by sample by sample by sample. If I collect 50 or 60 or 100 randomly selected men, they're going to have a different mean height than a different set of 100 or whatever randomly selected men. They're going to be different. They are going to vary. How we deal with that is very important.
- 6.35** See, this starts our need for something called **sampling distributions**. As we build this out, I've seen this for 20 years now, students and people really confuse what a sample distribution is to a **sampling distribution**, and they can be quite confusing.
- 6.36** I'm going to do three things for us to make this clear. I'm going to go over it at a high level what a sample distribution is, what a sampling distribution is. I'm going to go over it with a specific example. I think we use weights of 2-year-olds again because we're very comfortable with that at this point. Then we're going to show you on JASP exactly how to build one. You are going to see a sampling distribution with the data set that we have, our weights of 2-year-olds, 26.3 pounds, 2.5 for a standard deviation.
- 6.37** What a sample distribution is? Let's just use the number—I'm actually going to use 31, and I'm going to use it for a reason. Let's take a sample of 31 people. Let's say that we're collecting their heights. I go out, I randomly select 31 people. I take a measurement of their heights. Notice we are using continuous data here so that we're going to be building this into a normal distribution. That's why we're using measurements. I take these 31 heights and I measure each and every one of those 31 heights. We now have 31 pieces of information. Be with me.
- 6.38** We put those on a distribution. We should mark them down. Some are going to be over here, some are in the middle. Probably most in the middle, some over here. That is called a sample distribution. It is just a graph, pretty much a chart, a distribution of our sample data. You would have literal sample data, 71 inches, 71.5 inches, 72.75 inches. You would have the actual sample data on a distribution showing smallest to largest and where they were. That's what a sample distribution is. Remember, this is high level. What a sampling distribution is is very, very different.
- 6.39** A sampling distribution of a sample statistic does this. It says, take that first sample of 31, collect all the heights, maybe find the mean. That's a statistic. That's a statistic. All those 31 people would be averaged to how many results? One. One statistic. Now, you mark that down. Now, you go, you take another

sample of 31 people. It's got to be the same size. You find their heights and you average them. You're going to have one more sample statistic. Then you mark that down. You do that many, many, many, many, many, many, many, many times.

- 6.40** You don't have these individual values anymore. You actually have statistics of your samples that are in a distribution, a sampling distribution of the sample mean or proportion or variance, which are the big three that we typically use. We'll talk about the mean mostly because people understand the mean more than proportion and more than variance. But this is what a sampling distribution is, and I'll get much more specific about it in just a moment.
- 6.41** A sample distribution is just a distribution of the data in exactly one sample. It's literal figures of the people involved. We'll use people here, it can be anything else, but it's actual hard data. This is not what a sampling distribution does. A sampling distribution takes lots and lots of samples. It calculates a statistic for each of those samples. Then after that, after it computes that statistic, it puts those statistics in a distribution.
- 6.42** There's a question that should be in your head. Who cares? Why? Why do we care? Oh, fascinating things happen when you have sampling distributions, and you are going to see it today. You're going to see why this is so important because you have no idea how well your first sample distribution represents your population. You have no idea. But we do have an idea about how well sampling distributions represent the population, and they're remarkably well. We're going to start building this out.
- 6.43** Why sampling distributions actually matter? They matter because we never see a population, ever. We hardly have—okay, a census, yes, but that's very rare. Practicality, we never see a population. We only see samples. We have to know how that sample statistic behaves, how it behaves. Is it always gathered at the center? Is it really spread out? What's the spread look like? Does it matter on sample size? All of these things are incredibly, incredibly important to actually understand.
- 6.44** The common statistics that we do this with are sample mean, sample proportion, and sample variance. You also might be thinking, where's standard deviation? Why isn't that up there? It is not a great—it's a good, but it's not a perfect estimator. So sample mean points to population mean, sample proportion points to population proportion, sample variance points to population variance. Sample standard deviation is close. It's close, but it's not as good as these ones.
- 6.45** Let's talk about our baby weight example. I'm going to get much more specific. We're going to talk about how many samples are even possible. It's going to blow your mind, like internet-breaking stuff here. Crazy. Let's imagine that we take a look at our baby weight examples. We have a population, and there are 2 million 2-year-olds—I made that up, by the way. There are 2 million 2-year-olds in our country. I have no idea how they are. We're just using a round number: 2 million 2-year-olds in our country, and they're normally distributed. You randomly select 30 of them. Let's compute the sample mean weight.
- 6.46** Okay, so we have our 30 2-year-olds that we've collected. We sample their weight. You're going to get something like maybe 26 pounds. You put that down. Then you go out, you get another sample of 30 children. You sample their weight. You average their weight. You're going to get a sample mean of maybe 28 pounds. Then you go out again, you get another sample of 30 2-year-olds, and you find their weights, and you average their weights. You find the mean of their weights. That's another statistic. Maybe you get 24.3 pounds.
- 6.47** Then you do it again and again and again. What's your sample size? Oh, it never changes. Your sample size is always 30. Out of those hundreds of samples that you take, you take 30 kids, you find all their weights, you average their weights, it gives you one statistic, you put it on a graph, you do it again, you find one statistic, you put it on a graph, you do it again, you find one statistic, you put it on a graph, you start to build a sampling distribution of a sample statistic, which is our sample mean in this case.
- 6.48** But you don't do it once or twice or 5 times; you do it 10,000 times. You do it 100,000 times. What you would have is a full sampling distribution of the sample mean. Every possible sample of 30 2-year-olds across that 2 million, the size 2 million population of 2-year-olds. Do you know how many that is? That's a crazy amount.
- 6.49** If we take a sample of 30, and then another one, and then another one, and another one, we are going to have not individual results anymore. If we took 100 samples each of 30 kids, you would have now 100 sample means. Not 100 weights; 100 sample means, 100 sample averages of those weights. Those

should be pretty close to the population mean, actually. Each individual one might be different, but combined, it should be pretty close because the sample mean points to the population mean.

- 6.50** You might repeat this forever. You might repeat these samples forever. For all your 10,000 different samples of 30 kids averaging their weights, you might not get two of the same sample weights ever. Probably will, but they don't have to. They could go on forever and ever and ever. How many samples are possible? If you had two—just think about this. This is crazy. If you had 2 million 2-year-olds and you're taking samples of size 30, randomly selected, how many of those samples are actually possible?
- 6.51** This is called the combinatorics idea. Combination says if we take 2 million people, that's our n , that's the full size that we're picking from, and you're choosing r at a time, which is 30—we're not going to get into combinatorics, but I did show you a formula a while ago. It had some factorials.
- 6.52** The first time we saw a factorial, here's what this would say. There are this many groups. If you took 2 million and chose them without order, order doesn't matter here; whether you choose Johnny or Sue or Sue or Johnny, it doesn't matter. You're just averaging their weights anyway. Take 30 people, 30 of those 2-year-olds. Now take 30 different people. It could include some of these, actually. Now do it again and do it again. How many of those samples are possible?
- 6.53** This number, probably you're thinking, is very large already. It's over 170 digits long. Not 170, just 170 digits. You start writing like a 9, 5, write 168 zeros, and that's what you have. This is calculator-breaking. Your calculator won't handle this. This is an incredibly, incredibly, incredibly large number. This is how many samples are possible out of that population with a size of 30.
- 6.54** You're going to pick one. It's not going to perfectly represent, but if you took them all and if you found the sample statistic, sample mean in this case, and you organized that into a distribution, you are going to have a perfect sampling distribution of a sample mean, and it will be normally distributed. That's what we're going to see in just a moment. If you were to take all those samples, what's that look like?
- 6.55** We are going to learn all about what a sampling distribution looks like right now. This is the third. Remember, I did high level, we did specific, and now we're going to build one. We're going to take a look at exactly what a sampling distribution of the sample mean does.
- 6.56** This is under **central limit theorem** in JASP. It starts off with a normal distribution. This is our parent distribution. That's what we want. Over on the right-hand side, you can see it. That's our normal distribution. It has a mean of 0, standard deviation of 1. You know what? I would like to change that. I want to make this something that we understand because while we've been working with our 2-year-olds, let's do 26.3 pounds, and let's do a standard deviation of 2.5. This is going to change our parent distribution, and I'm going to fully explain what's going on here.
- 6.57** This is your population. This is what we had. We said, yes, the mean weight of the population of our 2 million 2-year-olds was 26.3 pounds, and the standard deviation is 2.5. It has built this particular probability distribution, a normal curve. It's normally distributed. It's built this from that data. Understand this is our population right now.
- 6.58** Now, we're going to be sampling from this population. We're going to build ourselves up to a sample size of 31. I'm going to do that for a reason. You'll see this in the central limit theorem, why that is. But what we're going to do now is we're going to start building samples. It's a remarkable program that lets you actually take samples. I'm going to put just a sample size of 5 and a total sample of, let's do 7, just so that we can see what happens.
- 6.59** What JASP is going to do, it is going to theoretically construct samples for you from this population. What's our sample size? Five. How many samples are we taking? Seven. Let's see what it does. Here's our parent population, just our normal curve, our distribution here that we're sampling from. Here are the samples that this computer constructed for you. This is going to put teeth behind what a sampling distribution is. Remember, it's one of the most confused topics amongst students. They rarely get this. You're going to get it. This is sample number one. These right down here are called rug marks. It's a funny little word that says this is your actual hard data.
- 6.60** Look at what this has done. This has taken a sample of 1, 2, 3, 4, 5 people. Your sample size was 5. These are the actual results of those 5 people, something a little bit below 24 pounds. Here's one right at 24 pounds. Here's one a little bit above 24 pounds, a little bit more than one almost to 28. This is a sample distribution. Not sampling yet. This is a sample distribution. It's just one sample and the

information you collected from one sample. Now, look up here at the yellow. That is the mean of those 5 data pieces. This is a statistic from that sample.

- 6.61** Now, we move on to the next one. Here's a second sample. Here's your 1, 2, 3, 4, 5 data entries for your second sample. This is a sample distribution. They've taken a mean. This is a sample statistic. This is your mean of that particular sample. And they've done it again. In fact, they've taken 7 different samples. Understand, get this: these are 7 distinct sample distributions. It's just your data. Each of them has 5 little marks in it. Here's one below 24, one above 24, and three that are clustered right below 28. This is yet another sample distribution. Each of them has a sample mean to it.
- 6.62** Now, take those 7 samples that you had. Each of those samples has exactly 5 pieces of information in it. That was 7 different sample distributions. You're looking at the last 3 right now. There's only 7. Each of them has 5. They've taken a mean of each of those samples. How many means should you have right now? Not 5. You should have 7. You should have 7 means. Take those means. Now, organize them into a distribution. This is a sampling distribution of the sample mean.
- 6.63** But you see, we only have 7 samples. How many samples were possible? Something that is more than 170 digits long. That's enormous. We only have 7. We're going to start to adjust this, and you're going to start to see a sampling distribution build. From this, you are going to identify the central limit theorem on your own. In your heads, I promise you, it will build in your heads right now. You'll identify it.
- 6.64** Here's what I want you to look for. Look at this number right here. That mean says this: I have taken all 7 of my sample statistics: 1, 2, 3, 4, 5, 6, 7. These are the 7 sample means that we had up here: 26.88, 24.64, 28.31, 23.78—that's a low one—26.18. Not a single one of those is the same. Not a single one of them is exactly 26.3. None of them are. Every sample had a different mean than your population. Every sample had a unique mean in itself. It's not the same as the rest of them. But now I'm grouping them, and now I'm averaging the average. This mean says take all 7 of your sample means, 1, 2, 3, 4, 5, 6, 7, the same ones we just saw, and now find the mean of that.
- 6.65** Pay close attention to this number and this number. Let's play with this a little bit more. It's okay. I don't want just 7 samples. I want 10 samples. I need to show you this one first to show you what's going to happen. Here, your samples have changed. It's retaken all these samples. There's now 5 new pieces of information. There are now 7 samples here, but this says until 10. So behind the scenes, it has 3 samples that you don't see. You don't see them; they're there. You don't know what the sample means are; they're there.
- 6.66** And now they've taken all those sample means. Now, you have 10 points of information, 10 sample statistics. You've put those 10 sample statistics in a distribution, and you have a sampling distribution of, in this case, a sample mean: 1, 2, 3, 4, 5, 6—those are pretty close to the mean—7, 8, 9, 10. You've taken the average of those. It's 25.97.
- 6.67** Let's now do this. Sample size of 5 is really small, real small. Let's take a sample size of maybe 10. Let's see what happens. I told you we're going to spend time on this, but it's really important that you get, I promise.
- 6.68** Now, you're still going to have 10 samples. Each of them now have no longer just 5 pieces of information, but this is a sample of size 10: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. There's 10 individual data values of these 10 2-year-olds that were selected. This is a sample of size 10. This is a sample of size 10. This is a sample—all of them are samples of size 10. Those 10 individuals, when we average their weight, the sample mean is right here, 26.43. Most of them are a little bit closer to your population mean already. Remember, 26.3.
- 6.69** Now, let's take all of those 10 samples. Let's find the sample mean for every one of them, all the way down to the 10 that you don't see. All of them, we'll put them here. This is a sampling distribution of those 10 sample means that you just saw. Look at the 26.32. That's actually pretty close to this. We're not even close to the right sample size. We're not even close to a whole bunch of samples. Let's see what happens now.
- 6.70** We're going to go ahead and we're going to put this up to 31. This is your sample size. Now let's take a sample—and this is the last time we'll go slowly through it, but I need to make sure you get this. If you get this, you'll get the rest of it. This is now samples of size 31. We're only taking 10 of them so you can see what's happening. But here's our parent distribution. This is our population. The population is normally distributed. It has a mean of 26.3 and a standard deviation of 2.5. We are sampling from this.

- 6.71** Check this out. Here's your first sample. There are 31 little rug marks down there, 31 individual pieces of data in this sample distribution. Those 31 pieces of information have a sample mean. That's a statistic. We do it again, and again, and again, and again, and again, all the way to 10. There are now 10 samples, each with 31 people, each that have a sample statistic. There should be 10 sample means. We organize those.
- 6.72** It's not pretty yet. I know it's not pretty yet, but we're going to start increasing the number of samples now. Here are our 10 sample means. One's below 26, two are real close to 26, one's a little above, a little above that, 26.2. Notice how the scale is really, really condensed now. It's tighter together. In fact, that sample mean is pretty close to our population mean. It's pretty close. We only have 10 samples of size 31.
- 6.73** What if we did 100? Here's the first 7 of those 100 samples. There are 93 more sample means that you don't see. Take a look at what's happening to our distribution. Look at all these little black lines down there. All those rug marks say this is 100 sample means that are organized in a distribution. That's a sampling distribution. Check out your mean.
- 6.74** Now, 100 is nowhere near, nowhere near the number of samples possible. Let's do 1,000. We're going to scroll quickly through this. If we did 1,000—whoa, that looks a lot prettier. Why is this all black down there? These are all your little rug marks. There's so darn many of them. It makes a black line. Where are they? Where are they clustered around? This is important. This is the point. Where are they clustered around? They are clustered around the population mean. They're clustered around the population mean.
- 6.75** In fact, if we do this, let's take 10,000. Understand that 10,000 is nowhere near the number of samples that we just talked about, the 170-digit-long number. But watch what happens when we do it. What this has done is taken amazingly 10,000 different samples of 31 individuals, found a sample mean, taken all 10,000 of those sample means, and organized those into a distribution. That is really, really, really, really, really close to a normal distribution already, and it's not even touching how many samples are possible. What's the mean? Exactly, exactly the same as your population mean.
- 6.76** This is the point. You have just identified what the central limit theorem is going to tell you, that while each individual sample is not going to equal your population mean, it can't. While they're going to be different from one another, when you distribute all of them, they will absolutely be central, central limit, centered around your population mean. In fact, if you average them, it is exactly your population mean. It's going to be approximately normally distributed as we see here. It's very, very, very close. You have a standard deviation called the **standard error**, which is predictable. We're going to be able to tell how this is actually spread out.
- 6.77** Why is that all black down there? That's where we're clustered. Our samples, our sample means are going to point to—they're going to cluster around our population mean.
- 6.78** Now, here's the question where we go from there. I'm prefacing something we're going to do next lesson. Would it be possible, possible, to get a sample mean that's way out here to the right? Yes, possible. Is it likely? No. Is it possible to get a sample mean way out here? Yes, but it's not likely.
- 6.79** If we do, one of two things is happening. Either we collected our sample randomly, and this was just a strange random occurrence which can happen, or something is different in the population you sampled from than you expected. Something's different. This is the whole premise of hypothesis testing. As you say, if the probability of that sample being around 24 is like 0.0005%, is that likely to actually happen? No, that's very rare by random standards. Then either that's a random chance, which is a possibility, or something has changed about this sample from our population than we expected. That says what we assumed is probably wrong. That's what we're going to go to in three lessons from now when we talk about hypothesis testing.
- 6.80** What the central limit theorem says is exactly what you are seeing right here. If you take a whole bunch of samples of size 30, greater than 30, 31 or more, and you find the statistics of each of those samples, and you organize them in a distribution, the sampling distribution of the sample mean will be normally distributed, pretty close, and its mean will be the population mean. That also happens for proportion. It will be normally distributed. It also happens for variance, but it will be skewed to the right and predictably skewed to the right.

- 6.81** What we talked about just now is that there are unbiased estimators that build predictable sampling distributions for us. Sample mean is one of them. We saw this. We just did a sampling distribution of the sample mean and the average of all those sample means was the population mean. We also saw that the sample means clustered. You saw the rug marks. They cluster around the population mean. It's an unbiased estimator. It's not going to over- or underpredict. The sample mean points to the population mean.
- 6.82** The same thing happens for sample proportion. You see this funny little symbol. You see this p with a— we call it p hat $[\hat{p}]$, actually. It has a little symbol on top of it. I told you a long time ago that with sample statistics, we have English letters, and they usually have at least some sort of an accent. This is how we represent sample proportion. The same thing, we're not going to go through it now, but the same thing that we just did with the sample mean happens with sample proportion. It targets, it points to the population proportion, and it creates a sampling distribution that is approximately normally distributed, and it's unbiased. It does point to it. It doesn't overestimate it, doesn't underestimate it.
- 6.83** The same thing happens with sample variance. Sample variance points to and estimates unbiasedly, not above or below, [population] variance. Median, range, and standard deviation, they're not unbiased. Standard deviation is very close. We use it, but we understand that it does have a tendency to underestimate it, and so we take that into account.
- 6.84** Now, here's our shapes. We just talked about it. We just saw it. Now you're seeing the teeth of this. That normal distribution is what your sample mean and your sample proportion create when you find a sampling distribution for those two statistics. You just saw it on the central limit theorem part of this. Sample variance doesn't do this. It creates something that is skewed to the right. We're actually going to study two different types of distributions that are based on variance. One is called the chi-squared distribution, and one is called the F distribution. They're actually both skewed right, just like this. They're predictable. We can find the measurements for these distributions.
- 6.85** Here's what our central limit theorem says. Our central limit theorem says that regardless of population shape, for a large enough n , the distribution of your sample means is approximately normal. You might not actually believe this. I'm going to show you this. We have to see this. You might not get this. That regardless of the population—that's a fantastic statement and maybe unbelievable right now—that if you sample from a distribution that is not normal, if your sample size is large enough, that your sampling distribution will still be normal. Check this out.
- 6.86** Let's go over here and say, "All right, central limit theorem, show me how it works." Let's make our parent distribution uniform. A uniform distribution looks like this. A uniform distribution says every outcome is equally likely. This is what it looks like. This is your parent distribution now. We're going to sample our 10,000 samples of size 31 from this. The samples are going to look a little bit funny. Actually, most of our means are still right around 26.3. They're all pointing to that. You're seeing the central limit theorem at work. You're seeing that these sample means cluster around your population mean. Actually, none of them so far, even out of 26, they're all right around 26.3.
- 6.87** Now, we do this 10,000 times. We organize it, and my goodness gracious, that's still a normal distribution. Regardless of your distribution shape of your population, if your sample size is large enough, your sampling distribution of the sample mean will look like this. Even if you say, "No, no, no, let's make it skewed and really strongly skewed to the [left] just because we want to." That's pretty strongly skewed left—actually, high skew left. Our sample means are a little bit worse, but our population is still exactly the same. It points to the same thing.
- 6.88** This is what the central limit theorem says, regardless of the shape of your population. If your sample size is large enough, the distribution of your sample mean is going to be approximately normal.
- 6.89** The law of large numbers says this. It says, as your sample increases, it's going to average out. Your samples are going to point to—your sample mean is going to point to your population mean. What the central limit theorem says is that it averages out predictably. In fact, what we mean by predictably is that the spread of the data is known.
- 6.90** You should have a question about why that is. Why is that there? Why is it affected by the sample size? Well, this is what the law of large numbers basically says to you. If your sample size goes up, is your sample mean going to be closer to your population mean? Yes, totally. You take a sample of 5 people or 500,000 people, which is going to have a more accurate sample mean? Well, the 500,000, if they're

randomly selected. It should be closer to your population mean. What that says is that as your sample size goes up, your sample means are going to cluster more and more strongly around your population mean.

- 6.91** Think about what happens to that fraction. As n gets bigger, that fraction gets smaller. That fraction is the spread of your means. If that fraction gets smaller, it's saying your sample means are clustering tighter and tighter and tighter around your population mean until you reach the population size and then it becomes 0. And then you have exactly a census.
- 6.92** This is what the central limit theorem says. Here's the parameters. Here's what you need to know. You need to know that your population is normal or, remember, this works for any population distribution. We just showed that. Or the sample size has to be greater than 30, like 31. If that happens, guys, honestly, all I'm doing is putting symbols to what you've just learned. That's all we're doing right now.
- 6.93** The symbols say this. Did you remember how, when we took our sample means, that's our $\bar{x}$, and how we found the average of those, that's the μ of the $\bar{x}$. So averaging our averages, taking the mean of our sample means. That's what that first symbol says. That's equal to our population. That's always going to happen. That's what the central limit theorem says for us.
- 6.94** Now we get to use the z-score again. You thought we were done. We're never done. We're never done with it. It's a good thing, too, because it makes it nice and easy. Our programs are pretty great.
- 6.95** I will say one thing, though. You're going to start to notice something in this z-score that now—that we talked about sampling—should concern you a little bit. It should concern you. You see, if we're going to calculate a z-score, we are now basing this on a population standard deviation that we rarely know. We're going to have to have something to deal with that by the next lesson. We have to have something to fix this problem. For right now, we're just working on sampling distributions.
- 6.96** The z-score looks similar, but there's some things up there. Instead of dealing with individual values, we're now dealing with sample means because we're talking about samples, and we average this sample; we find the mean of the sample. That statistic is now usable for us. What a z-score, according to our sampling distribution, does, it takes a sample mean. It subtracts our population mean. You're going to have a distance there, aren't you? They weren't all the same. However, that distance isn't very large most of the time, especially as the sample size goes up.
- 6.97** Then it's going to try to fit our standard error in that, the spread of our data, and say, how far away from your population mean is this? Now, get this. I know I'm jumping a little bit ahead here. It's important for you to build these bridges now so that when I get there, it makes sense. Here's what's going to happen. Your sample means should be around your population mean. It should be clustering there.
- 6.98** What if that distance is really big? What if your sample mean is very far away from your population mean? We divide by our standard error and say, "Whoa, this breaks the spread of what we would expect." Then something weird is going on. That's exactly what we're going to use a z-score for. It's saying if it's not where we expect it, something is bad about our claim or something is going wrong in our sample or, wow, that could be a random chance, but highly unlikely. Remember, less than 5% we would say would be highly unlikely. This is what this is going to be used for.
- 6.99** Let's do an example just to illustrate this concept, contrasting how we would find a z-score for an individual child—we all know this already—and how we would do it for a sample of 13 children, knowing we stem from a normal distribution.
- 6.100** We're going to click out of this, maybe scroll over just a little bit. Here's what this says. The probability of randomly selecting an individual child that weighs less than 25 pounds is about 30%. That's not unlikely. That's like, okay, that could totally happen by random chance. Now, think about what's going to happen next. I'm going to ask you a different question. What's the probability of randomly selecting 13 children, averaging their weight, and their weight is less than—average weight, their average weight is less than 25 pounds? Well, see, that average weight should be very near the population average weight, the population mean. If it's significantly different, something weird is going on.
- 6.101** It's very unlikely to get groups of people who average different than the population. Are you seeing the difference? Individuals, yes, they can range all over the place. But averages, when you aggregate the data, the law of large numbers says this sample mean should be close to the population mean. If it's not, that's pretty rare. Groups average to population, and as the groups increase, its average is closer.

- 6.102** So what would happen if we did this? This is what the second z-score we just learned about gives you. For that one child, that's about 30%. Now, let's talk about the sample of 13 we were just referencing. If we plug in now, not just 25 as an individual weight, but that 25 there, because that's based on a sample mean, this says, let's see what the probability of a sample of 13 children is. Look at the sample size up there. Under the square root, you have 13. That's your sample of 13 children. The 25 is their average weight. Let's find the probability that you would have an average weight of 13 individual students where you take their average mean and it's less than 25 pounds. That has a different z-score.
- 6.103** So this says the probability of randomly selecting 13 children, finding their sample mean, and it being less than 25 pounds is only 3%, remarkably less, because samples cluster around populations. And that 3% shows that to us, that your samples cluster around—sample means cluster around population mean. It's really unlikely to find a group of 13 children who's going to have an average weight of less than 25 pounds. Only 3%; that's less than 5%. That's fairly rare.
- 6.104** Why it matters? This is going to enable us to perform something called a confidence interval. Here's where we're going in the very next lesson. We're going to say something. We're going to say, okay, if we know we can take a sample of an appropriate size, and it's going to target the population mean. Appropriate sample, find the sample mean. We can do it with proportion; we can do it with variance, we'll do that later.
- 6.105** But if we focus now on the sample mean, we take that sample of an appropriate size. That sample mean is going to target the population mean. Now, you all know it's not going to be perfect. It's going to be above or it's going to be below. However, we have a very predictable spread, a standard error. We know what it is. We know what it should be around there to capture most of our data. We know that.
- 6.106** What we're going to do next is we're going to take that sample mean that we know targets our population mean. We're going to say, yes, I know it's not perfect. If it was perfect, it would be the entire population that I sampled. That would be called a census. I know it's not perfect. But I also know that a sampling distribution says, I can build an interval around that and capture a whole bunch of data. I am real, real, real sure. How sure? 90%, 95%, or 99% sure that this interval is going to capture the true population mean. That is what a confidence interval is.
- 6.107** That's what we're going to build next time, stemming from the central limit theorem that says we can build these based on the sampling distribution of the sample mean. It's also going to be the same exact logic for hypothesis testing. If we know we're supposed to be here and yet we're over here, something weird is going on. That's as simple as I can put it.

7. Confidence Intervals

At a glance

Confidence intervals use sample statistics to estimate population parameters. This lecture develops point estimates, margins of error, confidence levels, critical values, interpretation, assumptions, sample size, and the distinction between repeated-sampling reliability and claims about a fixed parameter.

- 7.1 Now, this is exciting. We get to start, finally, estimating **population parameters**. You have learned so much about populations, the continuous random variables that we are trying to estimate now. We've already talked about discrete random variables, and now we're going to use our **sample statistics** to estimate our **population parameters**.
- 7.2 Today, we're going to focus on mean. You see, we had these three estimators: we had sample mean for population mean, we had sample proportion for population proportion, and we had sample variance for population variance. Those last two are for our next lesson. Today, we're going to start building out the concept of a **confidence interval**.
- 7.3 Now, why do we need that? Well, sample statistics are used to estimate population parameters. That's true, but we have to know how good of an estimate that is. For instance, how could we possibly estimate the population mean? Are you going to know the population mean? No, not normally. Only once in a while, when we take a census, would you know this?
- 7.4 Maybe the thought crosses your mind, "Well, let's use the sample mean." We can totally do that. For instance, if we have a sample of maybe 20 2-year-olds, just a very small sample, and we say, "What's the best way that we could estimate our population mean just from the sample?"
- 7.5 Well, we would use our sample mean from that. We could take our sample data, we could go on our statistical program, whatever that might be. In our case, we're using JASP, and we take a look at 20 randomly selected weights of 2-year-olds. Someone is very ambitious about these weights. I think this is to the hundred-millionth of a pound, which would be incredibly weird to do, but that's what we did because it's me.
- 7.6 We're going to find the mean from this sample. Now, this is really easy to do in JASP. We click on Descriptives, we click on Descriptive Statistics, and it will immediately, when we click over, give us our mean, our arithmetic average. It'll also give us our standard deviation. Our mean for this sample is 25.87 pounds. Now, please remember, this is sampled from a population that has a mean of 26.3 pounds. Is this sample exactly the same as the population mean? No, that's okay. That's okay, but this is the point. That sample mean is a **point estimate**. It's the best estimator, single-value estimator that we would have for the population mean, but it is going to be different, almost certainly different.
- 7.7 How different? It depends on how big the sample is. The larger the sample, probably the less different that it's going to be. But we all understand that there is going to be a difference there. What we do is we say, okay, now that we understand that the sample mean is going to estimate the population mean, we're going to build some sort of an interval that says we are this confident that the true value, the thing we're trying to estimate, falls in here. That's the thrust of where we're going today. For now, our sample mean is 25.87. We might remember our sample standard deviation of 2.4 for later on in just a little bit. But if we had to use one single number, just one number, to estimate the population mean, then it would be the sample mean.
- 7.8 This is called a point estimate. A point estimate is a single sample statistic that we use to estimate a population parameter. We have a few of them. We have $\bar{x}$. We know that that's the sample mean. The sample mean will point to or estimate the population mean. And sample proportion, this $\hat{p}$, it's going to point to, it's going to estimate the population proportion.
- 7.9 If you're thinking back, what in the world does a proportion mean? We could say, "Hey, the proportion of males in this particular population is 47.3." A sample of that group of whatever random sample would be—maybe we had 48% were male from that sample. That would be a point estimate for our true population parameter. A proportion is just like a fraction or a decimal between zero and one, very much like a probability.

- 7.10** The last point estimate that we typically use is variance. Sample variance points to or it estimates our population variance. These are unbiased, so they do point to. They don't systematically overestimate; they don't systematically underestimate. They're the best single values that we could use.
- 7.11** But now we've built this whole thing for the past six lessons that while we can use sample statistics to estimate population parameters, we also know that there is error—that these values, just like we saw with the 25.87 pounds, that sample mean, 25.87 pounds, is not the same as the population mean. When we did our sampling distributions of the sample mean, we saw, in fact, that for most of those 10,000 samples, those sample means were close, but they weren't exact.
- 7.12** We're looking for something that tells us how close we actually are, how confident we can actually be that we have captured the true value of the population parameter that we're trying to estimate. This is exactly where we're going with confidence intervals. You see, sample estimates, they vary. They vary from sample to sample to sample. We saw that in our sampling distributions. That's called sampling variability. Different samples produce different point estimates. We saw that. Our sample mean is going to be different from the true population mean. We literally just saw that in JASP moments ago that that sample had a different mean than the population. That's okay.
- 7.13** But there is something that we have to overcome. See, there's an error there. If we looked at that error, 26.3 is the true population mean. We know it in this case, and 25.87 was the sample mean. There is a difference, an error in that. It's not because we messed up. It's just by the random selection that we have, by random chance.
- 7.14** Remember, a continuous random variable is a variable whose value is assigned by a chance process. In this case, the chance process is selecting random people. That is a random process, and you're going to get a variability there. There's going to be an error between our sample statistics and the true values of our population parameters. For sure, but that's okay.
- 7.15** You see, this is where a confidence interval comes in. What a confidence interval does is asks, if you can't really estimate the exact value of a population parameter with a sample statistic, if that's impossible, if we know that there's an error, we want to create some interval that we have a certain level of confidence that captures that true value of the population parameter.
- 7.16** The question is, how confident do you want to be? We have several common **confidence levels**. It would be nice to say, "Well, we want to be 100% confident," wouldn't we? Everybody wants to be 100% confident. Can you ever be 100% confident that we have captured the true population parameter? Sure, with a census. If your sample size is a census, you can be sure of this.
- 7.17** But see, what we're going to do is say, okay, if our sample statistic estimates this pretty well, and we know that the spread of this sampling distribution is knowable, we understand how it's distributed, we can use that backwards and create this spread, this interval of data, this interval that's based on our sampling statistics. It says, okay, if we use the error that we already know based on the sample statistic that we just found, and we know that this should really target the population parameter, let's build an interval based on a level of confidence that captures that, whether it's 90% or 95% or 99%. One hundred percent? Sure. Sure, but you've got to do a census.
- 7.18** The question becomes, how confident do you want to be? We typically answer how confident we want to be with one of three levels. Common confidence levels are 90%, 95%, and 99%. There's also something up there. You see this funny little symbol, don't you? There's another Greek symbol. Darn it, we got those all the time. What does that mean? That is a Greek lowercase letter alpha [α]. It's just an alpha. But what's important about it is this. I'm going to go way far in advance to try to build out where we're going with this stuff. Alpha is called a significance level. It's called a significance level, a 10% or a 0.10, 5% or 0.05, or a 1% or 0.01 significance level.
- 7.19** This is what's going to happen. The 90% tells you you are 90% confident that the interval you're about to create is going to capture the true population parameter. In our case today, it's the mean. Well, if you're only 90% confident that you're going to capture it, you're 10% confident that you're not. Where is that 10%?
- 7.20** Well, if you are building an interval, get this: here's your sample mean. It's right typically around the population mean. Now, you know it's not exact. And so what we're going to do is say, let's go out a little bit. It'll probably capture the population mean. How sure are we? Ninety percent. If we're 90% sure, we're 10% unsure. The 10% is in the two tails of that normal distribution that you just saw in the last

lesson. That 10%, 5% in each tail, is called your significance level. It is your alpha. It is the area in your tails. It is also the probability of what is called a Type I error in hypothesis testing. I'm going to explain that now, but I'll go into much more detail when we get there.

- 7.21** What a Type I error says is this. It says this. This is the probability that you are going to reject a true hypothesis. Well, whoa, how can that possibly happen? We all know that we can get data that's way apart from normal, but the odds of that happening are very small. How small? That small: 10%, 5%, or 1%.
- 7.22** Now, if we get some data out there, could it have occurred randomly? It could have, but the likelihood is very small. How small? Less than 10% or less than 5% or less than 1%. We say, if we meet this threshold past this amount, which we consider normal, then outside of here would be rare, that rare, 10%, 5%, or 1%, and that makes our judgment that, oh, what we based this whole problem on might be untrue.
- 7.23** This is the same idea that's going to hold for the rest of our class, the significance level. Again, we are estimating population parameters. We are typically creating an interval called a confidence interval that we are 90% sure or 95% sure or 99% sure captures the true value of the population parameter. You probably should have a question, why don't we just do 99% all the time? Why don't we do that all the time?
- 7.24** The more confident that you want to be, the broader your interval to the point where if you want to be 100%, it's infinity. It's the whole population. We have to limit it somewhere. Typically, the most common is 95%. That's why our rareness, when we talked about it, is usually 5%. They're the same thing. It's two sides of the same coin. Confidence intervals approach this problem by let's create an interval based on confidence, and a hypothesis test says, let's approach this by creating an area of unconfidence that says, if we make it here, well, that's so rare that what we said is wrong. That's where we're getting to.
- 7.25** Alpha is the probability of a Type I error that we just talked about, and we will really digest this when we get into the next part of statistics, Part 2, when we talk about hypothesis tests. Everything—can't say that—all but one thing in the next part of statistics is going to use a hypothesis test, and we're going to see an alpha of 0.01 and 0.05 an awful lot.
- 7.26** This is the Type I error. It's also the area in the tails that we just talked about, the area that says, okay, we know that we're 5% confident we haven't captured that true value, but we know that we're not going to be 100% confident. That would be a census. This is the area in the two tails.
- 7.27** The question in your head is, do you always have two tails? For a confidence interval, yes. The way that we do this, yes. A confidence interval always has two tails, always. Because you have this sample statistic, let's say sample mean, that's right by the population mean, you are creating an interval about that, so less than and greater than that. That means you're not going to capture part over here and part over here.
- 7.28** That alpha is split. That area under the curve is split between those two tails for a confidence interval. If you have a 90% confidence interval, 10% alpha, 5% in one tail, 5% in the other. If you have a confidence level of 99 %, your alpha is 0.01, 0.005 here, 0.005 on the other side. They're always two-tailed for a confidence interval.
- 7.29** Now, we're going to take a look at some of the **critical values** that show up, the critical z values. What in the world is a critical z value? The **critical values** for a confidence level are z-scores, which you're familiar with, that are based upon your confidence level. Here's the whole idea of what we're going to do. We're going to look at a normal distribution in just a minute. We're going to find these z-scores that mark out 90%, no problem, or 95%, or 99%, that's why that's our confidence level. Then backwards, we're going to take those z-scores and make up an interval from that. You're telling it already, "I want to be 95% confident." It's going to give you a z-score. You're going to use that z-score based on your sample mean to give an interval about that confidence. That's all that's going to happen.
- 7.30** It says if you want to be 95% sure that this interval is going to capture your true value, the population parameter, you are going to have to take that z-score of ± 1.96 . That will capture 95% of your normal curve. Therefore, we know with 95% confidence, they will capture the true value of the population parameter.

- 7.31** There's one more, isn't it? There's one more. The last level we have to find is 99%. Now, in your head, you should be thinking, this is going to be a larger number. This is going to capture more of my data. This is going to be further out into the tails. The more confident we want to be, the larger our interval. These z-scores are going to have to go further.
- 7.32** Here's how we use this. What we just learned is that if we have a 90% confidence level, we have basically a 10% significance level—this level that we know maybe we don't capture that true value population mean in this case—it gave us a z-score of ± 1.645 . That's not going to change. Every time you use a z-score, that is going to be the exact same for a significance level of a two-tail test or a confidence level of a two-tail test. You're going to have the same exact z values.
- 7.33** What it relates to—and you'll see this off to the right—this z of alpha over 2, all that's saying is on the right tail, that alpha should be half, half. So your significance level, this 10% should be half in your right tail and half in your left tail, 0.05 in the right and 0.05 in the left. That right there says, just for the positive—because we'll subtract it later anyway—we know that the z value that represents 5% on the right tail is 1.645. That's what that says.
- 7.34** We do the same thing for a 95%. We have already seen these numbers. We saw ± 1.96 . That says that the z value representing 0.025. Remember, $5\% \div 2$ would give you half here, half here. That's 0.025 in your right tail. That z value of 1.96 is what would give you that demarcation on that distribution.
- 7.35** Lastly, we would have just one more. For a 99% confidence interval, the 99% confidence level, and a significance level of 1%—I'm using these words a lot to get you used to them—we would have a probability of 0.005 in the right tail and 0.005 in the left tail. That z value that separates that is 2.575.
- 7.36** Now we get to use it. How are we going to use this? Oh, we're going to work backwards. We're going to take the fact that this z critical value is attributed to our confidence level. We know that they're always two-tailed. Here's what we're going to do. We're going to take our statistic. We're going to take our z critical value based on a confidence level, somehow work in the fact that we know how this distribution works, how it's spread out, the **standard error**, that that is a standard deviation over the square root of our sample size. Yes, this is based on sample size.
- 7.37** Oh, this should make sense. As our sample size goes up, our interval should shrink. Why? Well, as your sample size goes up, that statistic is going to be closer to your parameter. We don't need as big of a spread. It's going to shrink when our sample size goes up. Now we're going to build it going backwards. Let's go from our sampling distribution idea of the sample mean, and let's start building our confidence interval.
- 7.38** Here's the thought process. If you want to understand really what confidence intervals do, here's the thought process. Sample means cluster around the population mean. Yes, we know that. We also know that they're going to be slightly different from the population mean. They're not going to be exact. But if we take a certain size of them, over 30, then that sample mean is going to target the population mean.
- 7.39** But more than that, from our sampling distributions, we also know that that distribution of data is going to be very predictable. In fact, it's spread out—if our sample size is large enough, or we know the population is normally distributed—it's spread out by the standard error. It's the standard deviation divided by the square root of n. As n goes up, this goes down. As this goes down, things shrink up.
- 7.40** Let's use that. Let's use the spread. Let's talk about how confident we want to be, and let's relate that to our sample statistic to build our interval. Here's our conditions. If you have a random sample, if you know the population is normally distributed or the sample size is greater than 30, why do we have to know that? Here's why.
- 7.41** If you don't know that, you can't use the central limit theorem. This whole thing is based on the central limit theorem. The central limit theorem is what says that the mean of your sample means is the population mean. The central limit theorem is what says the standard error is the population standard deviation over the square root of n. If you don't have that, none of this works. You have to meet the requirements.
- 7.42** Remember how I told you at the very beginning of this class: statistics will let you do methods even if you fail the constraints and you can choose to do them or not. If you don't know this, this stuff don't work. If you use it instead, you're violating all these principles, and that does not make a lot of sense, and it's not integritous, which means you can lie with it. We don't want to do that.

- 7.43** We're always, always about, let's collect our data well, let's check our conditions. If they're met, then we move forward. If they're not met, are there ways to fix them? If there's not, we need a different method. What we don't do is try to fit methods to things that don't work. What we have to have is a random sample. What we have to have is a population that's normally distributed or sample size is more than 30 because then we get to use the central limit theorem. All of this stuff actually works that we talked about in the last lesson.
- 7.44** Finally, your population standard deviation has to be known. See, that's part of this whole z-score formula, wasn't it? You had to know that. Now, some of you right now in the back of your head are probably thinking, this is a problem. We'll talk about this when we get to the formula. But in the back of your mind, know that that right there should be triggering something. It will. If it doesn't, I'll make sure that it does. I will trigger you in a really good way. This will be a problem for us in just a bit.
- 7.45** Here's our formula. We saw it last time when we dealt with sampling distributions of sample statistics, specifically the mean, that we're going to take our sample mean, we're going to subtract our population mean. Theoretically, those should be pretty close together. If they're not, that's going to tell us something's wrong. We're going to divide by our standard error.
- 7.46** Look what happens when our sample size goes up, standard error goes down, the z-score explodes, saying, "This is very rare; this doesn't make sense. This is out of normal." This is where this is coming from. This is how we do this.
- 7.47** So, how can we now use this to build a confidence interval? Check this out. This is going to be cool. What is the one thing you're actually trying to estimate here? Well, you're not trying to estimate a z-score. You're not trying to estimate that. You're not trying to estimate your sample mean. The whole premise of this is that your population standard deviation is known, so you're not trying to estimate that. You're not trying to estimate your sample size because you're going to be the one collecting that.
- 7.48** This whole thing is trying to build a confidence interval for your mean. You're trying to find your mean. So, let's solve for the mean algebraically. If we were to solve for the mean—I promise you, once we do this, if you get this, you will perfectly understand a confidence interval. Because what we're trying to do is build something that captures our mean. Right now, we haven't solved for our mean. It's up there in the middle of this formula.
- 7.49** So let's do that. We're going to multiply both sides of this equation by the standard error, basically just the denominator. If we multiply both sides by the denominator, it's going to cancel on the denominator side, and it's going to give us z times that standard error. All we've done is multiply both sides by our denominator. Algebraically, we're good to go.
- 7.50** There's only really one more step that we got. We have to solve for that μ [μ], for that population mean. We're going to add the mean on both sides. We're going to subtract our z times standard error on both sides. What we're going to get is something that's solved for mean. Here's what this says. It says if you really wanted to find out your mean, here's what you would have to have.
- 7.51** If you wanted to know exactly what your population mean is, you would have to have your sample mean. You would have to know your z value of exactly where it is, which you don't know. You'd have to know your population standard deviation, which is a requirement to do this. You have to know that, and you'd have to have a sample size.
- 7.52** The only one of these you don't know is the z value. Where in the world are you going to get the z value? This is where the confidence level comes in. You pick that. You pick that. See, the sample mean you don't pick. That's from your sample. The sample size, you pick, but you need more than 31 to use this, or a normally distributed population. You have that. The population standard deviation is known.
- 7.53** You pick the z. Say, "How confident do you want to be?" "Well, Professor Leonard, I would like to be 90% confident." Fantastic. Then you're going to plug in 1.645. "I want to be 95% confident." Great. You're going to plug in 1.96. What's going to happen? You're going to add and subtract a slightly larger number. "I want to be even more confident. I want to be 99% confident." Perfect. Plug in 2.575. That's going to be even greater. That's going to be an even bigger interval.
- 7.54** Can you be 100%? Yes. But your z has to be infinity. Then you have something very, very large. It doesn't make sense mathematically. That's why we stop. We are using a z-score backwards. You are

telling this confidence interval how confident you want to be, taking a z-score that captures that amount of data and then building an interval from that.

- 7.55** If we want to be a level of confidence of whatever it is, 90%, 95%, or 99%, we're getting a z value from that, adding and subtracting this amount. Remember, we have to build the interval. You're going to add it and you're going to subtract it. That's where that $z_{\alpha/2}$ is because that's always a positive value.
- 7.56** You're going to add and subtract the 1.645 times standard error, or you're going to add and subtract your 1.96 times your standard error, or you're going to add and subtract your 2.575 times your standard error. You're going to do that around your sample mean. The sample mean targets the population parameter, your population mean. We say, "I want to be so confident," and it builds our interval. It builds our interval based on the spread of data that you know how it's going to be spread. The standard error is right there. Then you have this discernible interval that you say you are 90% confident or 95% confident or 99% confident captures the true value of the population mean. That's exactly what a confidence interval does, and that's all that it does. It's not magic. It's just really smart.
- 7.57** We call this last piece the thing that's added and subtracted. You'll often hear this. You'll especially hear it in our next lesson when we talk about polling for population proportion. What percentage of people are voting for candidate A? Forty-three percent, plus or minus, whatever. You've probably seen that. You have to see that in a poll. Otherwise, you start questioning that poll. That is called your **margin of error**. It will tell you how much around the mean you have to be at a certain confidence level to capture the true population parameter, in this case, the population mean. It mashes that all up and calls that E, a **margin of error**.
- 7.58** What are you going to do? Well, if you wanted to capture your mean, we would say we are 95%, 99%, 90% confident that the following interval will capture the population mean. We take our $\bar{x}$, our sample mean, we subtract our margin of error. Folks, that's exactly what we just did. All that E means is the z-score that you get from your confidence level times the standard error that you know how to do that. You subtract it from your sample mean, you add it to your sample mean, and it creates an interval. You're 95%, 99%, or 90% sure captures the true value of your population mean.
- 7.59** I think we're about ready for an example. You understand the theory behind it. Hopefully, right now, I've made a really good case for how and why we're able to use sampling distributions to build confidence intervals based on a z-score and a level of confidence, disregarding some area in the tails that we are fairly sure we're not going to be in any way. Pretty sure. Now, how do we do that?
- 7.60** Let's take a look at a 90% confidence interval for our weights of 2-year-olds. The only reason why we're using this so much is because we probably have remembered that the population mean is 26.3 pounds and the standard deviation is 2.5. Now, why I'm saying we have to remember that is because your z values that we use here and your margin of error has to know the population standard deviation. Right now, this should be concerning for you, and we'll talk about why in just a moment.
- 7.61** Let's suppose we collect a random sample of 20 children, 2-year-olds, from the population of 2-year-olds. Now, we know that this distribution is normal. Is that important? Oh, yes, because if we don't know that, we can't use this. See, you have to know that either the population you're sampling from is a normal distribution, or you have to have a sample size of more than 30. We don't. We only have 20. So we have to know one or the other.
- 7.62** Right now, I'm showing you that it's okay as long as we know the population is normal. Later on, I will show you that if our sample size is more than 30, we're okay. If we don't know either of these, we have to check for normality, and I will show you how to do that. It's not hard. I just have to show it to you. Right now, we're okay. We have a population that's normally distributed. We've sampled 20 weights from—yes, it's less than 30—but we know that we meet our criteria, and we also know the population's standard deviation.
- 7.63** You know what? I can't wait anymore. I can't trigger you anymore. Here it is. Are you going to know the population's standard deviation? If you did, you'd already know the mean. Because you have to know the mean in order to even find the standard deviation. It's in the formula. We took our sample data minus our mean. We added it up, we divided by n.
- 7.64** If you had to know your mean—get this; this is important. If you had to know your mean to find your standard deviation, why would we be doing any of this? You don't have to estimate a mean that you already know. You would never build a confidence interval around something you already know. You

don't have to be 99% sure you capture the true value of the mean if you already know the mean. If you have the population standard deviation, you should already have the mean.

- 7.65** This is going to mean something to us that there's something different we're going to have to do most of the time because most of the time, this is not satisfied. In just a bit, we're going to talk about it. This is just to get our feet wet to understand the idea, but we're going to modify this in just a moment. It's not going to be any harder, but it is going to be a little bit different.
- 7.66** Construct a 90% confidence interval for the population mean. Here's what we found from our descriptive statistics. We knew that the sample size was 20. I asked you to remember the sample mean was 25.87. I asked you to remember the sample standard deviation of 2.400. Notice this is easy to find. You can always find these things. You can always find your sample size, your sample mean, and your sample standard deviation. The whole thing that should be going on in your head is, okay, if I don't know the population standard deviation, is there a way I can use the sample standard deviation instead? Yes, yes, yes, we're going to do that. We know the population standard deviation is 2.5.
- 7.67** Let's go ahead and let's construct our 90% confidence interval. Here's how this would look. First, we'd find our critical value. Ninety percent, maybe already in your heads, is 1.645. For a normal distribution, that's true. If it was 95%, maybe in your heads, it would be 1.96. Maybe even in your heads right now, 99% would be 2.575. We're at 90%, so we're going to pick our z critical value of 1.645.
- 7.68** Now we're going to do something with it. We're going to take that z critical value that says, this is how confident I want to be. We're going to multiply by the spread of our data. Well, this is our standard error. This is our standard deviation over the square root of n. That's our margin of error. All we're doing here is taking our critical value from our confidence level. We're multiplying it by our standard error. This is the population standard deviation over the square root of n.
- 7.69** All it's saying is this: wherever you are in your sample, I want to be 90% sure. I'm going to multiply by my z-score, but I also have to take into account the spread of my distribution. That's your standard error. That's why we multiply those two things. We get a margin of error of 0.920 pounds. This is in the same units as our data, which is very convenient because now all we've got to do is say this. If you have a sample mean and you add 0.92 pounds to it and you subtract 0.92 pounds from it, it is going to create an interval. That interval is going to be a 90% confidence interval.
- 7.70** That interval is going to say, "Hey, I'm 90% sure that between this weight and this weight, this interval captures my population mean." I'm 90% sure that happens. That's exactly what we're going to do right here. We take our sample mean, we subtract that margin of error, we take our sample mean, we add that margin of error, and this, we are 90% sure captures the true value of our population mean. That's why it's right in the middle of that. This is exactly what a confidence interval does. Is it for sure going to capture that value? No. There's a 10% chance it's not, 5% on each side. We just talked about how that 2.5 up there, you're not going to know that because that's the population standard deviation.
- 7.71** I already prefaced this by saying, "I'm going to trigger you," and you're not going to—you can't do that because if you knew the population standard deviation, you already know this mean, and you would never want to estimate something you already know. It's a waste of time. Then I asked this question, is there going to be a way to use the sample standard deviation in lieu of the population standard deviation? The answer is yes.
- 7.72** See, there was a man named William Gosset, and William Gosset worked for Guinness. Yes, that Guinness. He came up with the same problem. He said, "Wow, I've drank way too much beer. These distributions just aren't making sense at all." Well, it wasn't his fault, and it even wasn't the beer for this one instance. No, it was that for small sample sizes, they don't fit the normal distribution. It doesn't fit. In fact, he found out that distributions are based on a sample size, especially for small samples.
- 7.73** This guy was such a genius. He built his own. He said, "Population standard deviation, it's rarely known. I don't know what it is. I have these small samples. They don't fit the normal curve. Nothing's coming out right." William Gosset, what he did is he built his own distribution. Now, we call it the Student's t-distribution. There's a common misconception that's because students use it. No, it's because William Gosset wrote under the pseudonym Student. He did all of his research under the pseudonym Student. And so it's called the Student's t distribution.

- 7.74** So what does this do? Essentially, it takes the place of a z distribution when you don't fit the criteria of a z distribution. The z requires something that we rarely know. The z requires us to know the population standard deviation. You all know that we rarely know this. He knew this also. He developed his own.
- 7.75** What a t distribution does is it gives you the same type of thing, but with a sample standard deviation instead of population. That's all that it does. What does a t distribution look like? It's quite similar, but it is based on sample size. Gosset—Student, the pseudonym—developed the t distribution. He discovered that it's dependent on sample size.
- 7.76** He talked about something called the degrees of freedom. I really honestly thought this was a super patriotic country song. It turns out it's not. It's not a super patriotic country song. It is this concept. The degrees of freedom is how much choice you have in a certain mathematical construct. That sounds like a lot of words for something very simple.
- 7.77** Imagine you had five numbers that added up to 100. Five numbers added up to 100. You could literally pick any four numbers that you wanted to for the first four. Anything. Anything you wanted. Twenty, 10, 50, 1 million, actually. You could pick anything you wanted. But if you had to end up at 100, your first four numbers determine your last. This has to be something to end up where you want.
- 7.78** We use this when the population standard deviation is unknown, which is pretty much all the time. In our course, we're pretty much only going to do this. The only time we're going to break that exception is for one reason. When we get to estimating proportions, we're going to be back in binomial land, and we're going to learn that binomial probability distributions are approximated by normal, not t, normal. And so we'll be using z for that. But other than that, we'll be using a t distribution.
- 7.79** Here's our formula for a confidence interval when your population standard deviation is not known. It looks really similar. It's calculated really, really, really similarly, but it uses a t distribution instead of a z. Let's take a moment right now and let's look at a t distribution. Let's take a look at what a t distribution looks like. We're going to go over to Learn Stats. We're going to go down to p values. We'll talk about what a p value is at length. But I want you to get a feel for what a t distribution looks like, and we're going to start at a sample size of just [two].
- 7.80** Here's what this is showing you. Because William Gosset discovered this t distribution, coined the phrase, we have a t distribution, he said, yes, this is built on the size of the sample. He said, we need to know degrees of freedom. So this would be a sample size of just two. Degrees of freedom is one. That's what we have over here on the far left. We have a degrees of freedom of one. This would be the shape, the shape of this distribution. You see that looks fairly normal. It's very peaky, and the tails are really big.
- 7.81** I want to show you one more thing here. We're going to highlight a test statistic, and I want to show you a p value of 0.10. Remember what this is? This would be the area in your tails for a 90% confidence interval. Let's take a look at this. This would be the area in the tails of a 90% confidence interval. Remember, 10% would be split amongst two tails, symmetrically, 5% on one side, 5% on the other side.
- 7.82** Now, you had a number in your head. You had a z value of 1.645 for a 90% confidence interval. Look at the t value. It's 6.3. That's way more. That's going to explode this interval for you. It's going to explode it. Why? Your sample size is so small. It doesn't fit the criteria that normal distribution did. You have to broaden that to gain that much confidence.
- 7.83** Well, let's see what happens as our sample size increases. Well, as the sample size increases, our degrees of freedom is naturally going to increase. Let's take a look at what happens when we have not just a sample of two. Let's say, what if we had a sample of four? We'd have degrees of freedom of three. It's going to get a little bit more narrow. Our test statistic is a little bit closer. We have a 2.353.
- 7.84** What if we said, how about 15? That would be degrees of freedom 15, sample size of 16. We start to see this thing really approach a normal distribution. If you take a sample size large enough, the t distribution approaches a normal distribution. It's never exact unless your sample size is extremely large. Even if we got to 30, which would be a sample size of 31, remember, we would be meeting the normal requirements. But because it's not a normal distribution, even the t-score, compared to a z-score, is going to be slightly larger. Your confidence interval is going to be a little bit bigger because of that.
- 7.85** Now, let's put on our own data. Our own data said, okay, we had a sample size of 20. Degrees of freedom for a sample size of 20 would be 19. Let's look at what the specific t distribution would do if we had a sample size of 20, degrees of freedom of 19. We'll put that in right now. This says, okay, we are

now working with our data that we have sample size of 20, degrees of freedom, 19. This is the look of our specific distribution that's based on sample size for our data of 20 collected weights of 2-year-olds. This is it.

- 7.86** Let's look at the t-score. The t-score is—hopefully you can see it up there—1.729. Remember, for a z-score, it was 1.645. This is a little bit broader. So what can we expect from this confidence interval versus our last one? It's going to be a little bit bigger. We have to make it a little bit broader to be as confident. Why? Because you're not as sure. You don't have as much of the data that you need in order to construct it like you would a z-score.
- 7.87** You don't know the true spread. This is the whole point. If you don't know the true spread, the true population standard deviation, you have to estimate it. Standard deviation of a sample underestimates the population standard deviation. You have to be more broad to account for that. That's exactly why that number is larger. What we're going to do is say, okay, we're going to do the same exact math. We have now a t-score, not a z-score, but the same math works for us.
- 7.88** Let's go back and let's now construct our confidence interval from our sample data. Our sample data said you have a sample size of 20. You have a degrees of freedom of 19. We have a sample mean of 25.87. We found that from descriptive statistics. We had to do no work for that. You have a sample standard deviation of 2.4. This is fantastic because we didn't know, theoretically, the population standard deviation, but we know that one; that's easy.
- 7.89** Now we found a t-score. This is the marker, the value, the critical value that's going to determine the width of our confidence interval. We said we want to be 90% confident. There was 5% in each side. That's the 0.05. But it's a different distribution. It's a little bit broader. Let's use that now and put this into our equation. The margin of error is calculated virtually the same way. We still take our critical value. This time it's t. It's not 1.645, it's now 1.729 because it's a t, not a z.
- 7.90** We have our sample standard deviation. Oh, this is great. This is fantastic. We don't know the population standard deviation, but this one we can always find. This is fantastic. We can always do this. This is why we use it. Then we divide by the square root of our sample size, which is 20. What's going to happen? What's going to happen if your sample size goes up? Your margin of error is going to go way down because this distribution is going to get closer to normal. Because we know that the sample mean is going to approach the population mean. It's going to really narrow.
- 7.91** If you remembered our margin of error for the z, for the z-scores that we just did, it is slightly smaller than this. This has given us just a little bit more spread. Still calculated the same. We're going to take our sample statistic, we're going to subtract our margin of error, we're going to take our sample mean, we're going to add our margin of error, and it still gives us an interval for our population mean.
- 7.92** The same exact interpretation. Same interpretation. It says we are 90% sure that the true value of the population mean is somewhere in this interval. The only issue is that it's a little bit larger than what we got from the z. Is it a big deal? No, it is what it is. We can rarely find a confidence interval knowing the population standard deviation. This one we can find all the time.
- 7.93** So t intervals, well, they're slightly wider than our z intervals. Just a little bit because it's a different distribution based on sample size. But this protects against overconfidence. You see, we want to be very confident in what we're presenting to people. We want to say, "If I'm 90% sure, I am truly 90% sure." Now, could you be more sure? Could you be 95% sure? Or 99% sure? Well, sure you could.
- 7.94** Think in your heads right now, what would happen? if we went into JASP and we said, "I now want to be 95% sure." What's going to happen to your tails? Your tails are going to get smaller. Your t value is going to get bigger. This interval is going to get wider. The more confident that you want to be, the wider your interval is.
- 7.95** Now, this is all fair and good, but let's look at some of them. Now, I know if you're anything like me, that you don't like to do a lot of the fancy pants of math, and you would rather have a computer program do it for you. That's exactly what's about to happen. See, JASP will do this for us.
- 7.96** If we take a look at our data, all 20 of our 2-year-old weights. We go to Analysis and we click on Descriptives and Descriptive Statistics. We just pull over our sample of 2-year-old weights. Oh, it'll give us our mean and it'll give us our standard deviation. That is exactly what we use to build this confidence interval you saw just a moment ago. It's the same thing.

- 7.97** We go, "Well, I want more. Computer program, can you tell me more?" Of course it can. Let's go right down here past all this stuff and say, "Hey, can you show me a confidence interval for the arithmetic mean?" That is exactly what we just found. Say, "Well, I don't want 95%. I'm really interested in a 90% confidence interval." Why don't we use normal? Why can't we do the normal? Well, this is a computer program. It doesn't know that you are working with a sample. It doesn't know that. There's no way that it can actually know the population standard deviation, and neither can you.
- 7.98** We're going to use a t model. We're going to say, "I want you to look at a Student's t distribution. I want you to put in the sample size that's appropriate for your data, [20]. I want you to do all the math for me. Give me a 90% confidence interval." And with respect to rounding, it is precisely exactly the same thing that you just found with a click of a button.
- 7.99** You go, "Well, I want to be more confident." What's going to happen? What's going to happen when we click 95%? Is this interval going to shrink or expand? It should get bigger because if we want to be 95% confident, your t-score has to capture more of your data. Your standard error is not going to change, but the t-score is. Smaller area in the tails, broader t-score. There, it's slightly larger.
- 7.100** What if you wanted to be 99% confident? What's going to happen to this interval? It has to broaden. It has to capture more of your population to be that confident. There we are. That would be a 99% confidence interval for the arithmetic mean. This is saying this, you're 99% sure, based on your sample, that the true value of the population mean, which you know absolutely nothing about, 99% sure it falls within this interval.
- 7.101** So what happens if it doesn't? What happens if you collect a sample and it's outside of that? Wow, that's pretty rare. How rare? Well, 1%. Is that rare enough to say that what you based your assumption on is wrong? Yes, this is the basis for hypothesis testing that we make it to in just a little bit.
- 7.102** The last thing that I would love to show you, and it's fantastic because it will really give you an impression of what confidence intervals are intended to do. I'm going to show you a confidence interval application on this. We're going to go here. We're going to learn about confidence intervals. We're going to start very simply. We're going to put in our data, our 26.3 for our mean. We're going to put in our 2.5 for our standard deviation. We're going to take samples of just size 10, and I want to start building some 90% confidence intervals.
- 7.103** I know this changed over here. I'm going to explain it to you right now. Here's what the computer is doing right now. It's taking a look and it's creating theoretical confidence intervals for you. It's saying, "I'm basing these intervals on your population mean and standard deviation. I'm collecting samples of size 10, and now I'm building a confidence interval from that." In fact, it's a 90% confidence interval. Here's what should happen. And this is the three confidence intervals, by the way. Each of them actually captures—that's this vertical line—each of them actually captures your population mean. The 26.3 is this population mean. All of them nailed it. You were 100% covered here. Every one of them nailed it.
- 7.104** Let's do 100. The computer has now done 100 confidence intervals. It's put them on this tree. That's a lot. That's very fast. The red ones miss. The red ones miss. Understand that with this technique, 10% of these confidence intervals are going to miss the true value of the population mean. That's why it's a 90% confidence interval. Ten percent are going to miss. It says 90% of the confidence intervals that we construct in this manner are going to capture the true value of the population mean. Ten percent are not.
- 7.105** Let's look at what happens. The red ones miss it. This one's too low, this one's too high. A few of them are too high, some of them are too low. Overall, 88 got it. Eighty-eight captured the true value, and [twelve] of them didn't. We were pretty close to 90.
- 7.106** Let's do 95. I would really love it if you would look at the intervals themselves and watch what happens when we change to 95%. They should expand. In expanding, more of them should capture the true value of the population mean. There's less red: 1, 2, 3, 4, 5, 6, 7. This has now captured 93 of these out of the 100, so 93% of them captured the true value of the population mean. Theoretically, it should be 95%. This is a little bit low.
- 7.107** Let's do 99. Now, our confidence intervals are fairly broad. They're really big. That means you have a much better chance of actually capturing this 26.3, the true population mean. Sure enough, we had a 99% confidence interval, and 99% of these confidence intervals, only one fails to capture the true value of the population mean.

- 7.108** This is exactly what confidence intervals show you. They say, "We are building this interval. This is around a sample statistic that we are 90%, 95%, or 99% sure captures the true value." The other way to say this is 90% of these confidence intervals, or 95% of these confidence intervals, or 99% of these confidence intervals are going to capture the true value that we create this way, are going to capture the true value of the population mean. That's all that this really says.
- 7.109** Where do we go from here? Now that we understand the confidence interval concept, we're going to take this and we're going to run with it. We're going to talk about a confidence interval for proportions. We're going to talk about a confidence interval for variance. The reason why is because those are the last two sample statistics that are unbiased estimators for our population parameters. We'll do that next time.

8. Confidence, Proportions, and Variance

At a glance

Confidence-interval methods extend beyond means. The lecture develops intervals for proportions and variance, introduces chi-squared distributions, compares assumptions and standard errors, and connects estimation to later goodness-of-fit, independence, and contingency-table applications.

- 8.1** Well, lesson eight means we are at the last lesson of the first part of statistics—Statistics Part 1. You should be feeling pretty good about lasting this long through statistics because a lot of times it can be very confusing, and we don't truly understand what it is that we're doing. When those two things go together, it's very hard to learn something. So hopefully, we've done a decent job of explaining what statistics is, modeling parameters, and now we're actually able to do that.
- 8.2** We took some time in the last lesson learning how we can estimate the population mean with something called a confidence interval. We're not going to be doing anything different than that for this lesson, but we have to learn some other things to estimate the last two parameters that we need to handle. We're going to estimate population **proportion**. We're going to estimate population **variance**.
- 8.3** Now, because we're not dealing with means like we did last time, we have a couple of different critical values that we're going to look at. One of them is going to be very familiar to you, the z value, and one is going to be a chi-squared $[\chi^2]$ test statistic. We're going to look at a different distribution. It's not a huge deal, but it is going to be different because a normal distribution and a t distribution don't work for variance, so we need something else.
- 8.4** But the whole thrust of this whole lesson is about the same thing as last time. We're creating a confidence interval. That's it. What we're going to be doing is taking our sample statistic—in these cases, sample proportion and sample variance. We're going to say, "Hey, we know that they're unbiased point estimates for our population parameters." Sample proportion points to population proportion. Sample variance points to population variance. But they're not going to be perfect. We know they're not going to be perfect. We know they vary sample by sample.
- 8.5** But we also know in certain conditions, we can build an interval around them to capture the true value of the population proportion in this case. Mean last time, proportion this time. Variance in the second case for this lesson. We'll be 90% or 95% confident or 99% confident that this interval captures that true population parameter. The other way that we say this is 90% of the confidence intervals created this way are going to capture the true value of the population parameter or 95 or 99%. What we're going for today are confidence intervals for the last two population parameters that we need to try to capture.
- 8.6** Now, why do we do this for a proportion? This might be one of the most valuable aspects of this first part for you. Proportions are in every single poll that you see to do with politics. When we talk about what candidate are people voting for, what is an approval rating—all they're going out and asking is one very simple question, are you voting for candidate A or not? Do you approve of candidate B or not? Do you like this policy, yes or no?
- 8.7** Notice what this is. This is already being constructed as a binomial outcome. You're not interested in knowing really all sorts of things when you're asking about candidate A. All you really want to know is, are you voting? Yes or no? Do you like it? Yes or no? Is it success or failure? That's what we're going to do here.
- 8.8** Here's what you're going to see with this first part of proportions. We are going to revisit binomial probability distributions. I told you when we did, we're going to make a case that binomial is going to transition to normal. You're going to see that today. That's why we were building that out. We're going to see that you're going to be using z -scores again, the ones you know and love, 1.645, 1.96, 2.575. Why did we spend so much time doing that? For this as well because we revisit it and we will approximate a binomial distribution with a normal curve, so that binomial stuff—proportions, yes or no, success or failure, candidate A, candidate B— now we can use a z -score for it. That is almost magic. I'm going to show you exactly why we can do this in just a moment.
- 8.9** But these are often things that we see, especially in election years, you're bombarded with all of these polls. Now, please don't let me make you cynical about these things. We can have a tendency to, after a statistics class, say, "I'm never trusting a single thing. They're lying to me." Not typically. Usually, a

pollster will collect a sample with integrity, and they will randomly select individuals. Because it's random, because they follow all of these procedures, they do a decent job of putting forth what the general population feels about this if they do it with integrity.

- 8.10** Sometimes in the private sector, sometimes when we see on the internet, these people who do research, but they won't post it, this is a little bit suspect. I just discovered 27% of the people think this. Well, 80% of statistics are made up just like that one. These are things that we become suspect of. Don't let it make you cynical when you see something on a verified source that has documented statistics or a psychological publication in some psych magazine or journal. Those are fairly well researched.
- 8.11** We see this often in election polls. Polls estimate what the population thinks in general as a whole. But obviously, we don't ask everybody, so we take a sample. These results are usually reported as a confidence interval. Now, you see them different. On the news, it's going to say something like 37% of the people think this. But then off to the right, it'll have a $\pm 2\%$, $\pm 1.5\%$, plus or minus—and it gives you a margin of error.
- 8.12** That's exactly what that is. Exactly what we did last time for mean is what we're going to do this time for proportions. Only it's going to be very relevant because you see this on a daily basis. If you look anywhere in the news and they do a survey or a poll. For us, it's going to be candidate A has 43% support, $\pm 9.7\%$. That's a margin of error. We're going to see exactly where that comes from today.
- 8.13** Before we go any further, we need to talk about proportion as a binomial probability. We prefaced that a little bit earlier in this lesson. But a proportion simply represents a part of a whole. It's a fraction. It's a percentage. It's represented as a decimal between 0 and 1. No, they can't be negative, and no, they can't be more than 1 because they are probabilities.
- 8.14** We talked a whole bunch about discrete random variables, and this is really what we're going to be modeling here, that if we are thinking about a poll as binomial outcomes, really what we're asking is the question, is it a success or a failure? Is it a yes or a no? Is it candidate A or not candidate A? Notice what we're not doing here is we're not asking, is it candidate A or candidate B or candidate C or candidate D? We're asking about one person right now. We're saying, "Are you voting for this candidate? Do you support this candidate or not?" That's only two outcomes. And so we fit this into a binomial probability distribution.
- 8.15** We say, "Okay, that's fantastic, but I can't really construct a confidence interval very well with what I know in that. Let's make a normal distribution fit over that. Let's see if that actually works." That is the whole idea for how we represent this binomial idea into a confidence interval using a normal approximation. The normal distribution approximates this. That is an outlandish claim. They go, "Wait, what? You're saying that this continuous distribution, a continuous distribution of a continuous random variable, is going to somehow fit over almost perfectly into a distribution of a discrete random variable. You're saying this one fits this one?" Yes. In certain circumstances, yes. I'm going to show you that right now.
- 8.16** From JASP, we're going to go over and look at the binomial probability distribution. To make this super clear, I'm going to adjust just one thing about it. We're going to talk about our probability of success being, [0.1] and we are going to look through what happens as we adjust our number of trials and see what happens to our distribution.
- 8.17** When we start out, here is what our probability distribution looks like. If you had only a 10%—that's what the [0.1] means—if you had only a 10% probability of success, thinking about what binomial means, they're based on Bernoulli trials, there's only two outcomes: success, failure; candidate A, not candidate A. That's what we're looking at. You see, the probability of success has to be fixed for every single one of those. In this case, it's just 10%— 10% chance of success, 90% chance of failure for each of these independent results. Are people independent in what they think? If they're randomly selected, yes. What you think should not influence what you think if you are randomly selected from the population. So they are independent.
- 8.18** Now, we're going to take a look at what happens as we start increasing our sample size. See, right now, there's only a sample size of 5. There's only 5 trials. This is very, very low. This, for you, should look nowhere near a normal distribution at all. This is strongly tail-right. Almost everything is around 0. The largest probability is 0. Why? Well, the probability of success is so small. It's only 10%.
- 8.19** Now, let's start adjusting this. Let's imagine what would happen if we have 10 trials. It still doesn't look normal, but look at where most of our data is. It's right around 1. Why is it 1? Why is it 1? Well, the

expected value for this was the number of trials times the probability of success. It was 10 in this case, times 0.1— $10 \times 0.1 = 1$. It should be centered around 1. That's what we would expect. Oh, that's our mean. That's exactly what we're talking about. This is our mean of this distribution— n times p , that was binomial expected value, that's also our mean.

- 8.20** Check this out. You're going to get the math chills here. I just know it. Let's do 100. Let's adjust this. Just a little bit more. That looks a little bit more normal to us, doesn't it? A little bit more normal. It's almost symmetric. In fact, most of our data here is centered around 10. Why is it centered around 10? Well, while our probability of success didn't change, we now increased our sample size, our number of trials, to 100— 100×0.1 , or divided by 10, this 10%, this is exactly 10. Our mean is $10—n \times p = 10$. It should be centered around 10. That's our mean.
- 8.21** Now, let's go even bigger. Let's go to 1,000. Actually, before I do it, because we're going to ruin the suspense if I don't, we're going to adjust our range from 100 to 150, and we'll take a look at this. Now, that looks really, really, really, really, really, really close to a normal distribution. Do you understand the magic of this? Because that's not a normal distribution. That's a binomial distribution built on discrete data.
- 8.22** But when our sample size, when our n , gets really, really, really large for probabilities of success that are not too close to 0—if you have more than 5 successes and more than 5 failures in your trials, which we do—it is going to approximate normal, strongly, strongly approximate normal. This is what allows us to build our confidence intervals based on a normal distribution. Binomial distributions are approximated by normal distributions. In fact, they're so predictable that we even have a formula for the mean. We have a formula for the standard deviation. It's no surprise because we literally did this just a few lessons ago. We know what the mean is, and we know what the standard deviation is.
- 8.23** For n large enough, for at least 5 successes, for at least 5 failures, exactly what you saw, 1,000 is pretty big. It's not nearly 10,000, but it's pretty big enough to show this. We find this. From a binomial distribution, we knew that for x successes, for n , your sample size, that's what we saw, for p , your probability of success, in our example we just did, it was 0.1, the probability of failure would be 0.9. We know this. The expected value was the sample size times the probability of success. This is what you saw. This is why that distribution was centered around 10% of your trials because the p was 0.1. It was just 10%.
- 8.24** But more than that, the standard deviation, the spread of this, what you're expecting across this distribution is knowable. It's the square root of n times p times q . So get this. If we were to take the n , which was 1,000 in our last example, we put in 1,000, it created that really nice normal-looking curve. If we put in 1,000 and we multiply it by 0.1, that's a probability of success, and we multiply it by 0.9, that's the rest of it, that's the q , the probability of failure, and took a square root, I've done this; it's about 9.
- 8.25** Here's what this means. If I took the mean, added 9, and subtracted 9, it should come up with something that looks familiar to us. Let's take a look at this. Here's our binomial distribution. In the very center of it, you can see it right here is 100. This is your mean. All this is, is $1,000 \times 0.1$. It's the very center of your data. It's the expected value. It is the mean. It's why this distribution is centered around 100. However, we know the spread. We know the standard deviation is the square root of n , that's 1,000, times p , that's 0.1, times q , which is 0.9, and then we take a square root. I've done that on the side. That's about 9.
- 8.26** Here's what this means. If I want to find the probability on an interval from 91 to 109, taking my mean of 100, subtracting 9, and adding 9, that would give me an interval of 1 standard deviation within the mean— 100 ± 9 . The mean minus the standard deviation plus the standard deviation. Let's see what that does. I'm going to find an interval from 91, that's the mean minus 1 standard deviation, to 109, that's the mean plus 1 standard deviation, and I want it to tell me the probability, and you get exactly, exactly 68%.
- 8.27** Why? Oh, that's like magic. This should be making connections in your head right now. One standard deviation of the mean on a normal distribution was exactly 68%. Here you have a binomial probability distribution that's approximated by normal. We had the mean of 100. You added the standard deviation, you subtracted the standard deviation, and you got exactly 68%. It's the same thing. The normal distribution approximates the binomial even down to the probabilities.
- 8.28** This is fantastic because now we can use a z-score to create a confidence interval, and this is what we're about to do. How? How are we going to build this confidence interval? We have to have a starting spot.

What I mean by that is we have to have a point estimate. Just like the sample mean was a point estimate for the population mean, we have to have a sample proportion, a point estimate.

- 8.29** We call this "p hat" $[\hat{p}]$. It's a funny little word, really, but it's a sample proportion. All it does is it says, "Look at your sample. Look at your sample. How many successes do you have?" Divide it by the total number of trials, your sample size, and you will have a proportion, just a fraction. Is this sample proportion a good estimate for your population proportion? It's the best one we have. It's the best single value we would have to estimate the population proportion.
- 8.30** Now, is it going to be exact? No. Just like the sample mean did not exactly estimate the population mean, the sample proportion is not going to exactly estimate the population proportion. It's our point estimate, but we don't know how exact it is, so we're going to build an interval around it exactly like we did for the sample mean. Are you seeing the similarity in what we did before and what we're doing? It's the same thing, but with a little bit different data. The point estimate, it's going to be an estimator for the population parameter, but we have to build an estimate based on confidence to say we're 90% confidence. This interval captures the true value of the population proportion.
- 8.31** Our normal approximation. Let's go through why we're able to do it. If x can be approximated, if the number of successes that we just saw on that binomial probability distribution that you just saw can be approximated by normal, if that's true, if the normal is based on a mean and a standard deviation, but now we know the mean. We know the mean is really just n times p , we know the standard deviation is really the square root of n times p times q .
- 8.32** You saw this, by the way. Right there on that JASP that we did, you saw that n times p was the center of that distribution. It was your mean. You saw that square root of n times p times q was the standard deviation. It gave you the 68% when I added it and subtracted it one time. It's the same thing.
- 8.33** You say, "This is fantastic, but really, I don't want to estimate the number of successes anymore." That's what binomial did. We don't have to go any further than that. Now, what I want to do is I want to estimate the proportion. Proportions are found by dividing the number of successes by your overall sample size. Let's divide everything by n . Well, if I do that across the board, x over n is going to be your sample proportion.
- 8.34** But now on the right-hand side, we see that I'm going to be able to simplify my n 's. When you divide a number by itself, it creates 1. I'm going to cancel out my n 's. It's a little bit fancier with a square root, but I'm going to pretend that my denominator is a square root of n squared. It's going to cancel one of those n 's, put the other one on a denominator inside of the square root, and give me this. This right there is the normal look for a binomial probability distribution based on a proportion. It's the same exact thing that we just saw. It even gives you a standard error at the very end of it.
- 8.35** How are we supposed to find a confidence interval? Well, just like before, we have to have a confidence level. These are going to look really familiar. We have a 90%, we have a 95%, we have a 99%, in your head, you're thinking, why are we using z ?
- 8.36** Why would that be the case that z is a good estimator? Well, because the binomial probability distribution is not estimated or approximated by t or chi-squared, it's by the normal. We just saw a normal curve, not a t curve. We saw a normal distribution fit that binomial. This is why we're able to use the z -score and why we studied it so much. Then you probably felt really bait-and-switched a while ago and said, "Hey, study the z , study the z , study the z ." T. You're like, "What the heck?" This is why. This is why. Because a normal curve approximates the binomial. Not a t , a normal. We use z -scores again.
- 8.37** Same exact confidence levels, same exact idea of a level of significance, this alpha $[\alpha]$, same stuff, same area in the tails, same two tails that we're going to have, and same z critical values that we're also going to have: 1.645 for 90%, 1.96 for 95%, 2.575 for 99%. Folks, this is basically it. All we have to do now is find a point estimate. Oh, we've got it, p hat. We've got to somehow come up with a margin of error. We're going to do that in just a moment.
- 8.38** We're going to take our typical z -score. We're going to manipulate it with our p and our n . Instead of having a true sigma $[\sigma]$, we'll have a standard error based on p and q and n . It's going to look a little different, but we've already done all the legwork on this. Just remember, these are always two-tailed, and now we can come up with a formula.

- 8.39** Our z formula that we had for confidence intervals from before was the first one. It was our take your mean—or, we're no longer working with mean. That doesn't make sense. We want to calculate an interval for $\hat{p}$, for our sample proportion. First one says take your sample mean, subtract your population mean, and then divide by your standard error. This is exactly what we transitioned to in the second part of it.
- 8.40** Instead of a sample mean, our point estimate is now sample proportion. Instead of the population mean, we have the parameter of population proportion. They're pointing to the same things— $\bar{x}$ points to μ , just like $\hat{p}$ points to p . It's the sample estimate, and then you subtract the true parameter from it. The bottom, the denominator, is just your standard error. It's the second. It's analogous to the same thing. It's how we built it.
- 8.41** From what we did before, working with the formula, changing our mean's normal distribution into a normal distribution that approximates a binomial for our $\hat{p}$, for our p —that's our standard error. We just found it. Now, the whole point is, let's go ahead and, just like we did before, let's build an interval that captures the p .
- 8.42** You see, you're going to know the z . You are going to tell it what you want it to be. If you want to be 90% confident, you're going to tell that z , "I need you to be 1.645." If you want to be 95% confident, you're going to tell that z , "I want you to be 1.96." We're going to know the p , we're going to know the q and the n and the $\hat{p}$. What we don't know is the true value of the p . What we're going to do is multiply both sides of this.
- 8.43** Margin of error looks really similar to what we did before. We're just transitioning into using p and q and n instead of standard deviation over the square root of n . If you see on the top equations there, our standard error, standard deviation over square root of n , changes into square root of p times q over n . We did that just a little bit before. Same idea, it's a standard error. It's giving you the spread of your distribution. That's all it's doing. It's just now it's from binomial, which is why it looks a little funny.
- 8.44** Our margin of error changes a little bit also. Notice the change is identical. σ over square root of n became square root of p times q over n . In our margin of error, σ over square root of n becomes square root of p times q over n . Now, unfortunately, we don't have the actual p . We don't have the actual q . We don't know the population. We're not going to know it, but we do have sample estimates. We have $\hat{p}$ from our sample, and we have $\hat{q}$.
- 8.45** Now, get this. This is for the next course. Because of this, because this is an approximation, and because of this right here that we're not going to really know p and q , the confidence interval approach is not as good as a hypothesis test approach for testing claims. For mean and for variance, it's going to be, but not for proportion, because of this right here. When we get there, I'm going to refer back to this. I'm going to say, "Hey, remember the binomial is approximated by normal?" I'm going to say, "Remember how we didn't really know the true value of p and q ?" This is why the confidence interval approach will not satisfy as well as a hypothesis test approach when we get to hypothesis testing for proportions. For means and for variance, yes, we're good.
- 8.46** Now we get to build this. The same exact idea holds true from sample mean to sample proportion. We're going to take that sample proportion. We're going to subtract the margin of error. It's going to give us a lower bound. We're going to take that sample proportion. We're going to add the margin of error. It's going to give us an upper bound. That margin of error is based on your critical value, that you tell it, "I want to be 90% confident," we're there. We're going to have a $\hat{p}$, that's from your sample. We're going to have a $\hat{q}$, that's from your sample. You're going to have an n , that's from your sample. We build this interval around our sample data that we're 90% or 95% or 99% confident captures the true value of our population proportion.
- 8.47** Let's say that we want to poll 100 people. We go out and we survey 100 people that are randomly selected. We ask them a very simple question, "Are you voting for candidate A?" Forty-three of them said, "Yes." We're using some very round numbers here, easy numbers so we can understand the math without having calculators, which is nice.
- 8.48** The n , our number of trials or sample size, same thing here, is 100. The x , the number of successes, whether you like candidate A or not, really doesn't matter. The success is just the interest that you're looking for. Forty-three percent satisfied the interest. That would be called success. Our sample proportion, it's really easy to find. All we're going to do is take our sample successes, divided by our

sample size, the number of trials, your people were your trials here, and you get 0.43, 43%. That's our sample proportion of success.

- 8.49** Well, the failure is pretty easy. If 43 said yes, well, then 57 said no, and we would have a 57% failure rate, or the $\hat{q}$ would be 0.57. This is the sample proportion of failure. Now we have to tell it, how confident do we want to be? I want to be 95% confident here so we can use some new numbers. Ninety-five percent is really standard. When you see a confidence interval for a poll or a proportion, a confidence interval for that on the news or something, typically they're using a 95% confidence interval. Unless they say otherwise, we pretty much assume that it's 95%. We're going to use that so we get very used to that.
- 8.50** Based on our data that we have, we have 100 for n , we have x is 43. We've got our sample proportion of success and our $\hat{q}$, and now we have our z critical value. This should look very familiar. This says that if you want to be 95% confident, you are going to have 5% in your tails, 2.5% on each side. Based on that, that z of 0.025, that critical value is 1.96. Same numbers we dealt with before, but now we're going to start building that interval.
- 8.51** Our margin of error. We start with our 1.96, that critical value. Remember, you're the one telling this formula what to do. You're telling it—well, I'm telling it, but I'm telling it, "I want to be 95% confident." That's the 1.96. More confident, larger z , larger interval. Then we take a square root of the sample proportion of success times the sample proportion of failure, divided by our sample size. Take a square root of that, we multiply by 1.96, and we get a margin of error of 0.097. Now, that doesn't look very large right now, but understand these are percentages. That would be a margin of error of 9.7%. A statistician would be rolling over in their graves right now. "You can't have a margin of error that big. That's crazy."
- 8.52** Well, now look at what would happen if you took a sample of more people. Think mathematically, what would happen if your sample size wasn't 100, but maybe 1,000? What's going to happen to the fraction? If the n was not 100, but 1,000, the denominator of your fraction is going to go way up. The value of your fraction is going to go way down. That interval is going to shrink. The margin of error is going to get smaller. If you want a better margin of error, what do you have to do? Take a bigger sample size. You can't change the z . You're not going to change your sample. You take a bigger sample size. Let's finish this, and then we'll talk about how to actually do it.
- 8.53** If we wanted to construct a 95% confidence interval based on our sample proportion of success, we have our margin of error. Now, it's a very simple matter. You say, "Here's my sample proportion. I know it's a point estimate for my population proportion, but I know it's not perfect. Based on a 95% confidence level, I'm going to construct a margin of error that subtracts and adds. This interval I'm constructing will contain 95% of the time for all these that I construct, the true value of the population proportion of people who in our nation would have supported candidate A." That's what we're saying here.
- 8.54** Let's take our 0.43. We'll subtract our margin of error. We'll take our 0.43, we'll add our margin of error. This has given us our interval, and we get this, 33 to 52%, 33.3 to 52.7%. Here's what this says. We are 95% sure that interval captures what the nation as a whole would think about the candidate. That is very, very broad.
- 8.55** Actually, you're not going to see it this way, are you? You're going to see it this way—43%, that's your point estimate, plus or minus your margin of error. This is typically how these are reported. You'd say, "Okay, hey, 43% of the nation thinks they're going to support candidate A, $\pm 9.7\%$. Oh, that's huge. That means that could be going either way. It crosses 50% even. If there's a candidate B, they could also be in the same exact interval."
- 8.56** How do you make this better, narrower? Well, we take a bigger sample size. The question is, how much? If this is what we mean, 95% of our confidence intervals that are constructed this way are going to contain our population proportion. In this case, that this interval from 33.3% to 52.7%, we're [95%] sure that's going to contain this true population proportion, we need a larger sample size. That's going to lower our margin of error. How do we do that? What if we just took our margin of error and we solved it for our sample size.
- 8.57** Why would you ever do that? This is truly what statisticians can do, what pollsters can do. They can tell it the margin of error that they want. They say, "Okay, here's what I would like to do. I would like to have a margin of error of no more than 2%. I'm going to know my z -score, aren't I? It's going to be 1.96 because it's going to be 95%. I might actually not know my sample probability of success or my sample probability of failure. I might not know that." But I can solve for n , and we can estimate those things.

- 8.58** Let's solve for n and see what this does. If we want to solve for n , we're going to do some fancy-pants math here. We're going to divide both sides by our critical value. Dividing both sides by our critical value, you'd have your margin of error divided by your critical value equal to your standard error. We're going to square both sides. That's how we get rid of a square root. The inverse of a square root is a square. We're going to square both sides. Taking both sides to the power two, we're going to get this whole expression, raised to the second power, equals our p hat times our q hat over n .
- 8.59** Remember what we're doing? We want to get the little n by itself. That's what we really want. Right now, it's not. Right now, it's at the denominator, the bottom of a fraction. This is no bueno. We're going to reciprocate. That means just flip over your fractions. Reciprocate both sides of our equation here. Our E is going to be moved to the denominator. Our critical value is going to be moved to the numerator. We're still going to square them. Our n is going to move up. Our p hat, q hat is going to move down.
- 8.60** Oh, there's only one more step. The last step is we just have to multiply by the denominator to cancel that out on both sides. We're going to multiply both sides by p hat, q hat. This is how you estimate for sample size. This is what this says. You start this typically before you even do your sample, and you tell it what I want my margin of error to be for my poll, for my survey. Say, "I want my margin of error to be 2%," that would be 0.02. I want my z -score to be 1.96. Why? Because it's a 95% confidence interval. If you wanted 99, you'd use 2.575.
- 8.61** Now, the p hat, q hat is a little concerning. Why is that a little concerning? Well, why would you find a sample size after you already have an estimate for your samples? Because those are from your samples. So either you do it twice or you start with no estimate. No estimate is the most conservative way to do this, and you just pick 50/50. That would give you the largest sample size possible. So I'm going to show you both of them.
- 8.62** Let's go ahead. We want to find a confidence interval that has at most a 2% margin of error. I also want a 95% confidence interval. Well, what that means, if I want a 95% confidence interval, my z critical value has to be 1.96. That's the numerator. My margin of error has to be 0.02. That's my denominator. I'm going to square that.
- 8.63** If I have a sample estimate, maybe from a previous sample, maybe from some other information I know, I can put that in. I say, "Oh, yeah, my probability of success was 0.43, probability of failure was 0.57," and it's going to tell you 2,353.9. You can't take 0.9. You can't survey 0.9 of a person. Even 0.1. So whatever this is, you round it up by one. You always round this up. So we would have to sample at least 2,354 people to have that margin of error.
- 8.64** Now, let's do the worst-case scenario. You have no estimate for your sample whatsoever. Use 50/50—0.5 for a p hat, 0.5 for a q hat. It will give you the largest sample size, and it would increase it just a little bit. Here's what this says. If you took a sample of 2,401 people and you wanted to construct a 95% confidence interval, you will have at most a 2% margin of error. You'll probably have less than that because your sample data will be less than that. It will probably not be 50/50, and so you will have a less-than-2% margin of error.
- 8.65** This is usually what people do before they even start their samples. They determine their sample size, they collect their sample size, then they build their confidence interval. That's a lot more than 100, isn't it? This is the nature of how you can use samples to estimate populations, though. Two thousand four hundred one people is not a small task to collect. If you had to randomly select these from a population, you're doing a lot of work already.
- 8.66** If you do randomly select them, and you do it well, and you maybe do a stratified sample, or you make sure that you survey every demographic of your population, those 2,400 people might actually represent your population fairly well. In fact, they're going to do it so well that there's only a 2% margin of error, that you can be sure that it would be 43% $\pm$ 2%. Now, you're in a relevant range. Now you say, "Okay, we're 95% sure that between 41 and 45% of the nation is going to support candidate A." That's much more meaningful. That is how you narrow the margin of error.
- 8.67** Now that we've conquered this, I'm going to talk very briefly about what's called the plus-four method. Then we're going to move on to our last confidence interval. The plus-four method is if you have very small sample sizes, that this technique can be inaccurate. It can be inaccurate because they are approximations. Oh, we talked about that, didn't we? We said it's approximately normal. We said the normal approximates it. We said you're not going to know your true value of your population parameters.

The plus-four method does something very basic. It just adds 4 outcomes. It adds 2 successes, and it adds 2 failures, and then it recalculates. By doing this, it makes it a little bit more accurate for very small samples.

- 8.68** Now we get to move on. We get to do our last confidence interval. Remember that we had really three sample statistics that pointed to our population parameters. We had sample mean, we found a confidence interval for that. We had sample proportion, we just finished that. Now we have sample variance. We're going to build a confidence interval for sample variance. We're going to build a confidence interval for population variance around our sample variance.
- 8.69** In order to do this, we have to have a different test statistic. We have to have a different critical value. We have to have something that looks like this. It's called the chi-squared critical value. Why do we need something different? Well, for means, we had a t value. We had a t value because that was symmetrical and it was very similar to a normal distribution. For proportion, we had a z value because the binomial was approximated by normal. That's symmetric. That's based on z.
- 8.70** Variance doesn't do this. In fact, when we looked at sampling distributions, we discovered that, yes, mean and proportion create these very nice bell-shaped curves that are symmetric. But variance did not do this. Variance had a distribution that was skewed to the right, tail-right. Always positive, never negative, but definitely not symmetric. Definitely not symmetric. Always skewed to the right. This is what the **chi-squared distribution** will show you.
- 8.71** You'll see it, you'll go, "Oh, my gosh, that's crazy." If you don't say, "Oh, my gosh, that's crazy," I'm going to feel really bad, by the way. Oh, my gosh, it's crazy. It's crazy. It's blowing my mind. Then all it does is one thing. It's a ratio of your sample variance to your population variance. That's it. Now, we want to estimate our population variance. We do not know our population variance. We're going to take this and we're going to manipulate this and we're going to build a confidence interval around this, but first we've got to understand this.
- 8.72** I want to go and take a look at JASP and see what this chi-squared distribution is all about. From our JASP program, if we go over to distributions, go all the way down. There's a whole bunch. Don't let it intimidate you. Some of these things we're not going to learn about in this class. That's okay. But we are going to learn about the F distribution. We'll get there another day. We're going to learn about the chi-squared distribution. We're going to do that right now.
- 8.73** We're going to take a look at what a chi-squared distribution looks like. We're going to start simply with just a sample size of 2. If you had a sample size of 2, your **degrees of freedom** would be 1. You just subtract 1. This is a chi-squared distribution for a **degrees of freedom** of 1. It looks nothing like a normal distribution. It doesn't even look like a—not at all, but you do notice a couple of things. It's never 0. It's never negative. It cannot be negative. It's always above the horizontal axis, and it tails right forever and ever and ever.
- 8.74** Let's start adjusting this. Let's look at a sample size of 3, degrees of freedom, 2. It starts to get a little flatter, and then something interesting happens, sample size of 4. It now has at least a little bit of a look of a distribution we're used to. As we go larger and larger, this is going to start to look more symmetric. This would be a sample size of 6. It's still very strongly tail-right. Here's a sample size of 10. I'm going to adjust our range so we can see it a little bit better. It's more symmetric, still tail-right. It's not normal.
- 8.75** Remember, it's not centered around a mean. It has nothing to do with the mean. This is a comparison of two different variances, sample to population based on a sample size. Let's look at 30. We'll adjust our range a little bit. It's getting a little bit more symmetric. We'll do one last one. We'll just go to a sample size of 61 and adjust this to 100 so we can see it. It's as far as I want to go, but that's about as symmetrical as we can make this. It's still tail-right, but it's definitely not a normal curve. Not yet. This is always going to be tail-right.
- 8.76** Now, let's start thinking about what we have. What we had was a sample size of 20. If we were thinking about our baby weights, if you remember that, we had a sample of 20 2-year-olds. We collected 20 weights from them. That would be a degrees of freedom of 19. Let's look at what this distribution would look like for a sample size of 20. I'm just going to adjust this to a range of 40 so we can get a better picture of it. What this means is that this is a chi-squared distribution for our specific sample. We have a sample size of 20, degrees of freedom, 19. This is what that looks like.

- 8.77** Now, here's what we're going to try to do. We're going to take our sample variance from that sample size and use it as a point estimate for a population variance and then build a confidence interval around it. But in order to do that, you know what this takes. You know that you have to have a margin of error, and that is going to be based on two critical values.
- 8.78** But the problem is this is not a normal curve. It's not even a t distribution. This is a chi-squared distribution. We're not going to be able to use 1.645 positive and negative. We're not going to be able to use 1.96. We're not going to be able to use t distribution values because that's not a t. That's not a z. That's not normal. That's not a Student's t distribution. That's a chi-squared distribution. It's not even symmetric. We're going to have to find two different values.
- 8.79** I'm going to show you. I got these from a table. I used other work to do this, but I'm going to show you what they are so you know where they're coming from. I want to show you the probability from a chi-squared critical value of 8.907 to 32.852. You're probably wondering why I'm putting them in backwards. If I don't do that, it gets out of order and it makes a big red sign on the screen that I messed up, and I don't want to mess up. I'm going to do it this way.
- 8.80** I want you to look at these two values. How much proportion, probability, or area of this curve do they capture? Ninety-five percent. If I take a chi-squared critical value of 8.907 and a chi-squared critical value of 32.852, that interval is going to capture 95% of the area under this curve, which is totally equal to 1.
- 8.81** You have no idea where those values come from. That is okay. This will do it for you. I got this from different technology. I'm telling you what these are for this particular distribution so that we can work with it, so I can show it to you. Then I'm going to show you how JASP will create it for you. These are coming from a table that I looked up so that we understand where this is working from.
- 8.82** Here's what you need to know. Based on our sample variance and these two chi-squared values, we're going to be able to build our confidence interval of 95%. Why 95? Because that's what those test statistics/critical values are telling us.
- 8.83** Now it's time to get to work. This chi-squared critical value, based on this, this is what forms the chi-squared distribution. It's what you just saw. On that distribution, we saw two values, 8.907, we saw 32.852. Now, let's go ahead and build this.
- 8.84** Understanding they're not symmetric, understanding that those two values are not going to be just one positive, one negative, they're not going to be opposites of each other—that's okay—understanding that this changes for every single sample size, as you have seen, understanding that we made this distribution based on our sample size of 20, degrees of freedom of 19, that it becomes more symmetric with size, we can now use this to estimate our population variance.
- 8.85** Let's take this critical value. It's also a test statistic for hypothesis testing, and we're going to solve for what we actually want to estimate. Folks, you've seen this now. This is the third time. We're taking this thing and solving for what we want to know. You are going to know the chi-squared. In fact, you already know it. In fact, you know two of them. You know the left one and the right one, the 8.907 and the 32.5, whatever it was.
- 8.86** You are going to know your sample size. You already know your sample size. It's 20. Minus 1 is 19. It's your degrees of freedom. You're going to know your sample variance. We'll be able to find that. It's super easy. What you don't know is the population variance. This is what you're trying to estimate. We're going to solve for it.
- 8.87** This is the easiest one to solve. We're ending on a high note. This is very easy to solve for. All we have to do is multiply both sides by the population variance, our denominator, and then divide both sides by chi-squared. It's just going to exchange them. That's all that's going to happen. Because you've now solved for our population variance, and because we have both of these test statistics—one big, one small, these critical values, one big and one small—let's bracket this. This right here is what's going to build your 95% or 90% or 99% confidence interval for your population variance. You'll know your sample size, you'll know your sample variance, your denominators. You see that there's two of them.
- 8.88** Why are there two of them? Well, we have to have a lower bound, and we have to have an upper bound. Which are you going to put where? We saw on our JASP just a moment ago that in order to contain 95%

of our probability or proportion, we had to have two test statistics, two critical values. One was 8.9-ish, one was 32 point, whatever it was-ish. Those bracket the 95%.

- 8.89** We need to have the bigger one on the left and the smaller one on the right. Why? Fractions work like this. The larger your denominator, the smaller the fraction. We need our larger of those critical values on the left bound and the smaller of those critical values on the right bound. Otherwise, our interval is going to be backwards. I don't know if you remember those numbers. I certainly had trouble with doing that, so I made sure I wrote them down.
- 8.90** Let's look at our sample one last time. Our baby weights sample, we had a sample of 20 kids. We found the sample mean; it was 25.87. We also found the sample standard deviation, and we had, just right now, found our chi-squared critical values for a 95% confidence interval based on that particular sample size, which changes the shape of the distribution. The left one was 8.907. The right one was 32.852. All we have to do is put them in the right places. We're going to be fine.
- 8.91** We'll take our degrees of freedom times our standard deviation squared or variance, if you have that, JASP will give it to you easily, and divide by the appropriate critical value in the appropriate denominator, big one on the left, small one on the right. That's all we've done here. Same exact stuff. What that's going to tell us is this. When all the math is done, this says that we are 95% sure that that interval from 3.331 to 12.28187 captures the true value of the population variance.
- 8.92** This is variance. The question in your head is, okay, if this is variance, how do I get to standard deviation? Because variance doesn't have the same units as what I want to see. I want to see the standard deviation. Just take a square root of everything. We take a square root of something squared, so the square root of variance is standard deviation. If we take a square root of all three of these segments of this confidence interval, we get 1.825 to 3.505. It says the same thing. We're 95% sure that that interval is going to capture the true value of the population standard deviation. Or 95% of the confidence intervals we make this way will capture the population standard deviation.
- 8.93** One last thing we want to talk about, and then we're going to be done. Now that we understand how this chi-squared distribution looks, where these critical values came from, what was to capture 95%, we're going to talk about how to actually do this practically. If you looked here, there was no way that you could just look there and say, "Of course, it's 8.9. Any idiot could see it's 8.9." No one does that. You have to have technology to do that or a table to do that. Don't feel like a dummy if you didn't know where those came from. That's okay. See, I came up with those ahead of time.
- 8.94** JASP will do this for you. If we take our data, take a look at it. Here is all of our data. We're pretty used to it at this point. We use it for a few examples now. There are 20 individual data values. Our degrees of freedom is for sure 19. But if we go to descriptive statistics, we start putting over our 2-year-old baby weights. It gives us our mean, it gives us our standard deviation. These are things we know. This is actually how we built this, but it will give us a whole lot more.
- 8.95** If we click on statistics, we scroll all the way down. Now, I'm not just talking about the confidence interval for the arithmetic mean. That's what we did last time. What we're doing now is, let's find a confidence interval for the variance. Yes, I do want it to be 95%. I do want it to use chi-squared. That's exactly what you just did. These are exactly the same numbers you have, disregarding rounding.
- 8.96** Let's make it even better. I want a confidence interval for the standard deviation. Yes, I do want 95%. I want it by chi-squared. That's exactly what you did. These are precisely the same numbers that you just found, and it's done in the blink of an eye. Now, to wrap things up, what's going to happen if you make this interval larger? Your interval here gets larger. More confidence, larger interval to capture that.
- 8.97** As we finish up with our chi-squared distribution, we're going to learn in the next course that it's used in lots of applications. We're going to use this in a **goodness-of-fit** test, an independence test, in **contingency tables**. We're going to use it, and now we have a really good feel for it. We understand that confidence intervals give ranges. They don't point to individual values, and they're a very good way to make inferences about our populations, which is exactly what we're trying to do. These larger estimates that we have are going to give us better estimates, which gives us a better inference. That's it.